GOOGLE WORKSPACE TRAINING



Google Sheets Advanced

Two-day course review draft

Analyse a fictional procurement register. Labs1–2 include67 captured stages covering source audit, conditional totals and counts, manual subset checks,21 logic cases and an isolated mixed-reference fault with verified repair. Labs3–12 still need author execution and captures; independent/reviewer checks and learner timing remain pending.

Delivered by Asraf · Microsoft Certified Trainer

Google Sheets Advanced

Two-day course review draft · Review draft, version 0.4

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Resources: Use the practice files supplied with this review package.

Google, Google Workspace, Gmail, Google Drive, Google Docs, Google Sheets, Google Slides, Google Sites, Gemini and NotebookLM are names associated with Google. This independent training guide is not published or endorsed by Google.

Screenshots show the product at the time of writing. Google updates its apps often, so labels, availability and layouts in your account may differ. This review draft still requires full lab validation.

About the author

Asraf is a Microsoft Certified Trainer and HRD Corp-certified IT trainer based in Malaysia. He has been teaching working professionals since 2008, helping teams at banks, pharmaceutical companies, construction firms, and retailers get real work done with Microsoft 365, Power Platform, Excel, Google Workspace, and, more recently, AI tools like Copilot.

He holds a Bachelor of Information Technology (Multimedia) from the University of Malaya, along with Microsoft certifications in Power BI (PL-300) and AI for business (AB-730). Outside the classroom he writes his own automation tools in Python and Node.js, mostly to save himself from doing the same task twice.

He runs his training practice solo, which means he has personally felt every inefficiency this book tries to fix.

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INTRODUCTION

Before you begin

This review book accompanies Google Sheets Advanced. Work through its chapters in order and keep the original practice inputs unchanged. The same notes and prompts also appear in the local course website.

Current Status

All chapters in this book are written drafts. Most have not yet been executed end to end in the application. The masterclass's reference brief chapter has a tested core path with a recorded recovery prompt; that does not validate the other exercises. Treat VERIFY notes and draft notices as production work still to complete.

What You Need

Use a desktop browser and the Google account approved for the class. Confirm app and AI access beforehand. Edition, administrator controls and language can affect availability.

Download the supplied fictional practice pack. No real client or personnel record is needed. Dataset week fields are relative practice weeks, not an agreed pilot start date.

How to Use the Exercises

Follow the numbered steps, attempt the output and then open the expected-result box. AI wording can vary; source facts and approval conditions must remain correct.

Values and prompts follow the notes. Replace source placeholders before submitting. Inspect proposed edits before applying them. Drafting does not authorise sending, inviting people or publishing.

Course Timing

This course is planned for 2 days, with 390 teaching minutes per day, excluding breaks and lunch. Estimates include practice, checks and discussion and need a classroom pilot.

Each course supplies independent starting inputs. Use a stated recovery file if an earlier output is missing.

Expanded Casework

Revision 0.4 adds casework, independent challenges and evidence gates to every chapter. The original short guided path alone is insufficient. Timings are targets, not measured learner results. Read each case input and keep baseline and scenario files separate.

01

CHAPTER 1

Audit the Source Register

In this chapter you will check types, blanks, duplicates and keys before analysis.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 45 minutes: 10 explanation/demonstration, 15 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 35 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: An audited source, preserved snapshot and empty Analysis workspace.

> **Status:** author reference with 30 staged application captures through 1.4. Native conversion, baseline diagnostics, preserved snapshot, empty Analysis, six raw intake cases, training contract and evidence-backed proposed dispositions checked. Upload required instructor handoff. Independent disposition/release-condition work, reviewer challenge and learner timing remain pending.

What You Will Learn

- Inspect dataset grain and keys.
- Check types and baseline controls.
- Preserve the source before analysis.

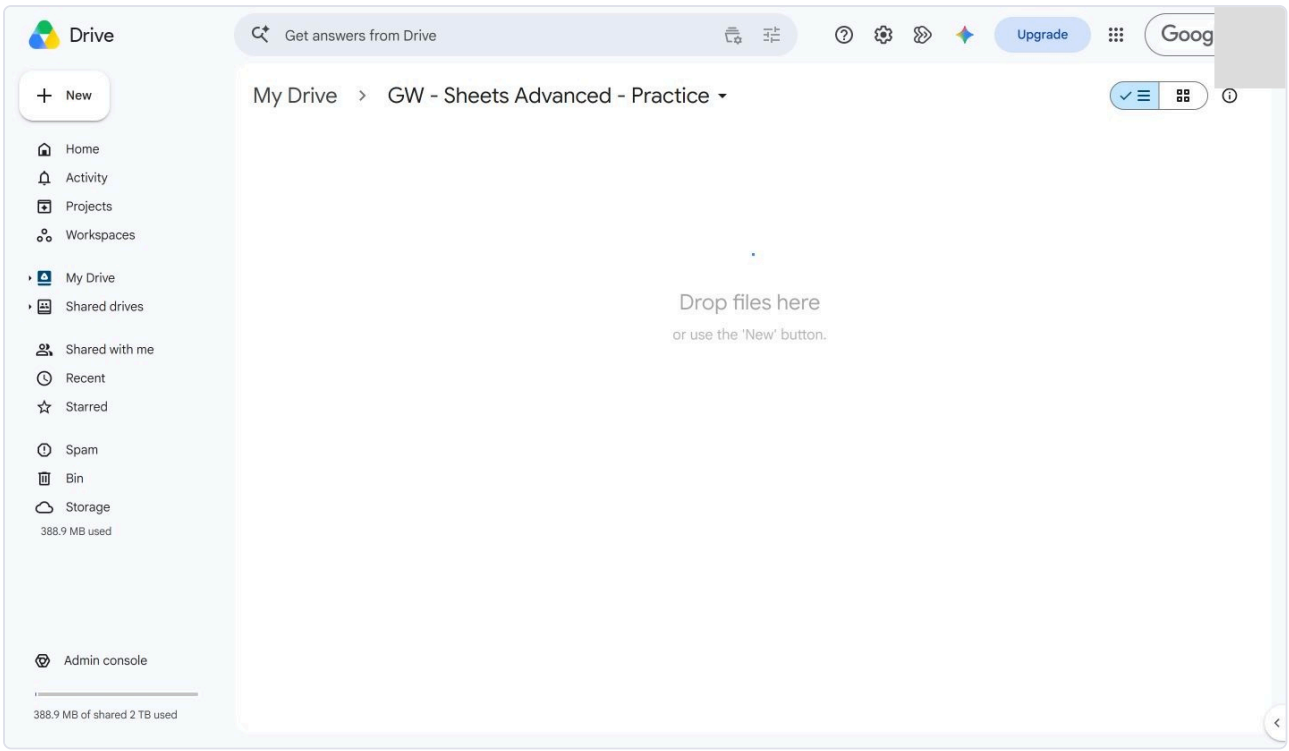
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

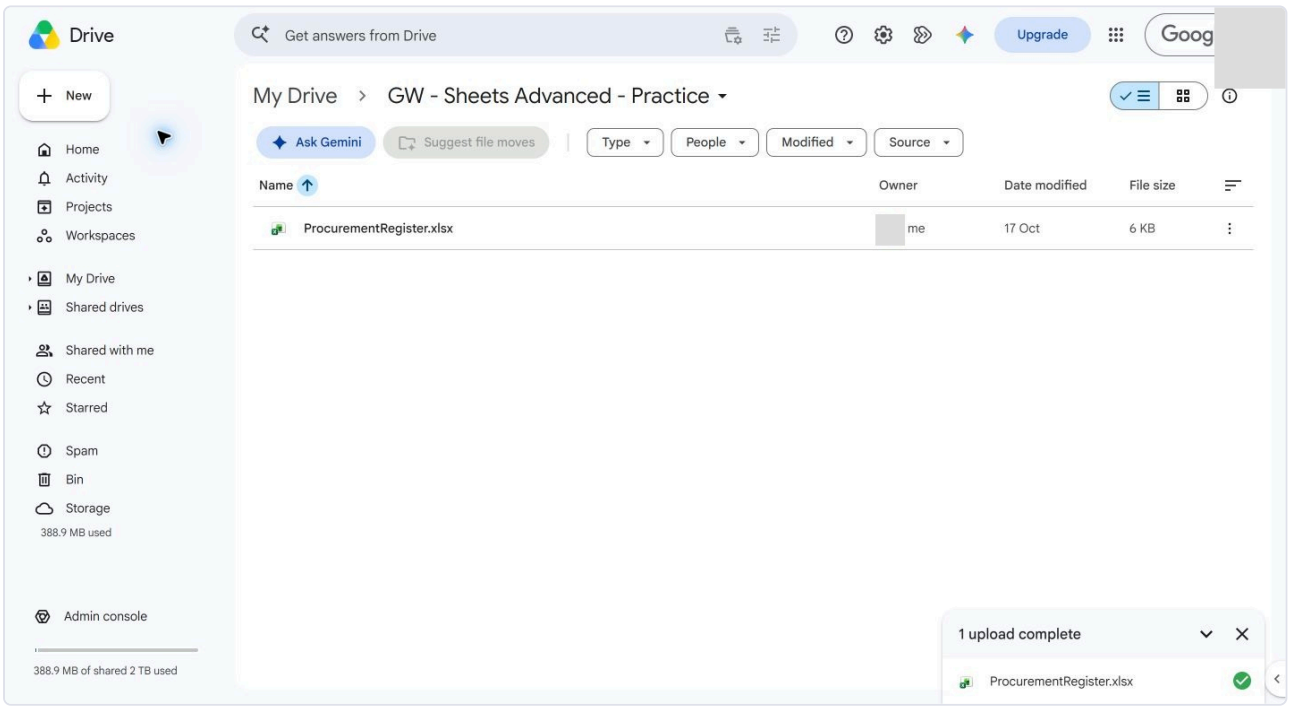
Download ProcurementRegister.xlsx. It contains 24 fictional order lines and a six-item rate table, separate from the pilot budget.

1.1 Prepare the Workbook

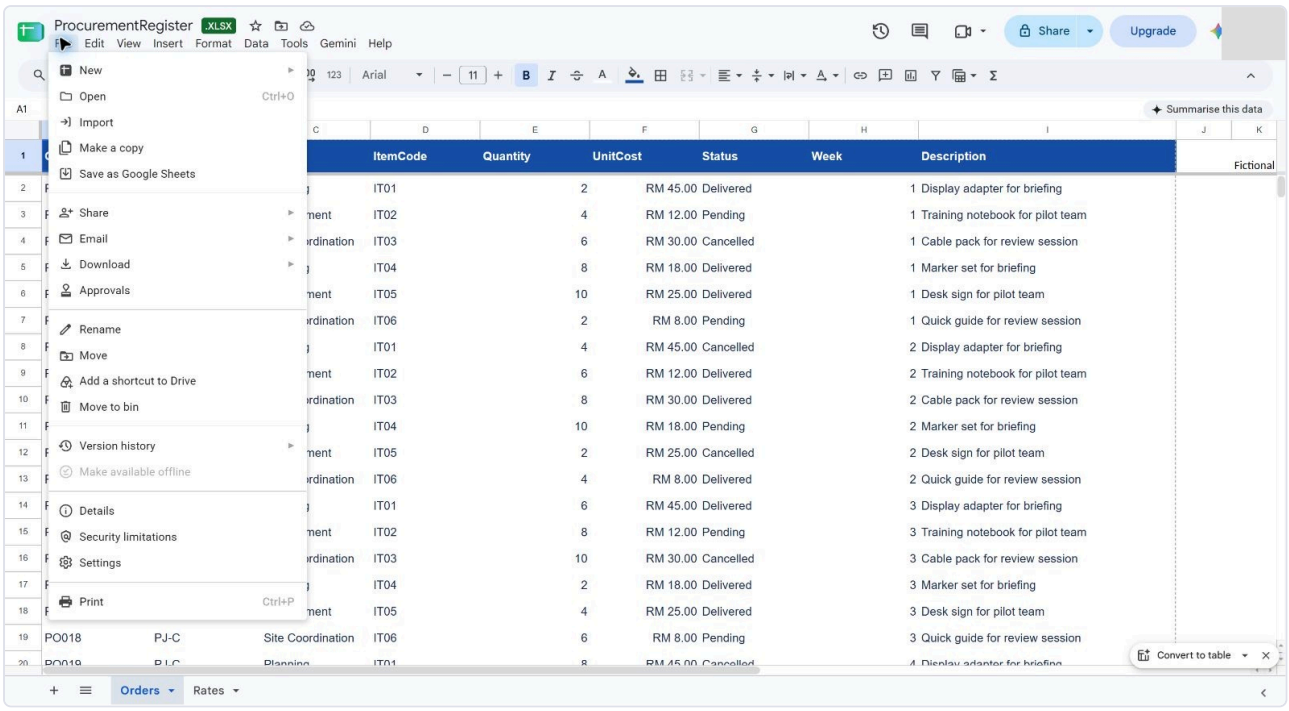
1. In the training account, create a separate **GW - Sheets Advanced - Practice** folder in My Drive. Keep it separate from the Foundation register.



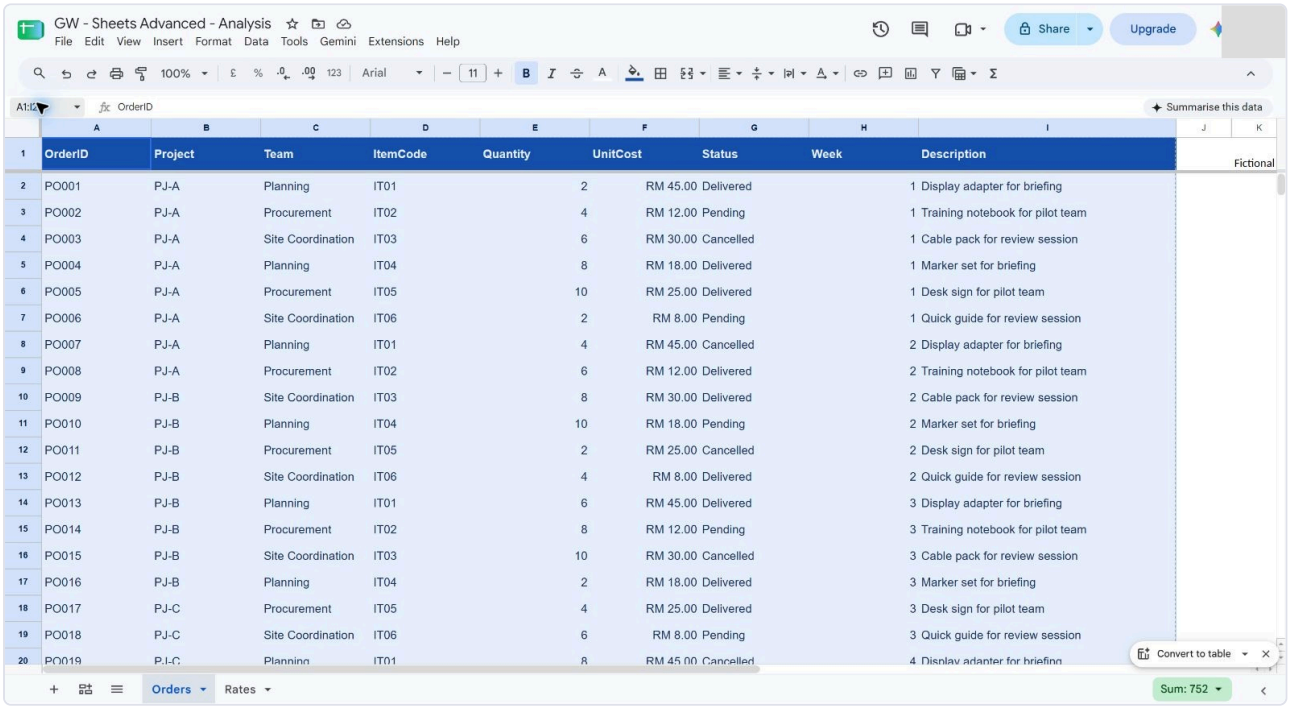
2. Upload **ProcurementRegister.xlsx** from the supplied practice files. In this author run the extension blocked local-file uploads, so the instructor uploaded the file manually. This is a handoff, not a verified automated upload.



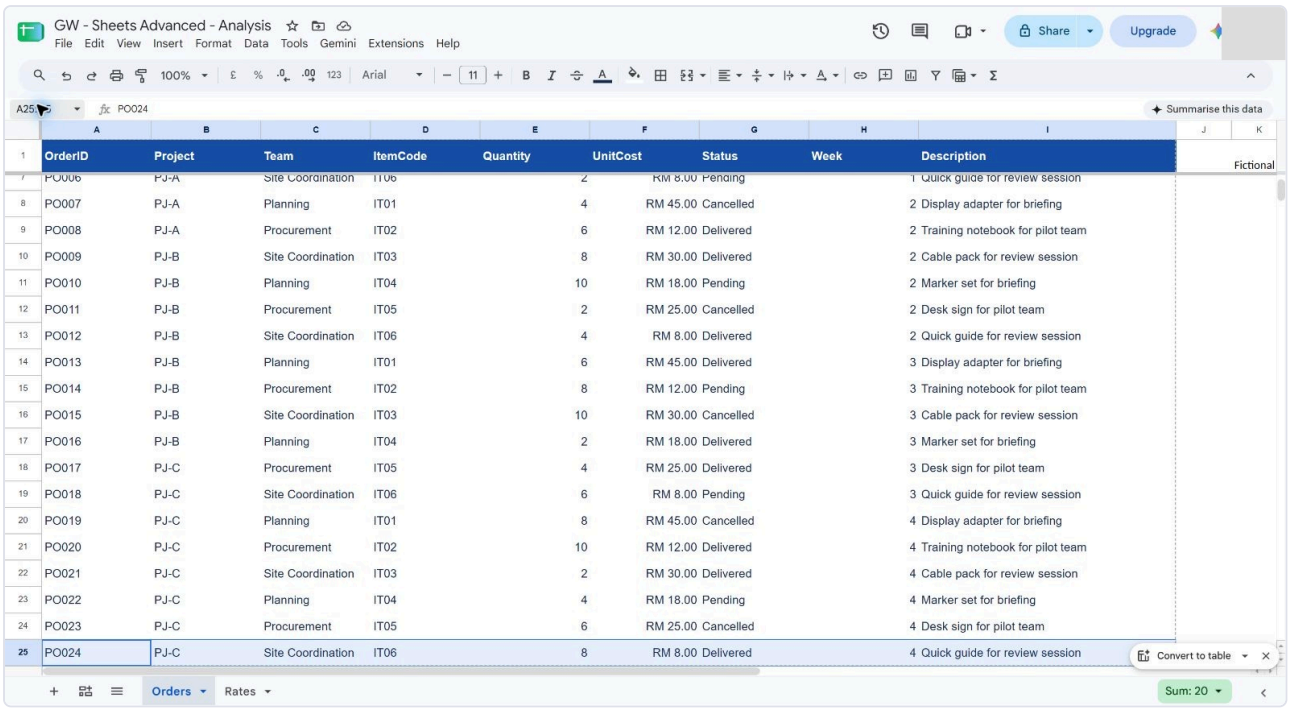
3. Open the Excel workbook and use **File > Save as Google Sheets**. Preserve the original XLSX. This creates a distinct native file; do not simply rename the Excel source.



4. Rename the native file **GW - Sheets Advanced - Analysis**. On **Orders**, confirm the nine working headers in **A1:I1** and inspect the records. The supplied **K1** fictional-data label is outside that working table.

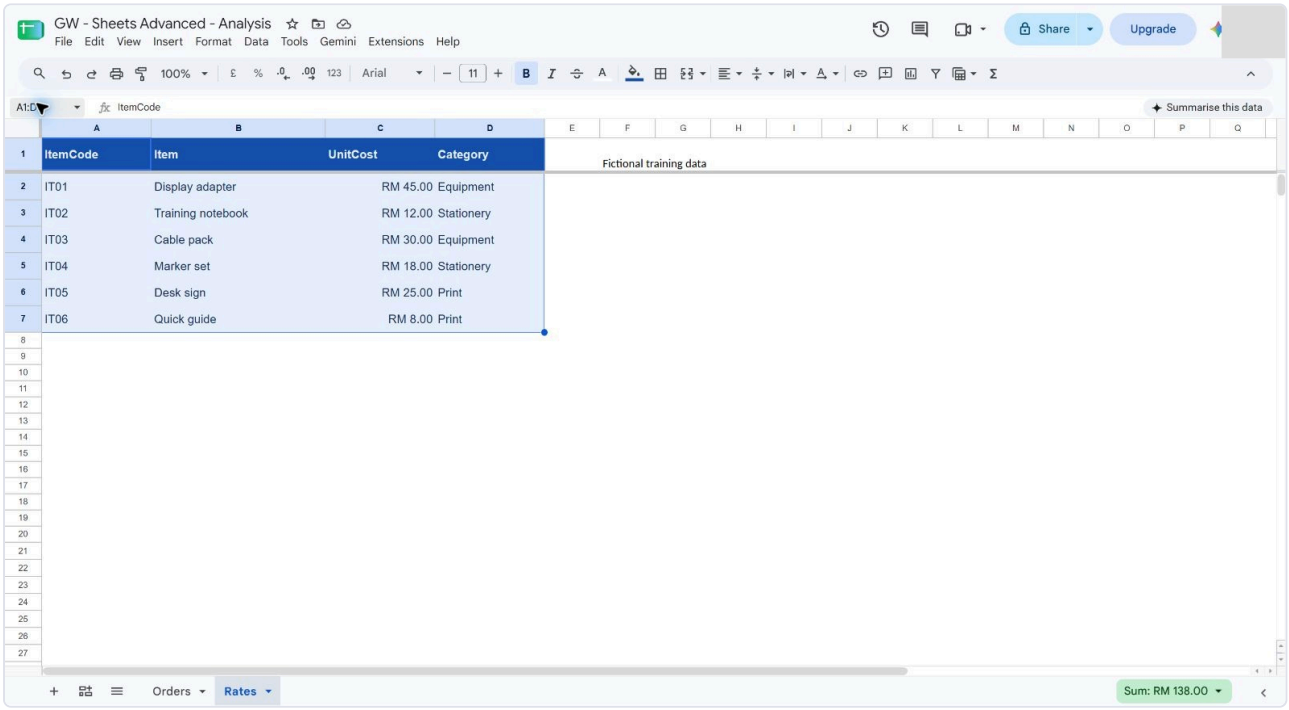


5. Navigate to **A25:I25**. Confirm **PO024** is the final supplied order, with 24 data rows in **2:25**. One row represents a fictional order line, not a project subtotal or purchase commitment.



OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description
PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session
PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing
PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team
PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session
PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing
PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team
PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session
PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing
PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team
PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session
PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing
PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team
PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session
PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing
PO020	PJ-C	Procurement	IT02	10	RM 12.00	Delivered	4	Training notebook for pilot team
PO021	PJ-C	Site Coordination	IT03	2	RM 30.00	Delivered	4	Cable pack for review session
PO022	PJ-C	Planning	IT04	4	RM 18.00	Pending	4	Marker set for briefing
PO023	PJ-C	Procurement	IT05	6	RM 25.00	Cancelled	4	Desk sign for pilot team
PO024	PJ-C	Site Coordination	IT06	8	RM 8.00	Delivered	4	Quick guide for review session

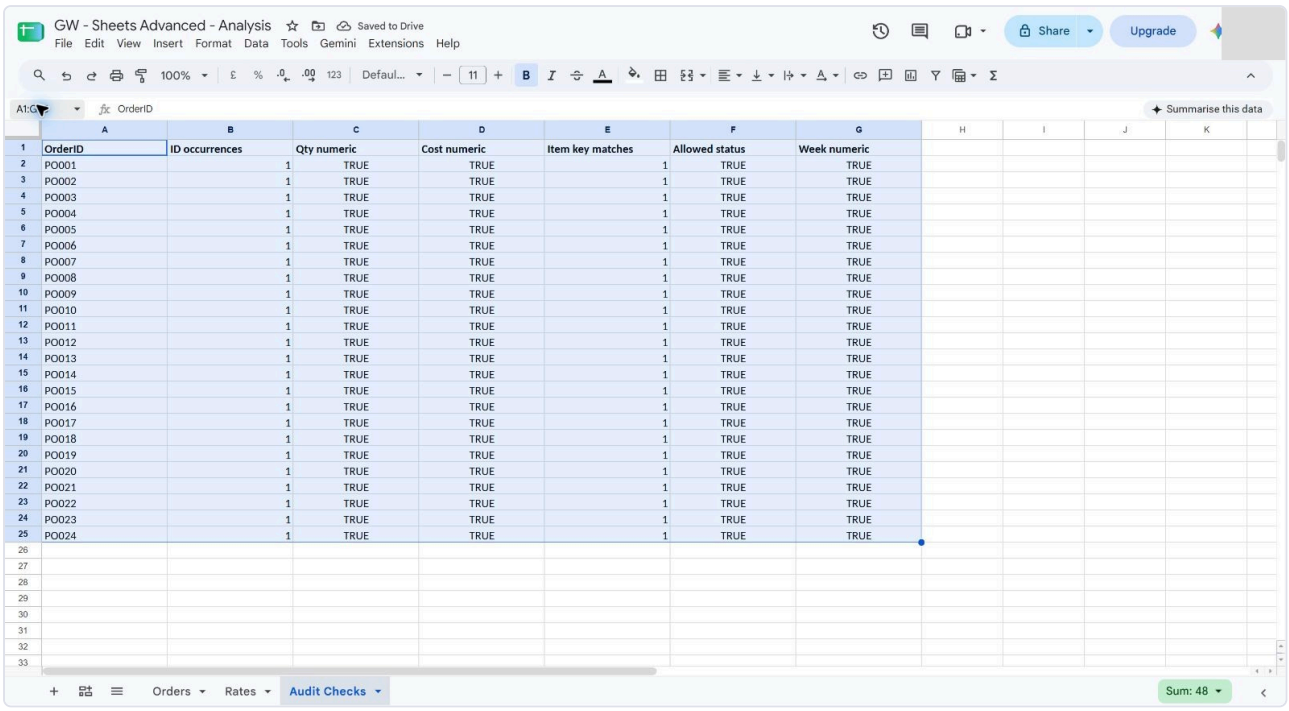
6. Open **Rates** and inspect **A1:D7**: six item codes **IT01–IT06** in A2:A7, with item, unit cost and category fields. Keep the supplied **F1** label outside the table.



ItemCode	Item	UnitCost	Category
IT01	Display adapter	RM 45.00	Equipment
IT02	Training notebook	RM 12.00	Stationery
IT03	Cable pack	RM 30.00	Equipment
IT04	Marker set	RM 18.00	Stationery
IT05	Desk sign	RM 25.00	Print
IT06	Quick guide	RM 8.00	Print

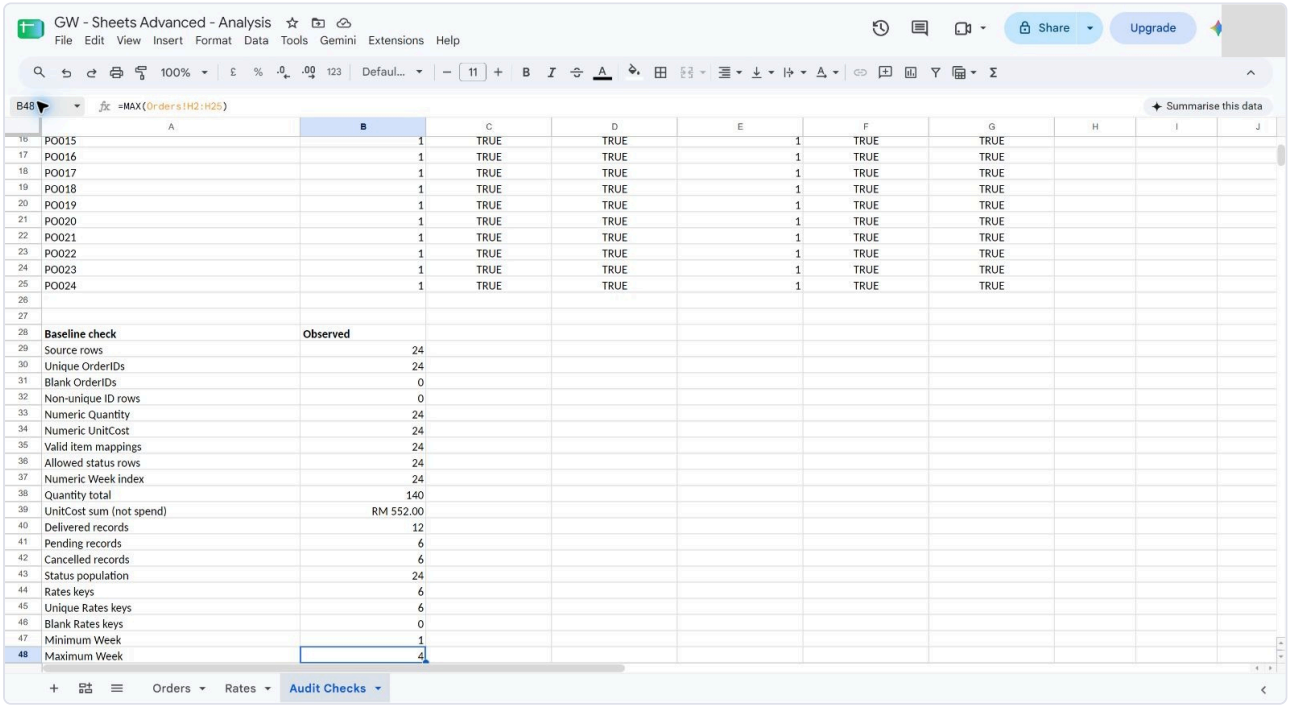
1.2 Audit Key Fields

1. Create **Audit Checks** so the diagnostics do not alter Orders. Paste the baseline diagnostic card from the copy-paste page into **A1:G25**. It references each source row and checks ID occurrences, numeric Quantity/UnitCost/Week, exact item-key counts and allowed statuses. Make A:G wide enough to read the headings; the first captured stage uses 155-pixel columns.



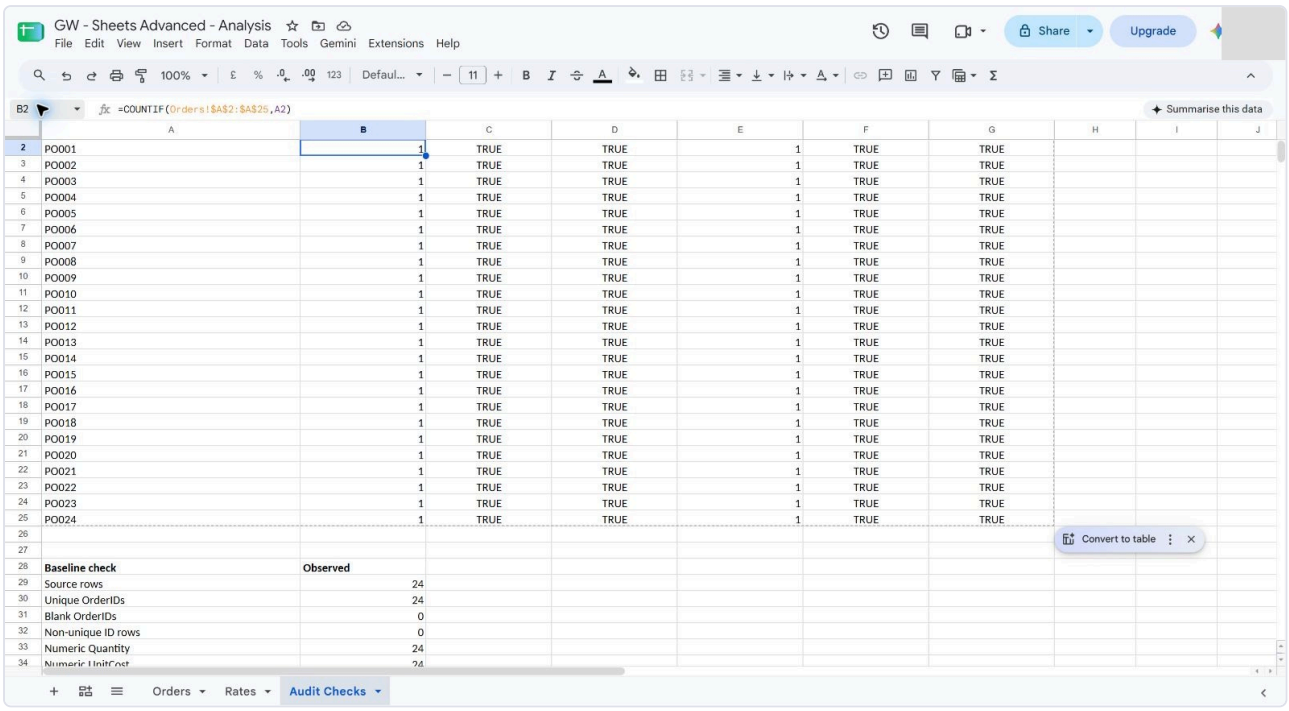
OrderID	ID occurrences	Qty numeric	Cost numeric	Item key matches	Allowed status	Week numeric
PO001	1	TRUE	TRUE	1	TRUE	TRUE
PO002	1	TRUE	TRUE	1	TRUE	TRUE
PO003	1	TRUE	TRUE	1	TRUE	TRUE
PO004	1	TRUE	TRUE	1	TRUE	TRUE
PO005	1	TRUE	TRUE	1	TRUE	TRUE
PO006	1	TRUE	TRUE	1	TRUE	TRUE
PO007	1	TRUE	TRUE	1	TRUE	TRUE
PO008	1	TRUE	TRUE	1	TRUE	TRUE
PO009	1	TRUE	TRUE	1	TRUE	TRUE
PO010	1	TRUE	TRUE	1	TRUE	TRUE
PO011	1	TRUE	TRUE	1	TRUE	TRUE
PO012	1	TRUE	TRUE	1	TRUE	TRUE
PO013	1	TRUE	TRUE	1	TRUE	TRUE
PO014	1	TRUE	TRUE	1	TRUE	TRUE
PO015	1	TRUE	TRUE	1	TRUE	TRUE
PO016	1	TRUE	TRUE	1	TRUE	TRUE
PO017	1	TRUE	TRUE	1	TRUE	TRUE
PO018	1	TRUE	TRUE	1	TRUE	TRUE
PO019	1	TRUE	TRUE	1	TRUE	TRUE
PO020	1	TRUE	TRUE	1	TRUE	TRUE
PO021	1	TRUE	TRUE	1	TRUE	TRUE
PO022	1	TRUE	TRUE	1	TRUE	TRUE
PO023	1	TRUE	TRUE	1	TRUE	TRUE
PO024	1	TRUE	TRUE	1	TRUE	TRUE

2. Paste the population-control card into **A28:B48**. Confirm **24 rows / 24 unique IDs / 0 blanks / 0 non-unique ID rows**, 24 numeric values in each required field, 24 matched item keys, and six unique nonblank Rates keys. The status population is **12 Delivered / 6 Pending / 6 Cancelled**, total 24. Quantity sums to **140**; the unit-price sum is **RM552**, which is a diagnostic checksum, not spend. Week is the numeric practice index **1–4**, not a date. Use a wider label column and navigate to B48 to see the complete control block.

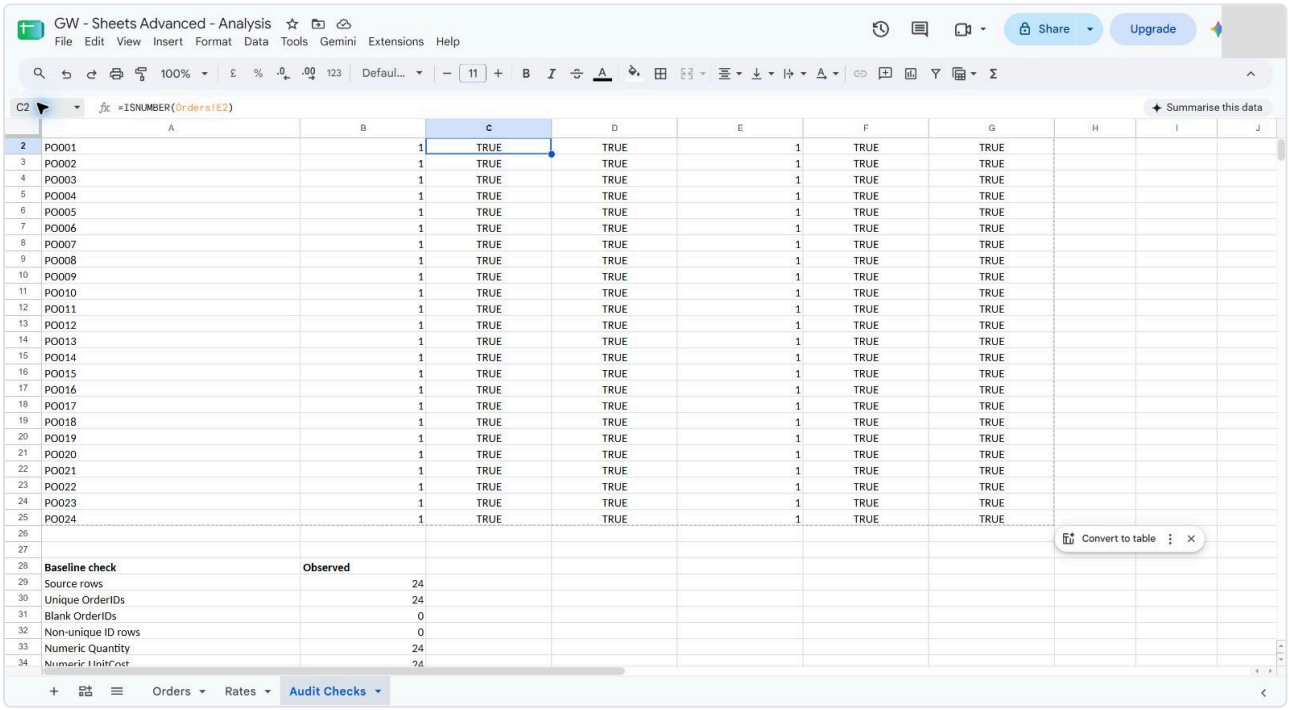


Baseline check	Observed
Source rows	24
Unique OrderIDs	24
Blank OrderIDs	0
Non-unique ID rows	0
Numeric Quantity	24
Numeric UnitCost	24
Valid item mappings	24
Allowed status rows	24
Numeric Week index	24
Quantity total	140
UnitCost sum (not spend)	RM 552.00
Delivered records	12
Pending records	6
Cancelled records	6
Status population	24
Rates keys	6
Unique Rates keys	6
Blank Rates keys	0
Minimum Week	1
Maximum Week	4

3. Inspect **B2 =COUNTIF(Orders!$A$2:$A$25,A2)**. Each baseline ID occurs once. The anchored source population stays fixed while each row tests its own relative ID. A count of one inside intake alone would not prove that an ID is new to Orders; the later casework checks both populations.

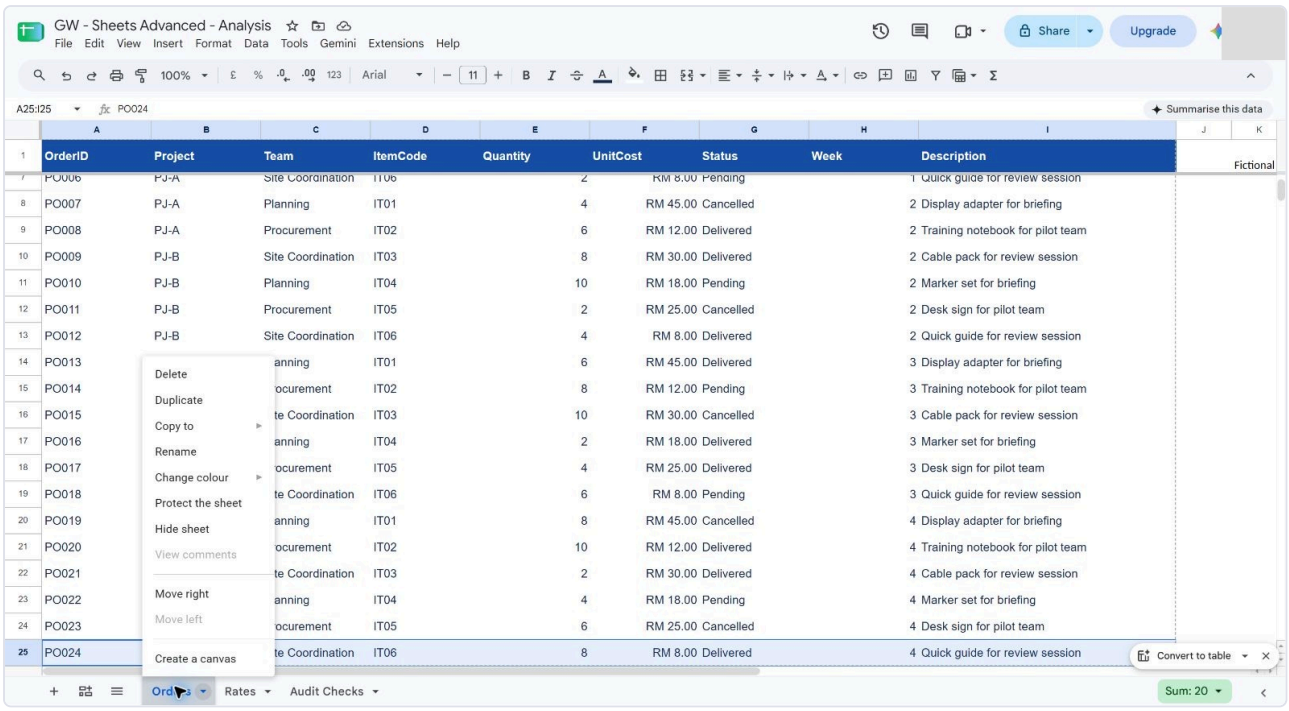


4. Inspect **C2 =ISNUMBER(Orders!E2)** and its results down to row 25. Verify Quantity and UnitCost are actual numbers. Keep Cancelled records visible; validity and the population chosen for a later question are separate decisions.

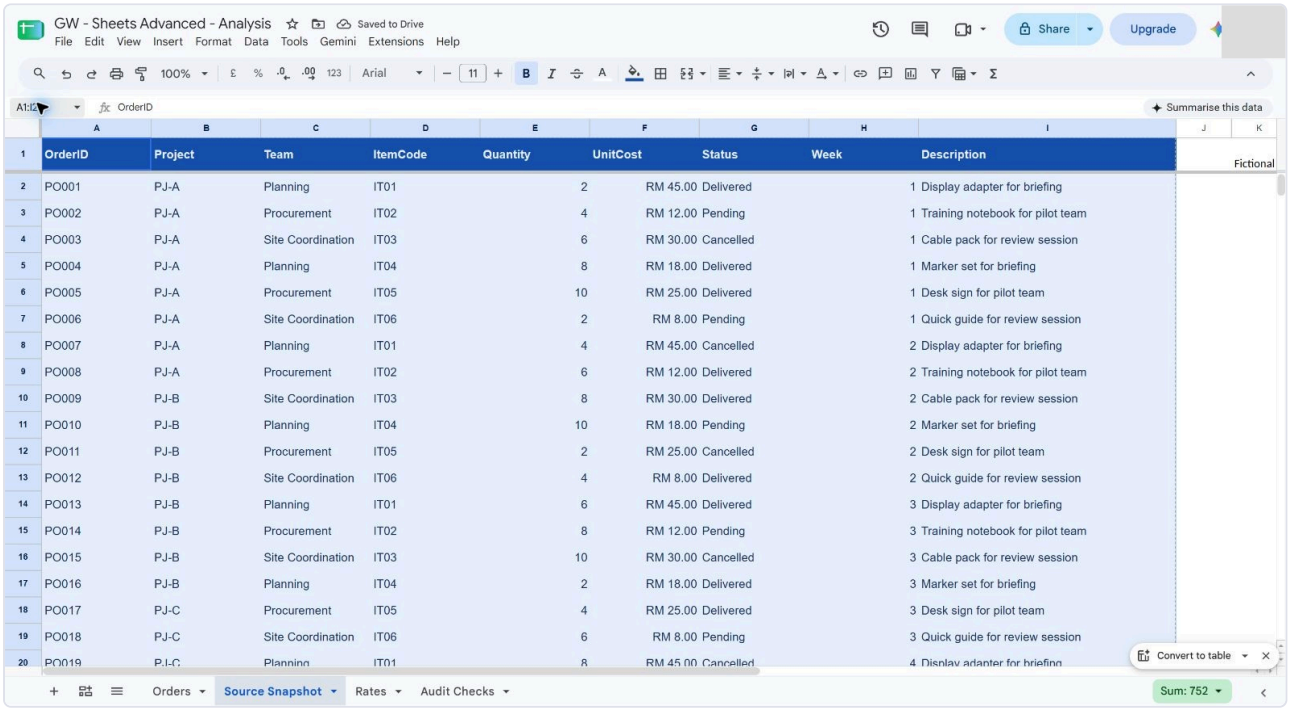


1.3 Protect the Analysis Trail

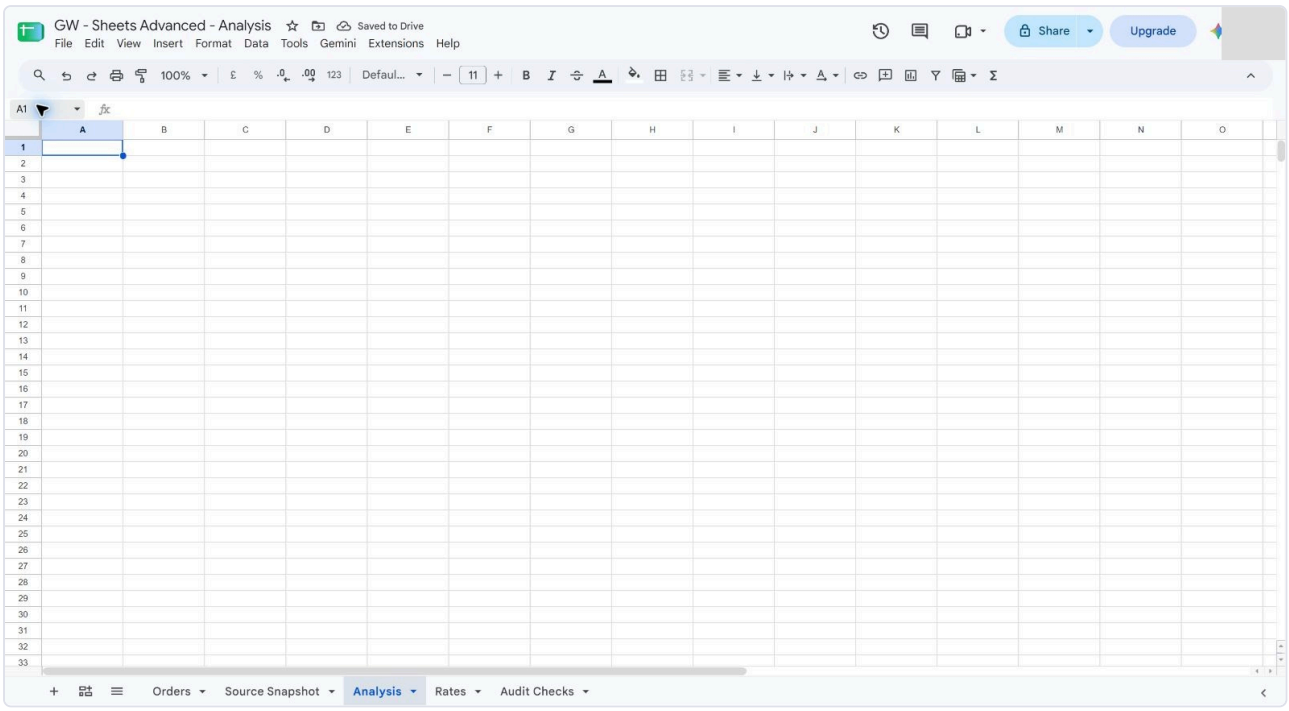
1. Return to **Orders**. Open its tab menu and choose **Duplicate** before adding any analysis calculation to that source.



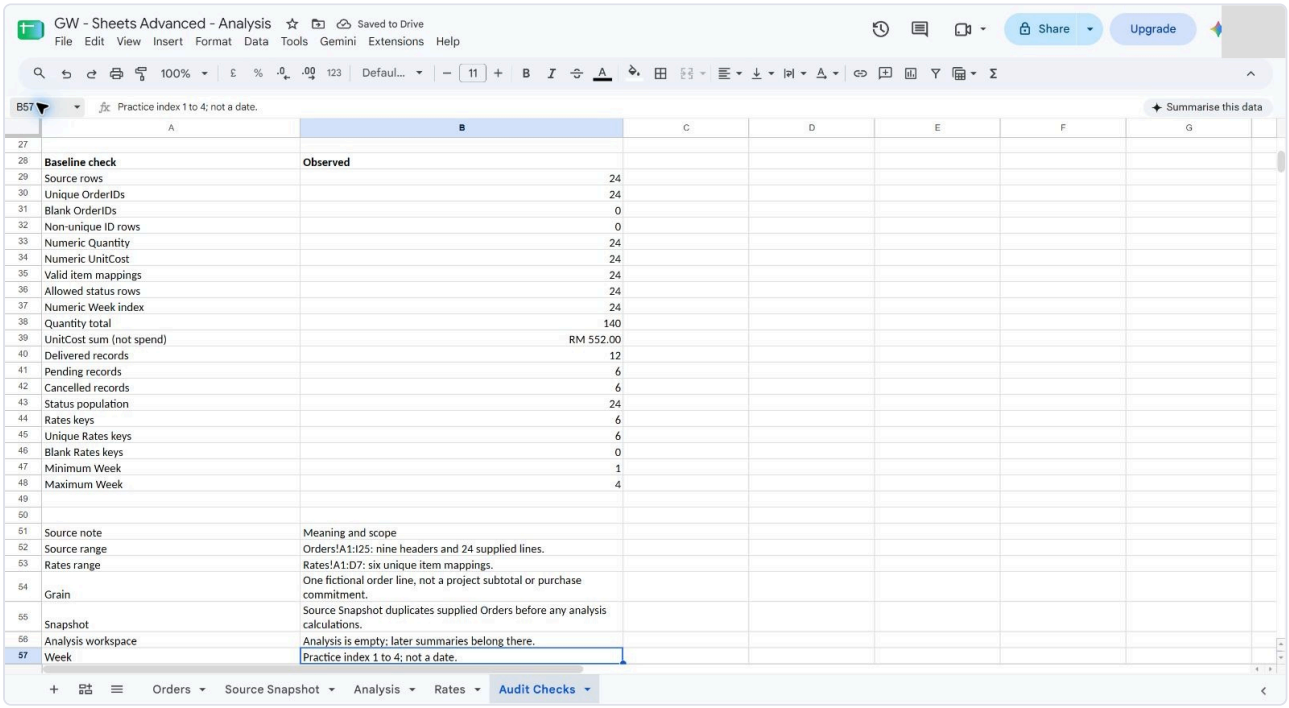
2. Rename the duplicate **Source Snapshot**. It retains the supplied Orders values and formatting. This is a preserved comparison copy; naming it does not create a protected-range permission or automatic version archive.



3. Create **Analysis** and leave it empty for later summaries. Keep diagnostics in Audit Checks rather than prebuilding later course outputs.



4. In **Audit Checks A51:B57**, record the source **Orders!A1:I25**, rate table **Rates!A1:D7**, order-line grain, snapshot purpose and empty Analysis workspace. Wrap the note text; in the captured workbook A is 320 pixels and B is 400 pixels. Source facts, training rules and unresolved questions must remain distinguishable.

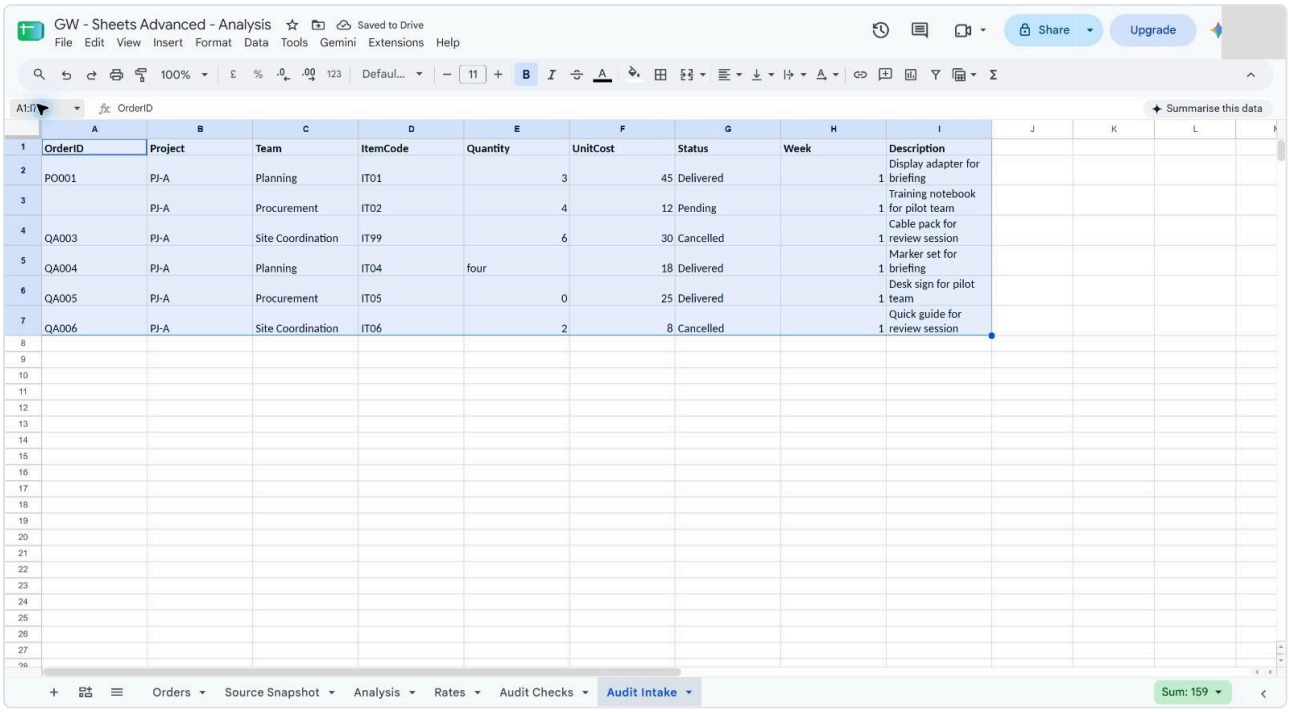


	A	B
27		
28	**Baseline check**	**Observed**
29	Source rows	24
30	Unique OrderIDs	24
31	Blank OrderIDs	0
32	Non-unique ID rows	0
33	Numeric Quantity	24
34	Numeric UnitCost	24
35	Valid item mappings	24
36	Allowed status rows	24
37	Numeric Week index	24
38	Quantity total	140
39	UnitCost sum (not spend)	RM 552.00
40	Delivered records	12
41	Pending records	6
42	Cancelled records	6
43	Status population	24
44	Rates keys	6
45	Unique Rates keys	6
46	Blank Rates keys	0
47	Minimum Week	1
48	Maximum Week	4
49		
50		
51	Source note	Meaning and scope
52	Source range	Orders!A1:I25: nine headers and 24 supplied lines.
53	Rates range	Rates!A1:D7: six unique item mappings.
54	Grain	One fictional order line, not a project subtotal or purchase commitment.
55	Snapshot	Source Snapshot duplicates supplied Orders before any analysis calculations.
56	Analysis workspace	Analysis is empty; later summaries belong there.
57	Week	Practice index 1 to 4; not a date.

1.4 Quarantine Bad Rows and Write a Data Contract

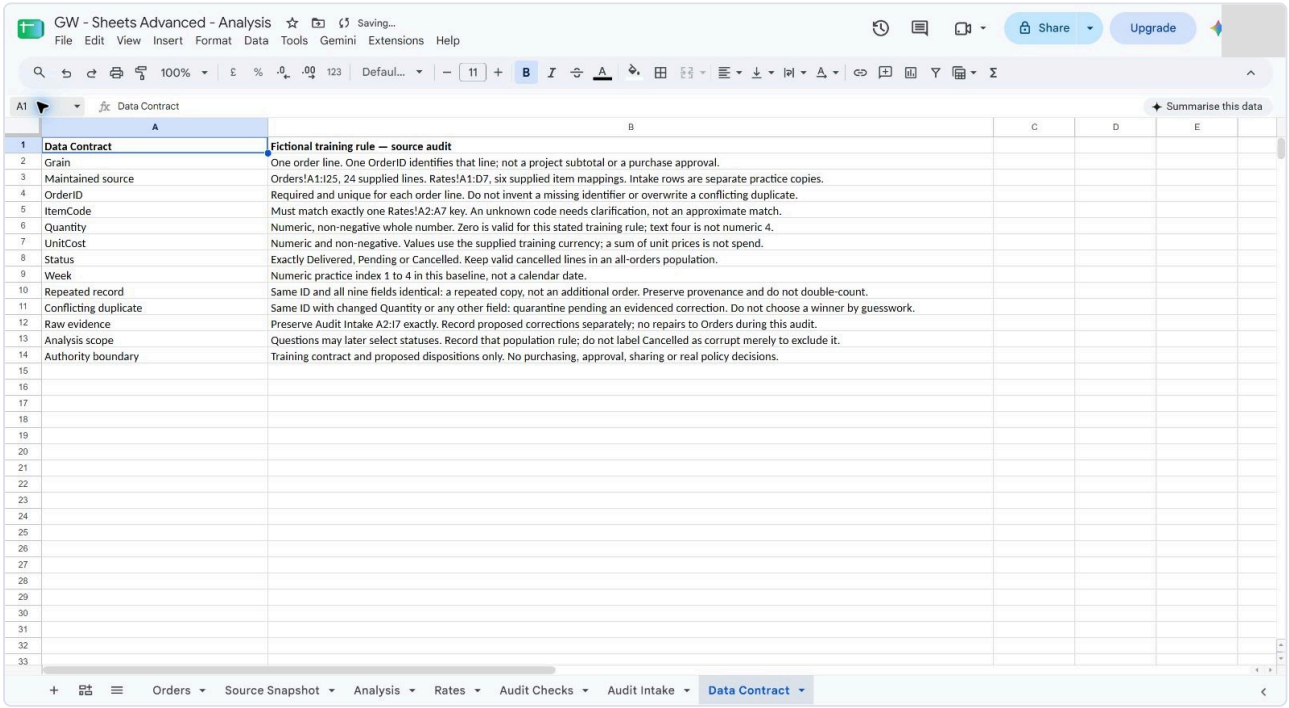
1. Create **Audit Intake** and paste the six-row raw card into **A1:I7**. These are labelled practice copies of PO001–PO006, altered only as described in the card. Preserve the blank ID, **IT99**, text **four**, numeric **0** and **Cancelled** exactly. Use the practice IDs QA003–QA006 only in intake. Add **Issue / Evidence /**

Disposition in J1:L1. Do not append any of these rows to Orders. The captured raw view uses 130-pixel A:I columns with wrapped Description cells.



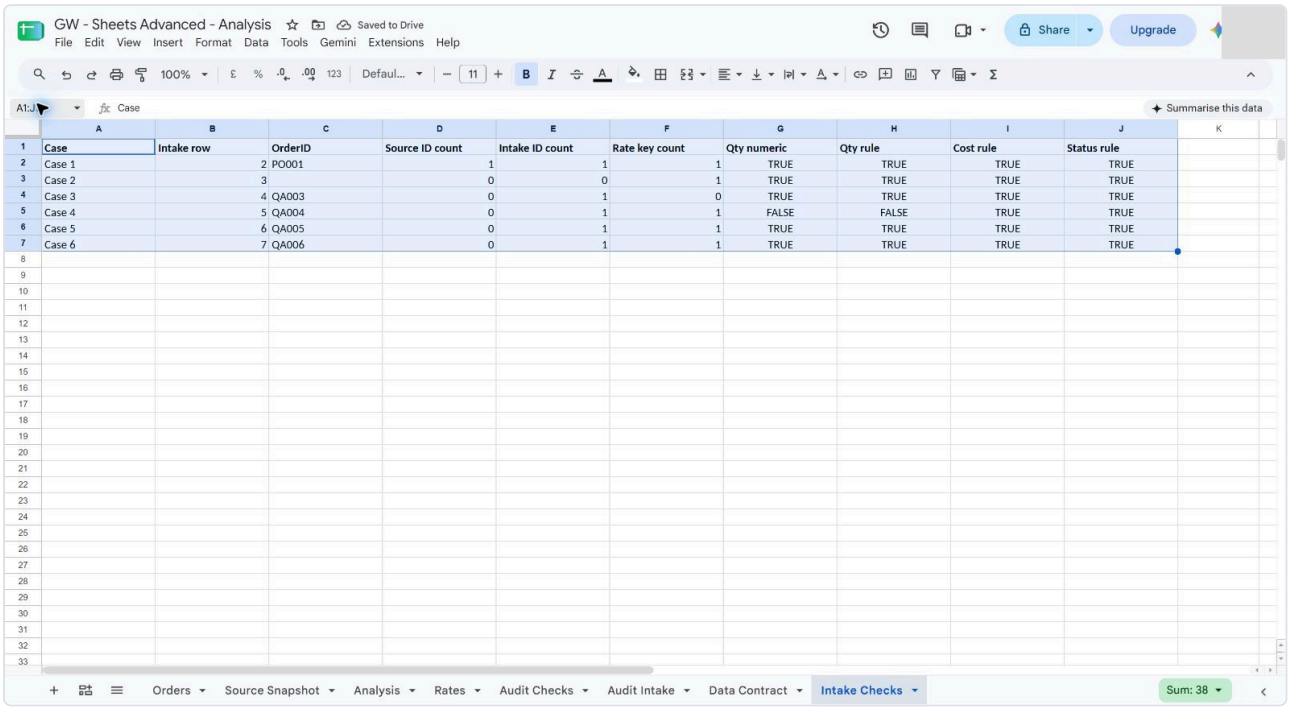
	A	B	C	D	E	F	G	H	I
1	OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description
2	PO001	PJ-A	Planning	IT01	3	45	Delivered	1	Display adapter for briefing
3		PJ-A	Procurement	IT02	4	12	Pending	1	Training notebook for pilot team
4	QA003	PJ-A	Site Coordination	IT99	6	30	Cancelled	1	Cable pack for review session
5	QA004	PJ-A	Planning	IT04	four	18	Delivered	1	Marker set for briefing
6	QA005	PJ-A	Procurement	IT05	0	25	Delivered	1	Desk sign for pilot team
7	QA006	PJ-A	Site Coordination	IT06	2	8	Cancelled	1	Quick guide for review session

2. Create **Data Contract** and record the stated training rules: one order line per required unique OrderID; exactly one Rates mapping; numeric non-negative whole-number Quantity; numeric non-negative UnitCost; statuses Delivered, Pending or Cancelled. **Zero Quantity is explicitly valid for this fictional training rule.** Distinguish a repeated identical row from the same key with a conflicting field. Preserve raw evidence and record proposed corrections separately. This is a training contract, not a real purchasing policy. Use A about 280 pixels and B about 900 pixels, with wrapped text.



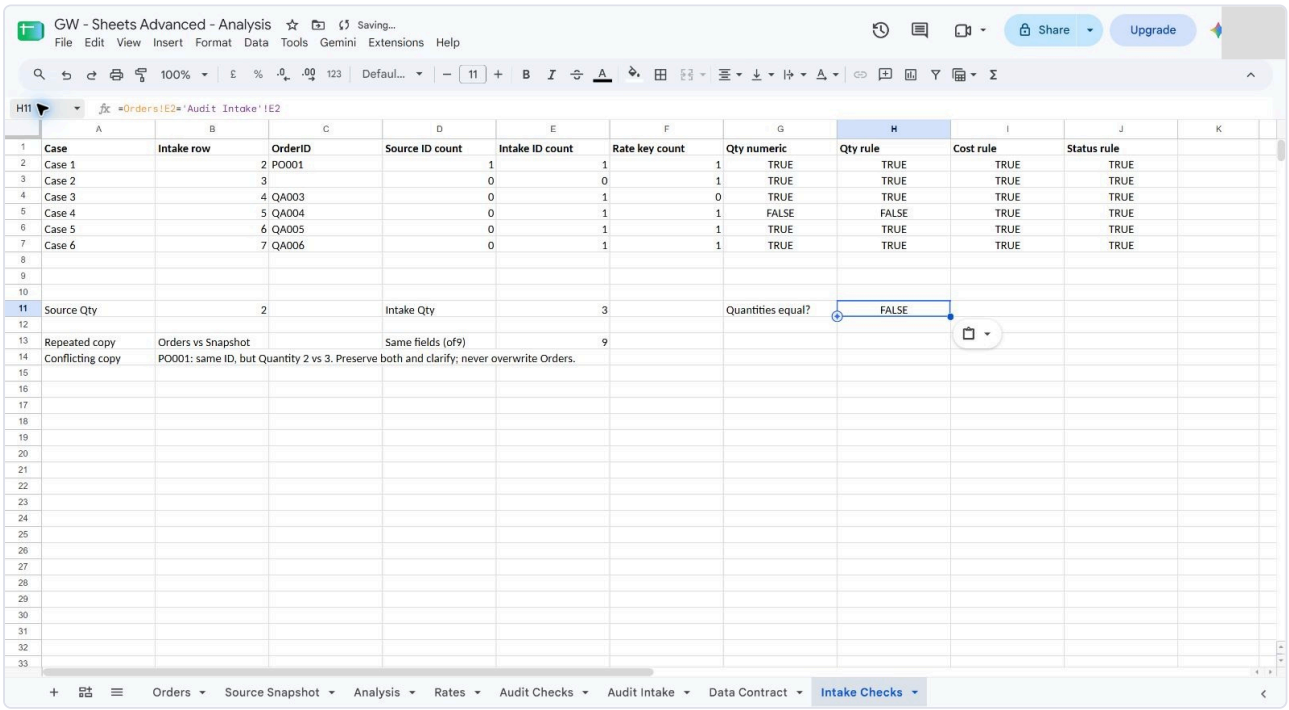
	A	B
1	Data Contract	Fictional training rule — source audit
2	Grain	One order line. One OrderID identifies that line; not a project subtotal or a purchase approval.
3	Maintained source	Orders!A1:I25, 24 supplied lines. Rates!A1:D7, six supplied item mappings. Intake rows are separate practice copies.
4	OrderID	Required and unique for each order line. Do not invent a missing identifier or overwrite a conflicting duplicate.
5	ItemCode	Must match exactly one Rates!A2:A7 key. An unknown code needs clarification, not an approximate match.
6	Quantity	Numeric, non-negative whole number. Zero is valid for this stated training rule; text four is not numeric 4.
7	UnitCost	Numeric and non-negative. Values use the supplied training currency; a sum of unit prices is not spend.
8	Status	Exactly Delivered, Pending or Cancelled. Keep valid cancelled lines in an all-orders population.
9	Week	Numeric practice index 1 to 4 in this baseline, not a calendar date.
10	Repeated record	Same ID and all nine fields identical: a repeated copy, not an additional order. Preserve provenance and do not double-count.
11	Conflicting duplicate	Same ID with changed Quantity or any other field: quarantine pending an evidenced correction. Do not choose a winner by guesswork.
12	Raw evidence	Preserve Audit Intake A2:I7 exactly. Record proposed corrections separately; no repairs to Orders during this audit.
13	Analysis scope	Questions may later select statuses. Record that population rule; do not label Cancelled as corrupt merely to exclude it.
14	Authority boundary	Training contract and proposed dispositions only. No purchasing, approval, sharing or real policy decisions.

3. Create **Intake Checks** and paste its diagnostic card into **A1:J7**. The source-ID count tests Orders, the intake-ID count tests the separate six-row intake, and the rate-key count tests Rates. Inspect both numeric type and the non-negative whole-number rule. A zero key count or FALSE result is evidence to investigate; a set of TRUE checks does not by itself approve a record for release.



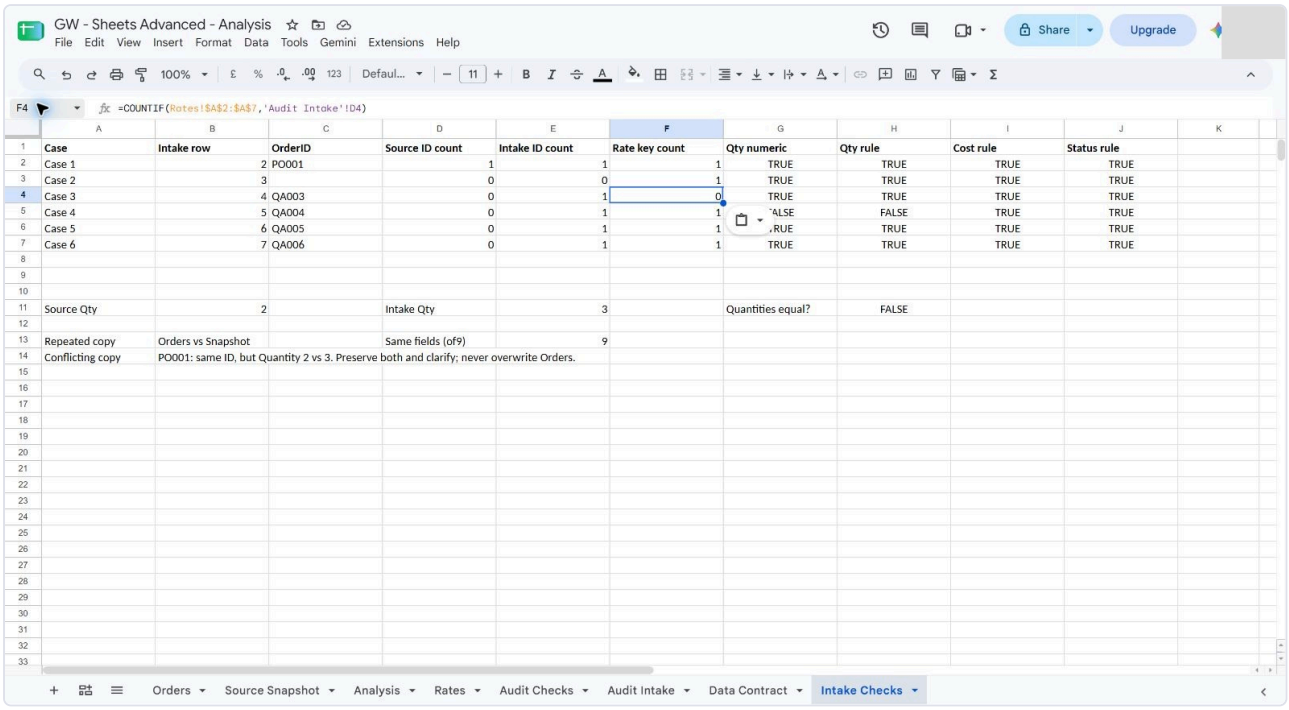
	A	B	C	D	E	F	G	H	I	J
1	Case	Intake row	OrderID	Source ID count	Intake ID count	Rate key count	Qty numeric	Qty rule	Cost rule	Status rule
2	Case 1	2	PO001	1	1	1	TRUE	TRUE	TRUE	TRUE
3	Case 2	3		0	0	1	TRUE	TRUE	TRUE	TRUE
4	Case 3	4	QA003	0	1	0	TRUE	TRUE	TRUE	TRUE
5	Case 4	5	QA004	0	1	1	FALSE	FALSE	TRUE	TRUE
6	Case 5	6	QA005	0	1	1	TRUE	TRUE	TRUE	TRUE
7	Case 6	7	QA006	0	1	1	TRUE	TRUE	TRUE	TRUE

4. Compare the conflicting **PO001**: source Quantity **2**, intake Quantity **3**, equality **FALSE**. The supplied Orders row and its unaltered Source Snapshot have **9 of 9 identical fields**, illustrating a repeated copy. Preserve the conflicting intake row and request evidence; do not overwrite the baseline or count both as separate orders. The supporting comparison cells are on the copy-paste page.



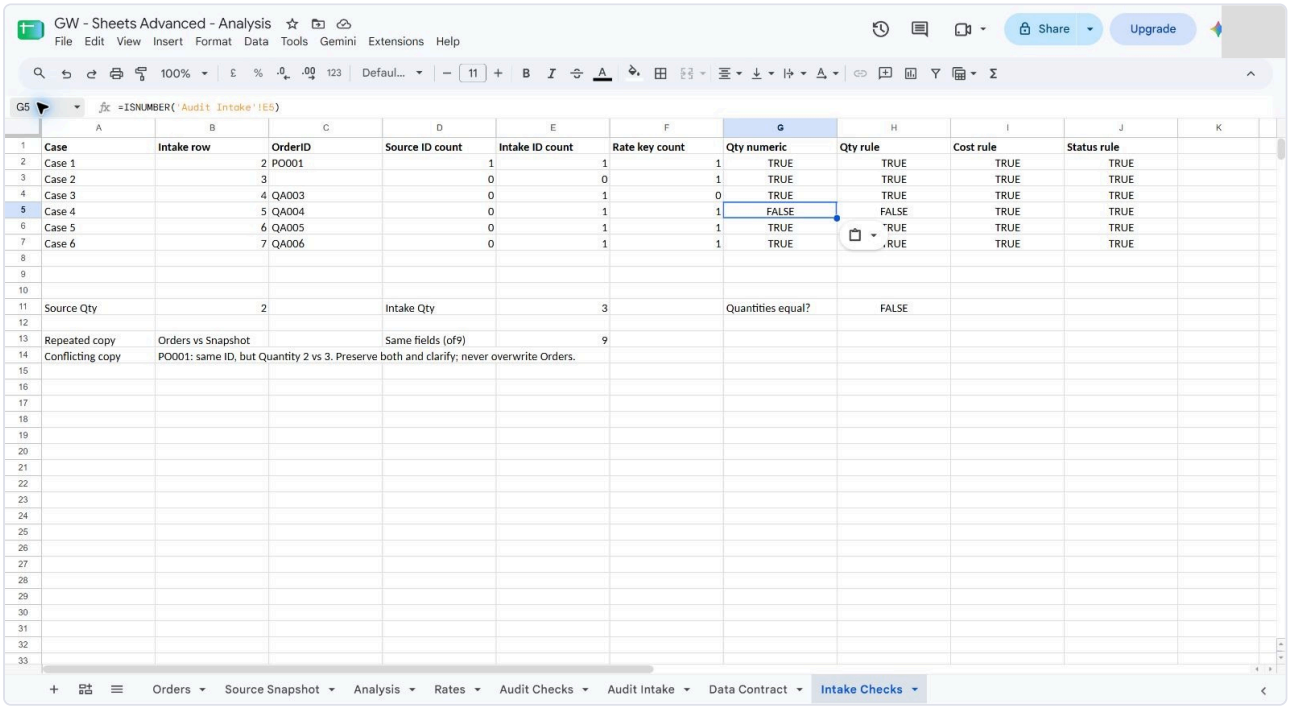
	A	B	C	D	E	F	G	H	I	J
1	Case	Intake row	OrderID	Source ID count	Intake ID count	Rate key count	Qty numeric	Qty rule	Cost rule	Status rule
2	Case 1	2	PO001	1	1	1	TRUE	TRUE	TRUE	TRUE
3	Case 2	3		0	0	1	TRUE	TRUE	TRUE	TRUE
4	Case 3	4	QA003	0	1	0	TRUE	TRUE	TRUE	TRUE
5	Case 4	5	QA004	0	1	1	FALSE	FALSE	TRUE	TRUE
6	Case 5	6	QA005	0	1	1	TRUE	TRUE	TRUE	TRUE
7	Case 6	7	QA006	0	1	1	TRUE	TRUE	TRUE	TRUE
11	Source Qty	2		Intake Qty	3		Quantities equal?	FALSE		
13	Repeated copy	Orders vs Snapshot		Same fields (of9)	9					
14	Conflicting copy	PO001: same ID, but Quantity 2 vs 3. Preserve both and clarify; never overwrite Orders.								

5. Inspect **F4 =COUNTIF(Rates!$A$2:$A$7,'Audit Intake'!D4)**. **IT99** has **0** exact matches. Its Cancelled status is valid, but that does not repair the unknown item key. Keep it quarantined pending clarification.



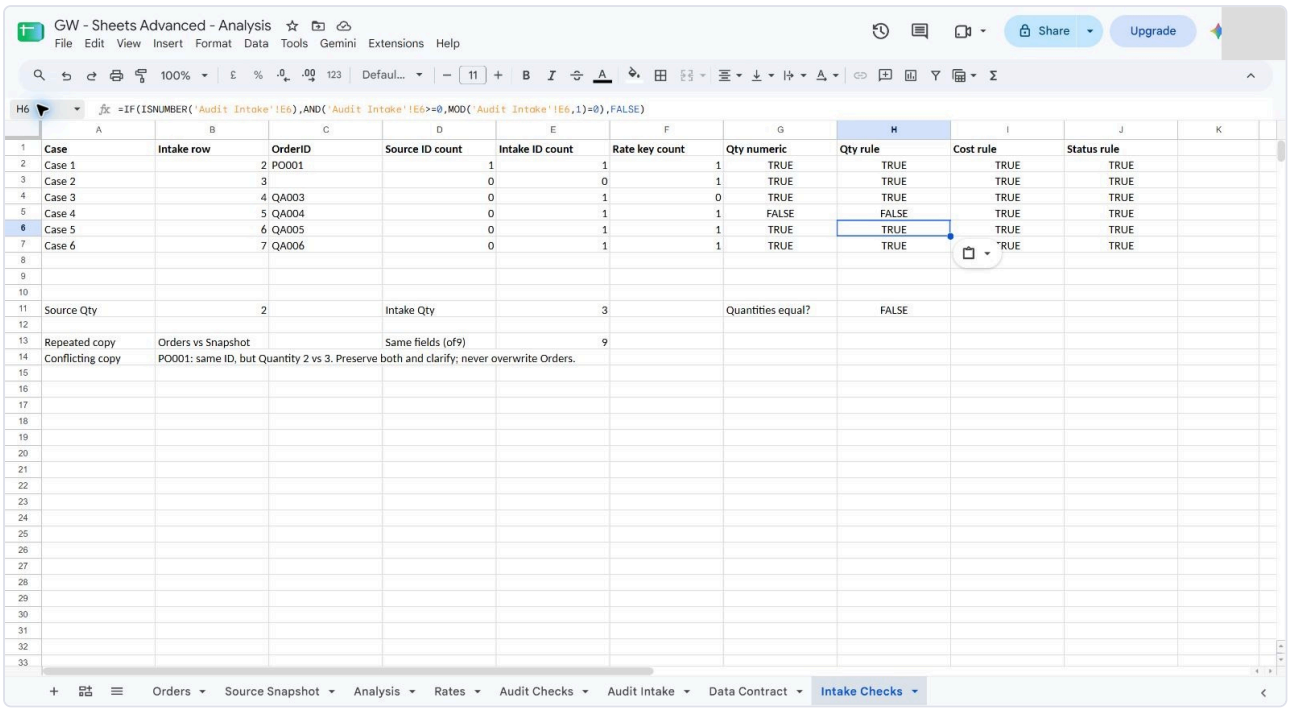
	A	B	C	D	E	F	G	H	I	J	K
1	Case	Intake row	OrderID	Source ID count	Intake ID count	Rate key count	Qty numeric	Qty rule	Cost rule	Status rule	
2	Case 1	2	PO001	1	1	1	TRUE	TRUE	TRUE	TRUE	
3	Case 2	3		0	0	1	TRUE	TRUE	TRUE	TRUE	
4	Case 3	4	QA003	0	1	0	TRUE	TRUE	TRUE	TRUE	
5	Case 4	5	QA004	0	1	1	FALSE	FALSE	TRUE	TRUE	
6	Case 5	6	QA005	0	1	1	TRUE	TRUE	TRUE	TRUE	
7	Case 6	7	QA006	0	1	1	TRUE	TRUE	TRUE	TRUE	
8											
9											
10											
11	Source Qty	2		Intake Qty	3		Quantities equal?	FALSE			
12											
13	Repeated copy	Orders vs Snapshot		Same fields (of9)	9						
14	Conflicting copy	PO001: same ID, but Quantity 2 vs 3. Preserve both and clarify; never overwrite Orders.									

6. Inspect **G5 =ISNUMBER('Audit Intake'!E5)**: text **four** returns **FALSE**, as does its Quantity rule. Do not replace it with 4 without source evidence. Record the proposed next action separately.



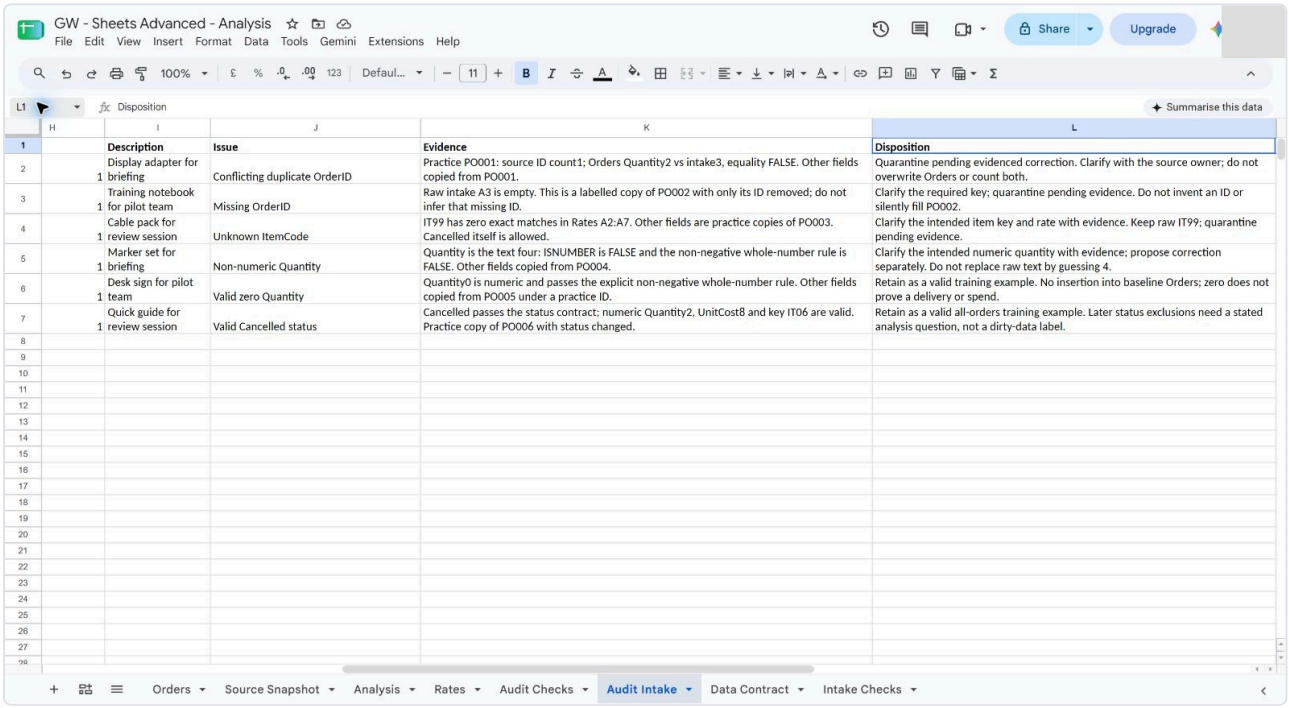
	A	B	C	D	E	F	G	H	I	J	K
1	Case	Intake row	OrderID	Source ID count	Intake ID count	Rate key count	Qty numeric	Qty rule	Cost rule	Status rule	
2	Case 1	2	PO001	1	1	1	TRUE	TRUE	TRUE	TRUE	
3	Case 2	3		0	0	1	TRUE	TRUE	TRUE	TRUE	
4	Case 3	4	QA003	0	1	0	TRUE	TRUE	TRUE	TRUE	
5	Case 4	5	QA004	0	1	1	FALSE	FALSE	TRUE	TRUE	
6	Case 5	6	QA005	0	1	1	TRUE	TRUE	TRUE	TRUE	
7	Case 6	7	QA006	0	1	1	TRUE	TRUE	TRUE	TRUE	
8											
9											
10											
11	Source Qty	2		Intake Qty	3		Quantities equal?	FALSE			
12											
13	Repeated copy	Orders vs Snapshot		Same fields (of9)	9						
14	Conflicting copy	PO001: same ID, but Quantity 2 vs 3. Preserve both and clarify; never overwrite Orders.									

7. Inspect the zero-quantity row: ISNUMBER and the non-negative whole-number rule are both **TRUE**. Retain it as a valid training example under the stated contract; do not infer a delivery or purchasing decision from a zero.



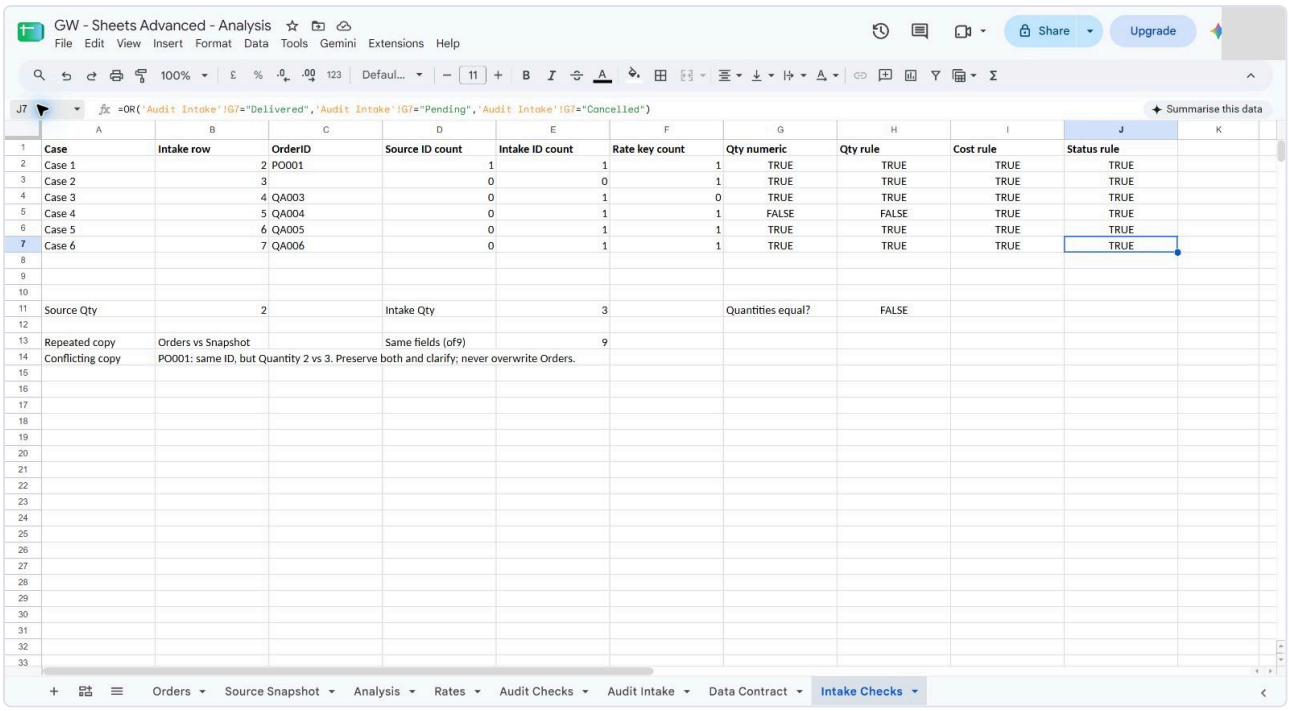
Case	Intake row	OrderID	Source ID count	Intake ID count	Rate key count	Qty numeric	Qty rule	Cost rule	Status rule
Case 1	2	PO001	1	1	1	TRUE	TRUE	TRUE	TRUE
Case 2	3		0	0	1	TRUE	TRUE	TRUE	TRUE
Case 3	4	QA003	0	1	0	TRUE	TRUE	TRUE	TRUE
Case 4	5	QA004	0	1	1	FALSE	FALSE	TRUE	TRUE
Case 5	6	QA005	0	1	1	TRUE	TRUE	TRUE	TRUE
Case 6	7	QA006	0	1	1	TRUE	TRUE	TRUE	TRUE
Source Qty	2		Intake Qty	3		Quantities equal?	FALSE		
Repeated copy	Orders vs Snapshot		Same fields (of9)	9					
Conflicting copy	PO001: same ID, but Quantity 2 vs 3. Preserve both and clarify; never overwrite Orders.								

8. Complete **Audit Intake J2:L7** with each Issue, supporting cell evidence and proposed Disposition. The first four cases remain quarantined pending evidence; the last two are valid training cases retained separately. Clarification requests are recorded as next actions, not sent. Wrap the text, widen J/K/L to about **260/560/500 pixels**, and navigate to **L1** to see the complete disposition column.

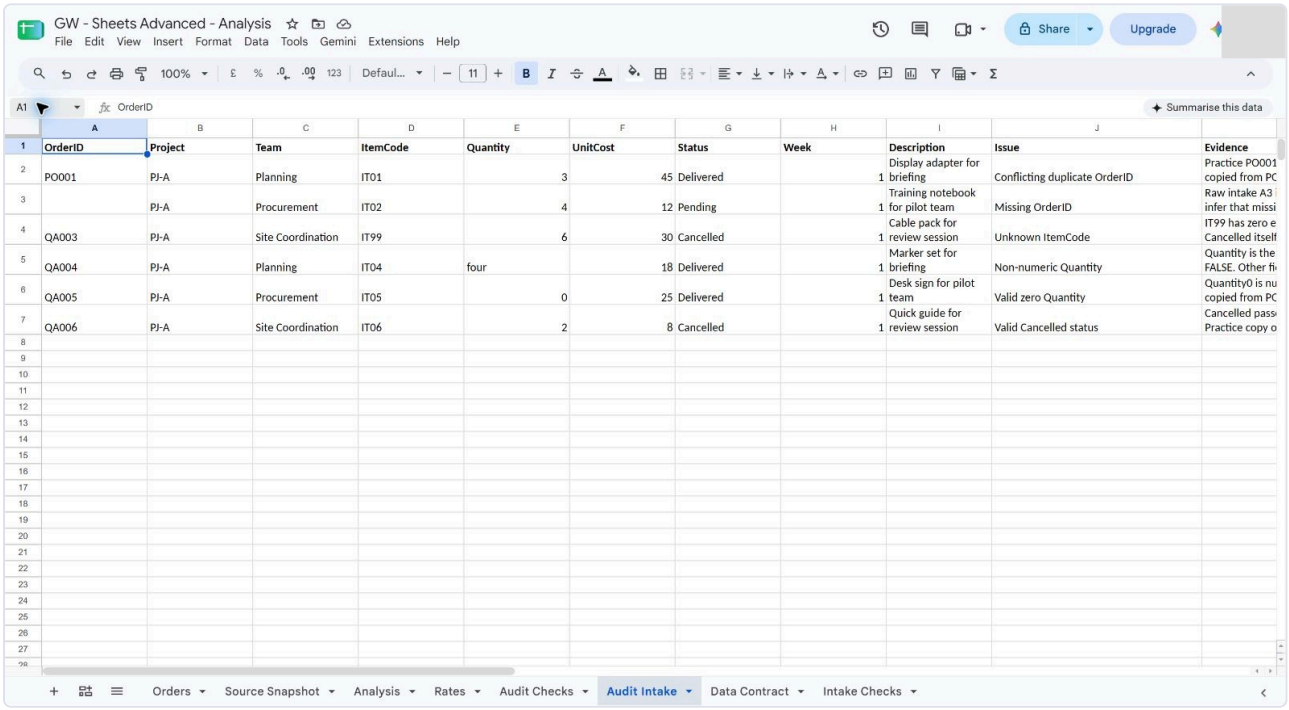


	Description	Issue	Evidence	Disposition
1	Display adapter for briefing	Conflicting duplicate OrderID	Practice PO001: source ID count1; Orders Quantity2 vs intake3, equality FALSE. Other fields copied from PO001.	Quarantine pending evidenced correction. Clarify with the source owner; do not overwrite Orders or count both.
1	Training notebook for pilot team	Missing OrderID	Raw intake A3 is empty. This is a labelled copy of PO002 with only its ID removed; do not infer that missing ID.	Clarify the required key; quarantine pending evidence. Do not invent an ID or silently fill PO002.
1	Cable pack for review session	Unknown ItemCode	IT99 has zero exact matches in Rates A2:A7. Other fields are practice copies of PO003. Cancelled itself is allowed.	Clarify the intended item key and rate with evidence. Keep raw IT99; quarantine pending evidence.
1	Marker set for briefing	Non-numeric Quantity	Quantity is the text four: ISNUMBER is FALSE and the non-negative whole-number rule is FALSE. Other fields copied from PO004.	Clarify the intended numeric quantity with evidence; propose correction separately. Do not replace raw text by guessing 4.
1	Desk sign for pilot team	Valid zero Quantity	Quantity0 is numeric and passes the explicit non-negative whole-number rule. Other fields copied from PO005 under a practice ID.	Retain as a valid training example. No insertion into baseline Orders; zero does not prove a delivery or spend.
1	Quick guide for review session	Valid Cancelled status	Cancelled passes the status contract; numeric Quantity2, UnitCost8 and key IT06 are valid. Practice copy of PO006 with status changed.	Retain as a valid all-orders training example. Later status exclusions need a stated analysis question, not a dirty-data label.

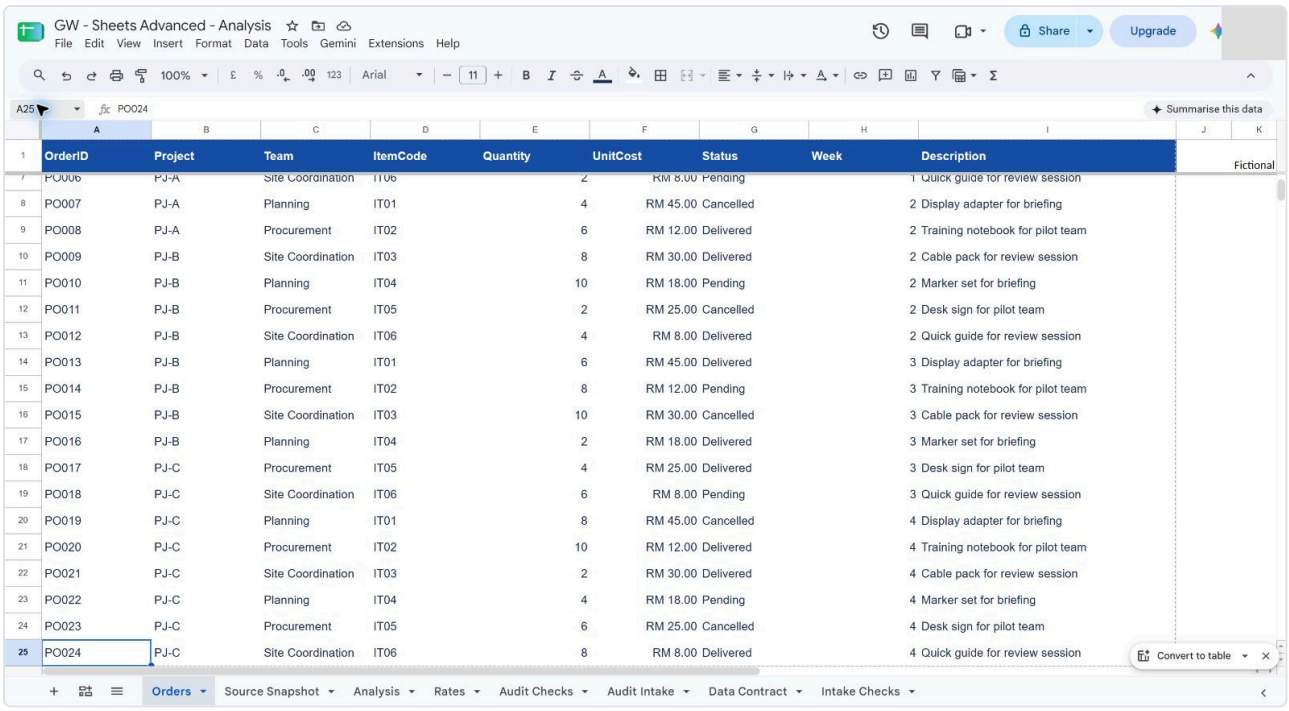
9. Inspect **J7** on Intake Checks. The QA006 Cancelled example passes the allowed-status test, as do its numeric and key checks. A later analysis may choose a status population, but a valid Cancelled record is not automatically corrupt.



10. Reopen **Audit Intake A1:I7** and verify every raw value remains intact beside the recorded issues. No repair has been applied to the raw examples or maintained source.

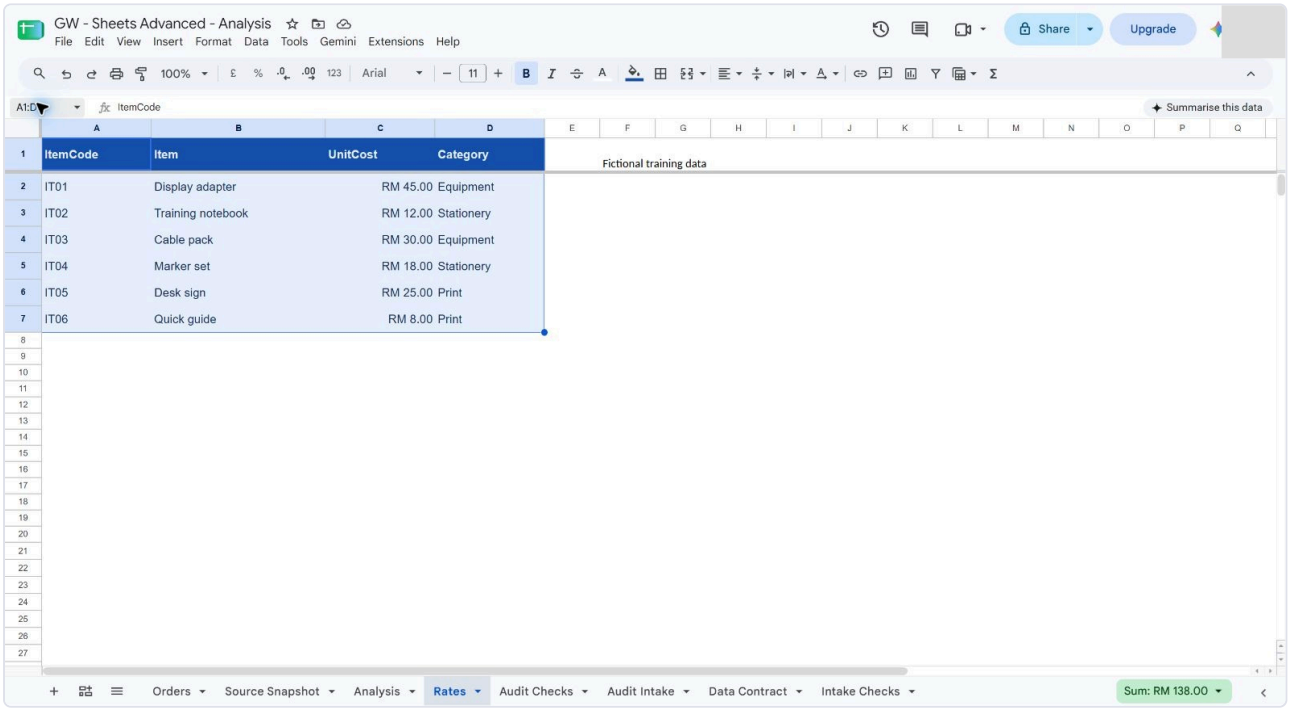


11. Return to **Orders** and recheck its original 24 records through PO024. Native readback confirms its supplied values and formatting are unchanged.



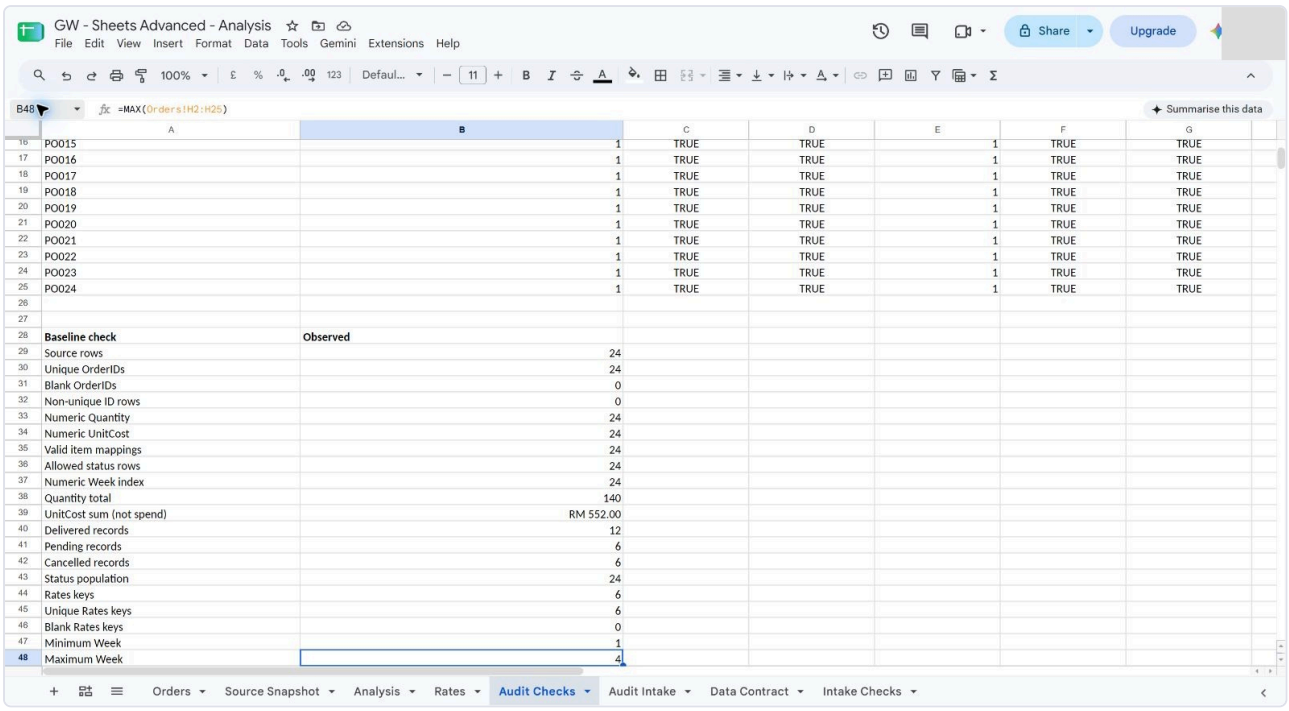
OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description
PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session
PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing
PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team
PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session
PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing
PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team
PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session
PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing
PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team
PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session
PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing
PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team
PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session
PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing
PO020	PJ-C	Procurement	IT02	10	RM 12.00	Delivered	4	Training notebook for pilot team
PO021	PJ-C	Site Coordination	IT03	2	RM 30.00	Delivered	4	Cable pack for review session
PO022	PJ-C	Planning	IT04	4	RM 18.00	Pending	4	Marker set for briefing
PO023	PJ-C	Procurement	IT05	6	RM 25.00	Cancelled	4	Desk sign for pilot team
PO024	PJ-C	Site Coordination	IT06	8	RM 8.00	Delivered	4	Quick guide for review session

12. Recheck the original six **Rates** keys and supplied values. The unknown intake code has not been added to the reference table to make a test pass.

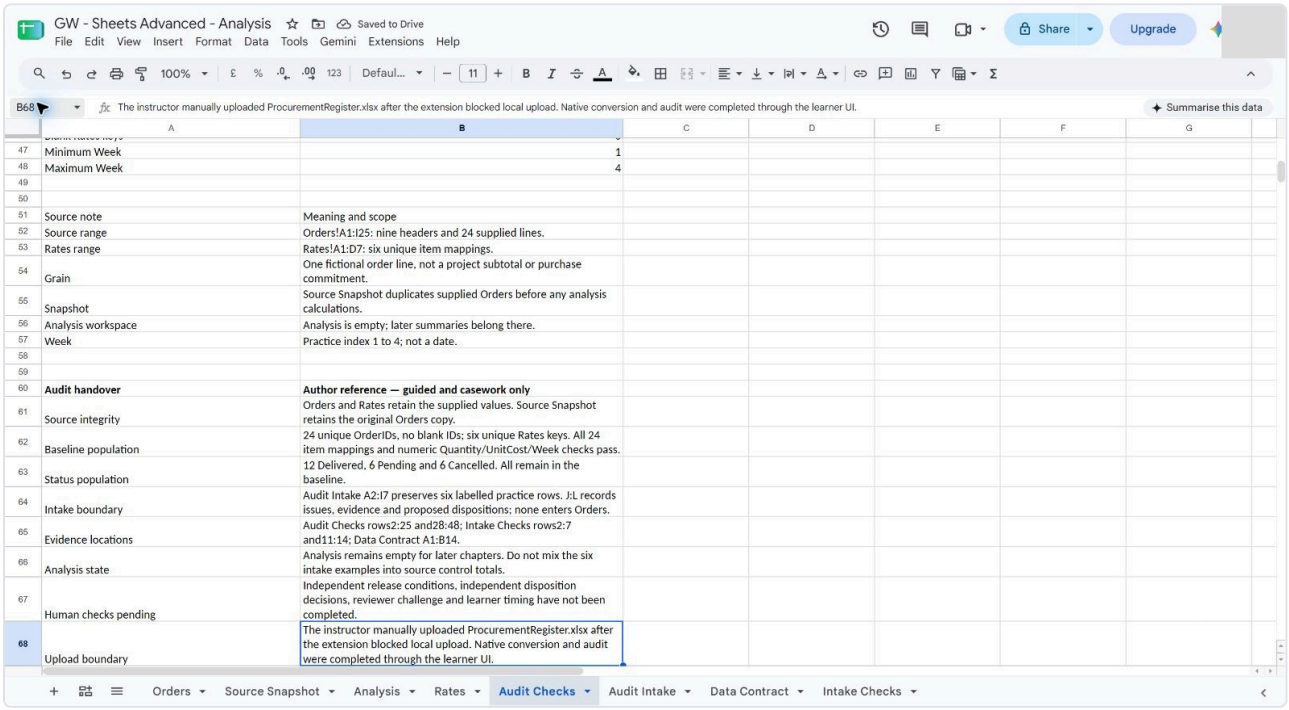


ItemCode	Item	UnitCost	Category
IT01	Display adapter	RM 45.00	Equipment
IT02	Training notebook	RM 12.00	Stationery
IT03	Cable pack	RM 30.00	Equipment
IT04	Marker set	RM 18.00	Stationery
IT05	Desk sign	RM 25.00	Print
IT06	Quick guide	RM 8.00	Print

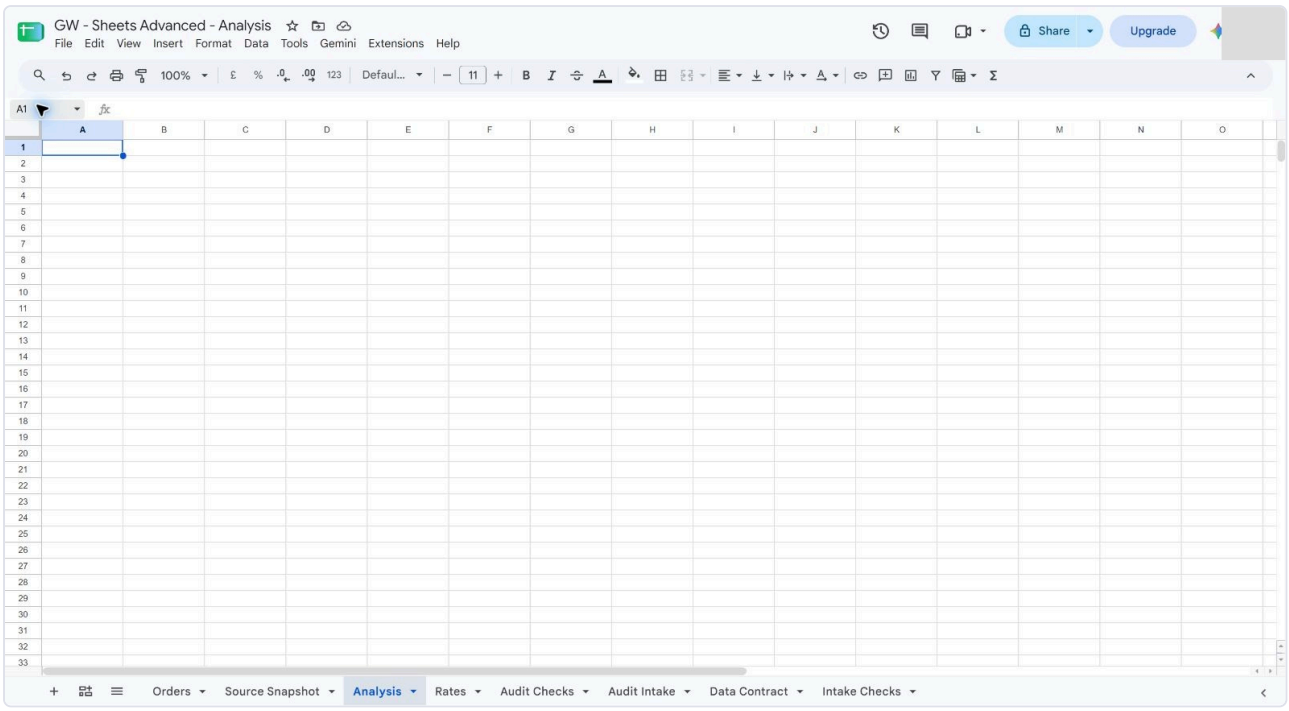
13. Reopen the baseline population controls. Counts, types, key coverage, status population, Quantity 140 and the unit-price checksum 552 remain unchanged. Intake anomalies are excluded from these bounded controls.



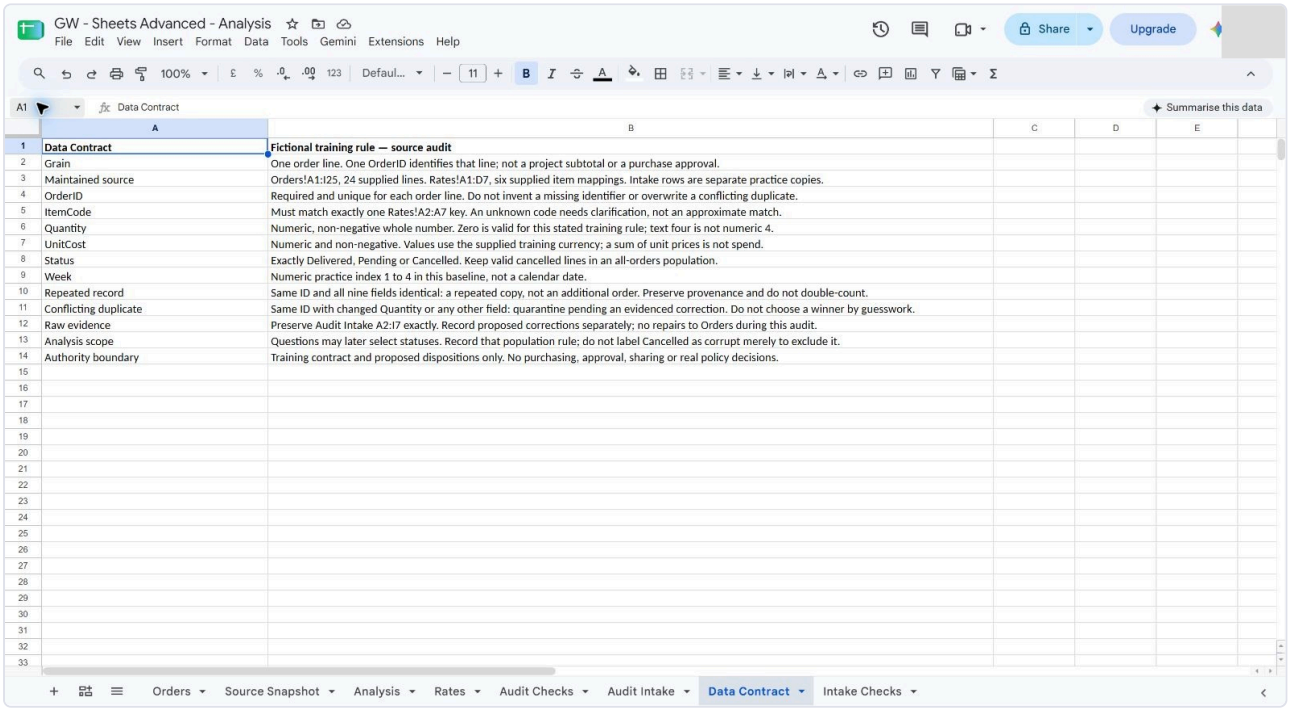
14. Record the current source integrity, evidence locations and intake boundary in **Audit Checks A60:B68**. Label independent disposition decisions, release conditions, reviewer challenge and learner timing as pending. The author walkthrough does not substitute for those checks.



15. Confirm **Analysis** remains empty. Leave the current workbook ready for conditional calculations in the next lab, with no later summary or dashboard prebuilt.



16. Leave the saved workbook with its **Data Contract**, isolated intake and diagnostics available. The source snapshot and original XLSX remain available for comparison.



> **Validation boundary:** guided 1.1–1.3 and author casework 1.4 are checked in the learner UI, supplemented by bounded read-only native-cell verification. Upload used an instructor handoff. Independent release conditions and disposition decisions, reviewer challenge and learner timing are not completed.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Independently decide whether each anomaly should enter the analysis, remain pending or be retained as a valid case. Justify every decision against the contract.
2. Write a release condition for the source: what must be true before conditional calculations and lookups can be trusted?
3. Have a reviewer challenge one disposition and resolve it using source evidence or an explicit unresolved label.

Evidence to Hand Over

- Data contract and six-row anomaly audit with preserved raw values.
- Baseline uniqueness/type/key checks and a justified quarantine policy.
- Independent release conditions and a recorded challenged decision.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A cancelled row is removed as dirty data	Separate data validity from the inclusion rule of the analysis question.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

An analysis begins with data meaning, types and source coverage.

Check yourself: What does one row represent, and which field identifies it?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
GW - Sheets Advanced - Analysis
```

```
Source Snapshot
```

```
Analysis
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Separate intake anomalies: duplicate PO001 with changed quantity; missing OrderID; ItemCode
IT99; Quantity “four”; valid zero quantity; valid Cancelled status. Do not inject these into
Orders. Other fields can be copied from a source row only in Audit Intake and labelled as
practice.
```

Captured Source-Audit Reference

The supplied source has 24 unique OrderIDs PO001–PO024, six unique Rates keys IT01–IT06, Quantity total140 and unit-price checksum552 (not spend). All24 Quantity/UnitCost/Week entries are numeric, all24 item keys map, statuses are12 Delivered/6 Pending/6 Cancelled. Source tables retain their supplied values and formats; Analysis is empty. Manual upload handoff was required; native conversion and audit were executed in the application.

Baseline row diagnostics — Audit Checks A1:G25

Paste this bounded card into the named diagnostic tab, not Orders. It supplies the exact formulas observed in the author run; inspect representative formulas and explain the anchored population.

```
OrderID ID occurrences  Qty numeric     Cost numeric    Item key matches        Allowed status
Week numeric
=Orders!A2      =COUNTIF(Orders!$A$2:$A$25,A2)  =ISNUMBER(Orders!E2)    =ISNUMBER(Orders!F2)
=COUNTIF(Rates!$A$2:$A$7,Orders!D2)
=OR(Orders!G2="Delivered",Orders!G2="Pending",Orders!G2="Cancelled")    =ISNUMBER(Orders!H2)
=Orders!A3      =COUNTIF(Orders!$A$2:$A$25,A3)  =ISNUMBER(Orders!E3)    =ISNUMBER(Orders!F3)
=COUNTIF(Rates!$A$2:$A$7,Orders!D3)
=OR(Orders!G3="Delivered",Orders!G3="Pending",Orders!G3="Cancelled")    =ISNUMBER(Orders!H3)
=Orders!A4      =COUNTIF(Orders!$A$2:$A$25,A4)  =ISNUMBER(Orders!E4)    =ISNUMBER(Orders!F4)
=COUNTIF(Rates!$A$2:$A$7,Orders!D4)
=OR(Orders!G4="Delivered",Orders!G4="Pending",Orders!G4="Cancelled")    =ISNUMBER(Orders!H4)
=Orders!A5      =COUNTIF(Orders!$A$2:$A$25,A5)  =ISNUMBER(Orders!E5)    =ISNUMBER(Orders!F5)
=COUNTIF(Rates!$A$2:$A$7,Orders!D5)
=OR(Orders!G5="Delivered",Orders!G5="Pending",Orders!G5="Cancelled")    =ISNUMBER(Orders!H5)
```

```
=Orders!A6      =COUNTIF(Orders!$A$2:$A$25,A6)  =ISNUMBER(Orders!E6)    =ISNUMBER(Orders!F6)
=COUNTIF(Rates!$A$2:$A$7,Orders!D6)
=OR(Orders!G6="Delivered",Orders!G6="Pending",Orders!G6="Cancelled")    =ISNUMBER(Orders!H6)
=Orders!A7      =COUNTIF(Orders!$A$2:$A$25,A7)  =ISNUMBER(Orders!E7)    =ISNUMBER(Orders!F7)
=COUNTIF(Rates!$A$2:$A$7,Orders!D7)
=OR(Orders!G7="Delivered",Orders!G7="Pending",Orders!G7="Cancelled")    =ISNUMBER(Orders!H7)
=Orders!A8      =COUNTIF(Orders!$A$2:$A$25,A8)  =ISNUMBER(Orders!E8)    =ISNUMBER(Orders!F8)
=COUNTIF(Rates!$A$2:$A$7,Orders!D8)
=OR(Orders!G8="Delivered",Orders!G8="Pending",Orders!G8="Cancelled")    =ISNUMBER(Orders!H8)
=Orders!A9      =COUNTIF(Orders!$A$2:$A$25,A9)  =ISNUMBER(Orders!E9)    =ISNUMBER(Orders!F9)
=COUNTIF(Rates!$A$2:$A$7,Orders!D9)
=OR(Orders!G9="Delivered",Orders!G9="Pending",Orders!G9="Cancelled")    =ISNUMBER(Orders!H9)
=Orders!A10     =COUNTIF(Orders!$A$2:$A$25,A10) =ISNUMBER(Orders!E10)   =ISNUMBER(Orders!F10)
=COUNTIF(Rates!$A$2:$A$7,Orders!D10)
=OR(Orders!G10="Delivered",Orders!G10="Pending",Orders!G10="Cancelled") =ISNUMBER(Orders!H10)
=Orders!A11     =COUNTIF(Orders!$A$2:$A$25,A11) =ISNUMBER(Orders!E11)   =ISNUMBER(Orders!F11)
=COUNTIF(Rates!$A$2:$A$7,Orders!D11)
=OR(Orders!G11="Delivered",Orders!G11="Pending",Orders!G11="Cancelled") =ISNUMBER(Orders!H11)
=Orders!A12     =COUNTIF(Orders!$A$2:$A$25,A12) =ISNUMBER(Orders!E12)   =ISNUMBER(Orders!F12)
=COUNTIF(Rates!$A$2:$A$7,Orders!D12)
=OR(Orders!G12="Delivered",Orders!G12="Pending",Orders!G12="Cancelled") =ISNUMBER(Orders!H12)
=Orders!A13     =COUNTIF(Orders!$A$2:$A$25,A13) =ISNUMBER(Orders!E13)   =ISNUMBER(Orders!F13)
=COUNTIF(Rates!$A$2:$A$7,Orders!D13)
=OR(Orders!G13="Delivered",Orders!G13="Pending",Orders!G13="Cancelled") =ISNUMBER(Orders!H13)
=Orders!A14     =COUNTIF(Orders!$A$2:$A$25,A14) =ISNUMBER(Orders!E14)   =ISNUMBER(Orders!F14)
=COUNTIF(Rates!$A$2:$A$7,Orders!D14)
=OR(Orders!G14="Delivered",Orders!G14="Pending",Orders!G14="Cancelled") =ISNUMBER(Orders!H14)
=Orders!A15     =COUNTIF(Orders!$A$2:$A$25,A15) =ISNUMBER(Orders!E15)   =ISNUMBER(Orders!F15)
=COUNTIF(Rates!$A$2:$A$7,Orders!D15)
=OR(Orders!G15="Delivered",Orders!G15="Pending",Orders!G15="Cancelled") =ISNUMBER(Orders!H15)
=Orders!A16     =COUNTIF(Orders!$A$2:$A$25,A16) =ISNUMBER(Orders!E16)   =ISNUMBER(Orders!F16)
=COUNTIF(Rates!$A$2:$A$7,Orders!D16)
=OR(Orders!G16="Delivered",Orders!G16="Pending",Orders!G16="Cancelled") =ISNUMBER(Orders!H16)
=Orders!A17     =COUNTIF(Orders!$A$2:$A$25,A17) =ISNUMBER(Orders!E17)   =ISNUMBER(Orders!F17)
=COUNTIF(Rates!$A$2:$A$7,Orders!D17)
=OR(Orders!G17="Delivered",Orders!G17="Pending",Orders!G17="Cancelled") =ISNUMBER(Orders!H17)
=Orders!A18     =COUNTIF(Orders!$A$2:$A$25,A18) =ISNUMBER(Orders!E18)   =ISNUMBER(Orders!F18)
=COUNTIF(Rates!$A$2:$A$7,Orders!D18)
=OR(Orders!G18="Delivered",Orders!G18="Pending",Orders!G18="Cancelled") =ISNUMBER(Orders!H18)
=Orders!A19     =COUNTIF(Orders!$A$2:$A$25,A19) =ISNUMBER(Orders!E19)   =ISNUMBER(Orders!F19)
=COUNTIF(Rates!$A$2:$A$7,Orders!D19)
=OR(Orders!G19="Delivered",Orders!G19="Pending",Orders!G19="Cancelled") =ISNUMBER(Orders!H19)
=Orders!A20     =COUNTIF(Orders!$A$2:$A$25,A20) =ISNUMBER(Orders!E20)   =ISNUMBER(Orders!F20)
=COUNTIF(Rates!$A$2:$A$7,Orders!D20)
=OR(Orders!G20="Delivered",Orders!G20="Pending",Orders!G20="Cancelled") =ISNUMBER(Orders!H20)
=Orders!A21     =COUNTIF(Orders!$A$2:$A$25,A21) =ISNUMBER(Orders!E21)   =ISNUMBER(Orders!F21)
=COUNTIF(Rates!$A$2:$A$7,Orders!D21)
=OR(Orders!G21="Delivered",Orders!G21="Pending",Orders!G21="Cancelled") =ISNUMBER(Orders!H21)
=Orders!A22     =COUNTIF(Orders!$A$2:$A$25,A22) =ISNUMBER(Orders!E22)   =ISNUMBER(Orders!F22)
=COUNTIF(Rates!$A$2:$A$7,Orders!D22)
=OR(Orders!G22="Delivered",Orders!G22="Pending",Orders!G22="Cancelled") =ISNUMBER(Orders!H22)
=Orders!A23     =COUNTIF(Orders!$A$2:$A$25,A23) =ISNUMBER(Orders!E23)   =ISNUMBER(Orders!F23)
=COUNTIF(Rates!$A$2:$A$7,Orders!D23)
=OR(Orders!G23="Delivered",Orders!G23="Pending",Orders!G23="Cancelled") =ISNUMBER(Orders!H23)
=Orders!A24     =COUNTIF(Orders!$A$2:$A$25,A24) =ISNUMBER(Orders!E24)   =ISNUMBER(Orders!F24)
=COUNTIF(Rates!$A$2:$A$7,Orders!D24)
=OR(Orders!G24="Delivered",Orders!G24="Pending",Orders!G24="Cancelled") =ISNUMBER(Orders!H24)
=Orders!A25     =COUNTIF(Orders!$A$2:$A$25,A25) =ISNUMBER(Orders!E25)   =ISNUMBER(Orders!F25)
```

```
=COUNTIF(Rates!$A$2:$A$7,Orders!D25)
=OR(Orders!G25="Delivered",Orders!G25="Pending",Orders!G25="Cancelled") =ISNUMBER(Orders!H25)
```

Baseline controls — Audit Checks A28:B48

```
Baseline check  Observed
Source rows     =COUNTA(Orders!A2:A25)
Unique OrderIDs =COUNTUNIQUE(Orders!A2:A25)
Blank OrderIDs  =COUNTBLANK(Orders!A2:A25)
Non-unique ID rows      =COUNTIF(B2:B25,">1")
Numeric Quantity        =COUNT(Orders!E2:E25)
Numeric UnitCost        =COUNT(Orders!F2:F25)
Valid item mappings     =COUNTIF(E2:E25,1)
Allowed status rows     =COUNTIF(F2:F25,TRUE)
Numeric Week index      =COUNT(Orders!H2:H25)
Quantity total  =SUM(Orders!E2:E25)
UnitCost sum (not spend)        =SUM(Orders!F2:F25)
Delivered records       =COUNTIF(Orders!G2:G25,"Delivered")
Pending records =COUNTIF(Orders!G2:G25,"Pending")
Cancelled records       =COUNTIF(Orders!G2:G25,"Cancelled")
Status population       =SUM(B40:B42)
Rates keys      =COUNTA(Rates!A2:A7)
Unique Rates keys       =COUNTUNIQUE(Rates!A2:A7)
Blank Rates keys        =COUNTBLANK(Rates!A2:A7)
Minimum Week    =MIN(Orders!H2:H25)
Maximum Week    =MAX(Orders!H2:H25)
```

Raw practice intake — Audit Intake A1:I7

Rows2–7 are practice copies of source PO001–PO006. Changes: PO001 Quantity2→3; PO002 ID removed; PO003 practice ID QA003 and ItemCode IT99; PO004 practice ID QA004 and Quantity text four; PO005 practice ID QA005 and numeric Quantity0; PO006 practice ID QA006 and status Cancelled. Other fields are copied only as fictional practice context. Do not insert these into Orders or replace the blank with PO002.

```
OrderID Project Team    ItemCode        Quantity        UnitCost        Status  Week
Description
PO001   PJ-A    Planning        IT01    3       45      Delivered       1       Display
adapter for briefing
        PJ-A    Procurement     IT02    4       12      Pending 1       Training notebook for
pilot team
QA003   PJ-A    Site Coordination       IT99    6       30      Cancelled       1       Cable
pack for review session
QA004   PJ-A    Planning        IT04    four    18      Delivered       1       Marker set for
briefing
QA005   PJ-A    Procurement     IT05    0       25      Delivered       1       Desk sign for
pilot team
QA006   PJ-A    Site Coordination       IT06    2       8       Cancelled       1       Quick
guide for review session
```

Training contract — Data Contract A1:B14

```
Data Contract   Fictional training rule — source audit
Grain   One order line. One OrderID identifies that line; not a project subtotal or a purchase approval.
Maintained source       Orders!A1:I25, 24 supplied lines. Rates!A1:D7, six supplied item mappings. Intake rows are separate practice copies.
OrderID Required and unique for each order line. Do not invent a missing identifier or overwrite a conflicting duplicate.
ItemCode        Must match exactly one Rates!A2:A7 key. An unknown code needs clarification, not an approximate match.
Quantity        Numeric, non-negative whole number. Zero is valid for this stated training rule; text four is not numeric 4.
UnitCost        Numeric and non-negative. Values use the supplied training currency; a sum of unit prices is not spend.
Status  Exactly Delivered, Pending or Cancelled. Keep valid cancelled lines in an all-orders population.
Week    Numeric practice index 1 to 4 in this baseline, not a calendar date.
Repeated record Same ID and all nine fields identical: a repeated copy, not an additional order. Preserve provenance and do not double-count.
Conflicting duplicate   Same ID with changed Quantity or any other field: quarantine pending an evidenced correction. Do not choose a winner by guesswork.
Raw evidence    Preserve Audit Intake A2:I7 exactly. Record proposed corrections separately; no repairs to Orders during this audit.
Analysis scope  Questions may later select statuses. Record that population rule; do not label Cancelled as corrupt merely to exclude it.
Authority boundary      Training contract and proposed dispositions only. No purchasing, approval, sharing or real policy decisions.
```

Intake diagnostics — Intake Checks A1:J7

```
Case    Intake row      OrderID Source ID count Intake ID count Rate key count  Qty numeric
Qty rule        Cost rule       Status rule
Case 1  2       ='Audit Intake'!A2      =IF(C2="",0,COUNTIF(Orders!$A$2:$A$25,C2))
=IF(C2="",0,COUNTIF('Audit Intake'!$A$2:$A$7,C2))       =COUNTIF(Rates!$A$2:$A$7,'Audit
Intake'!D2)     =ISNUMBER('Audit Intake'!E2)    =IF(ISNUMBER('Audit Intake'!E2),AND('Audit
Intake'!E2≥0,MOD('Audit Intake'!E2,1)=0),FALSE)        =AND(ISNUMBER('Audit
Intake'!F2),'Audit Intake'!F2≥0)       =OR('Audit Intake'!G2="Delivered",'Audit
Intake'!G2="Pending",'Audit Intake'!G2="Cancelled")
Case 2  3       ='Audit Intake'!A3      =IF(C3="",0,COUNTIF(Orders!$A$2:$A$25,C3))
=IF(C3="",0,COUNTIF('Audit Intake'!$A$2:$A$7,C3))       =COUNTIF(Rates!$A$2:$A$7,'Audit
Intake'!D3)     =ISNUMBER('Audit Intake'!E3)    =IF(ISNUMBER('Audit Intake'!E3),AND('Audit
Intake'!E3≥0,MOD('Audit Intake'!E3,1)=0),FALSE)        =AND(ISNUMBER('Audit
Intake'!F3),'Audit Intake'!F3≥0)       =OR('Audit Intake'!G3="Delivered",'Audit
Intake'!G3="Pending",'Audit Intake'!G3="Cancelled")
Case 3  4       ='Audit Intake'!A4      =IF(C4="",0,COUNTIF(Orders!$A$2:$A$25,C4))
=IF(C4="",0,COUNTIF('Audit Intake'!$A$2:$A$7,C4))       =COUNTIF(Rates!$A$2:$A$7,'Audit
Intake'!D4)     =ISNUMBER('Audit Intake'!E4)    =IF(ISNUMBER('Audit Intake'!E4),AND('Audit
Intake'!E4≥0,MOD('Audit Intake'!E4,1)=0),FALSE)        =AND(ISNUMBER('Audit
Intake'!F4),'Audit Intake'!F4≥0)       =OR('Audit Intake'!G4="Delivered",'Audit
Intake'!G4="Pending",'Audit Intake'!G4="Cancelled")
Case 4  5       ='Audit Intake'!A5      =IF(C5="",0,COUNTIF(Orders!$A$2:$A$25,C5))
=IF(C5="",0,COUNTIF('Audit Intake'!$A$2:$A$7,C5))       =COUNTIF(Rates!$A$2:$A$7,'Audit
Intake'!D5)     =ISNUMBER('Audit Intake'!E5)    =IF(ISNUMBER('Audit Intake'!E5),AND('Audit
Intake'!E5≥0,MOD('Audit Intake'!E5,1)=0),FALSE)        =AND(ISNUMBER('Audit
Intake'!F5),'Audit Intake'!F5≥0)       =OR('Audit Intake'!G5="Delivered",'Audit
Intake'!G5="Pending",'Audit Intake'!G5="Cancelled")
Case 5  6       ='Audit Intake'!A6      =IF(C6="",0,COUNTIF(Orders!$A$2:$A$25,C6))
=IF(C6="",0,COUNTIF('Audit Intake'!$A$2:$A$7,C6))       =COUNTIF(Rates!$A$2:$A$7,'Audit
Intake'!D6)     =ISNUMBER('Audit Intake'!E6)    =IF(ISNUMBER('Audit Intake'!E6),AND('Audit
Intake'!E6≥0,MOD('Audit Intake'!E6,1)=0),FALSE)        =AND(ISNUMBER('Audit
Intake'!F6),'Audit Intake'!F6≥0)       =OR('Audit Intake'!G6="Delivered",'Audit
Intake'!G6="Pending",'Audit Intake'!G6="Cancelled")
Case 6  7       ='Audit Intake'!A7      =IF(C7="",0,COUNTIF(Orders!$A$2:$A$25,C7))
=IF(C7="",0,COUNTIF('Audit Intake'!$A$2:$A$7,C7))       =COUNTIF(Rates!$A$2:$A$7,'Audit
Intake'!D7)     =ISNUMBER('Audit Intake'!E7)    =IF(ISNUMBER('Audit Intake'!E7),AND('Audit
Intake'!E7≥0,MOD('Audit Intake'!E7,1)=0),FALSE)        =AND(ISNUMBER('Audit
Intake'!F7),'Audit Intake'!F7≥0)       =OR('Audit Intake'!G7="Delivered",'Audit
Intake'!G7="Pending",'Audit Intake'!G7="Cancelled")
```

Supporting comparison cells: B11 **=Orders!E2** gives2; E11 **='Audit Intake'!E2** gives3; H11 **=Orders!E2='Audit Intake'!E2** givesFALSE. E13 **=SUMPRODUCT(--(Orders!A2:I2='Source Snapshot'!A2:I2))** gives9 identical fields in the preserved copy. A repeated copy is not an additional order; the conflicting PO001 intake remains quarantined.

Proposed exception register — Audit Intake J1:L7

These are guided author-reference dispositions, not completed independent decisions or messages to a source owner.

```
Issue   Evidence        Disposition
Conflicting duplicate OrderID   Practice PO001: source ID count1; Orders Quantity2 vs intake3,
equality FALSE. Other fields copied from PO001. Quarantine pending evidenced correction.
Clarify with the source owner; do not overwrite Orders or count both.
Missing OrderID Raw intake A3 is empty. This is a labelled copy of PO002 with only its ID
removed; do not infer that missing ID.  Clarify the required key; quarantine pending evidence.
Do not invent an ID or silently fill PO002.
Unknown ItemCode        IT99 has zero exact matches in Rates A2:A7. Other fields are practice
copies of PO003. Cancelled itself is allowed.   Clarify the intended item key and rate with
evidence. Keep raw IT99; quarantine pending evidence.
Non-numeric Quantity    Quantity is the text four: ISNUMBER is FALSE and the non-negative
whole-number rule is FALSE. Other fields copied from PO004.     Clarify the intended numeric
quantity with evidence; propose correction separately. Do not replace raw text by guessing 4.
Valid zero Quantity     Quantity0 is numeric and passes the explicit non-negative whole-number
rule. Other fields copied from PO005 under a practice ID.       Retain as a valid training
example. No insertion into baseline Orders; zero does not prove a delivery or spend.
Valid Cancelled status  Cancelled passes the status contract; numeric Quantity2, UnitCost8 and
key IT06 are valid. Practice copy of PO006 with status changed. Retain as a valid all-orders
training example. Later status exclusions need a stated analysis question, not a dirty-data
label.
```

Independent release conditions, independent disposition reasoning, reviewer challenge and learner timing remain pending. The six intake examples never enter baseline controls. A preserved Source Snapshot is a copy, not a protected-range permission or automatic archive.

02

CHAPTER 2

Build Conditional Calculations

In this chapter you will revise mixed references and apply IF, SUMIF, COUNTIF, SUMIFS and COUNTIFS with test cases.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 75 minutes: 15 explanation/demonstration, 32 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 60 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Checked line amounts, a project/status summary grid and tested logic bands.

> **Status:** author reference with37 staged application captures through2.4. Row formulas, mixed-reference grids, distinct populations, manual six-ID checks,21 expected-versus-actual logic cases and isolated anchor-fault repair verified. Original input fields and Lab1 evidence preserved. Independent threshold summary,250/restoration test, reviewer and learner timing remain pending.

What You Will Learn

- Build row calculations.
- Apply conditional totals with mixed references.
- Test a nested rule at its boundaries.

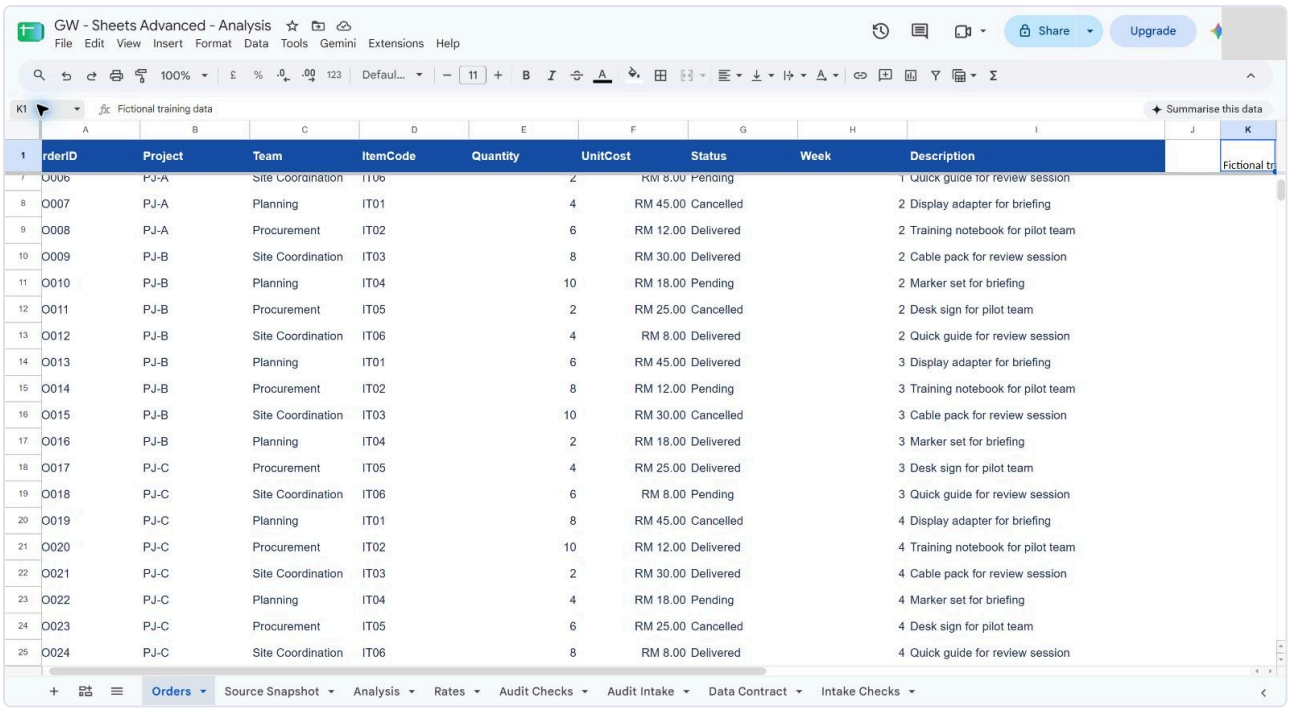
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Orders A1:I25 and the Analysis tab from Chapter 1.

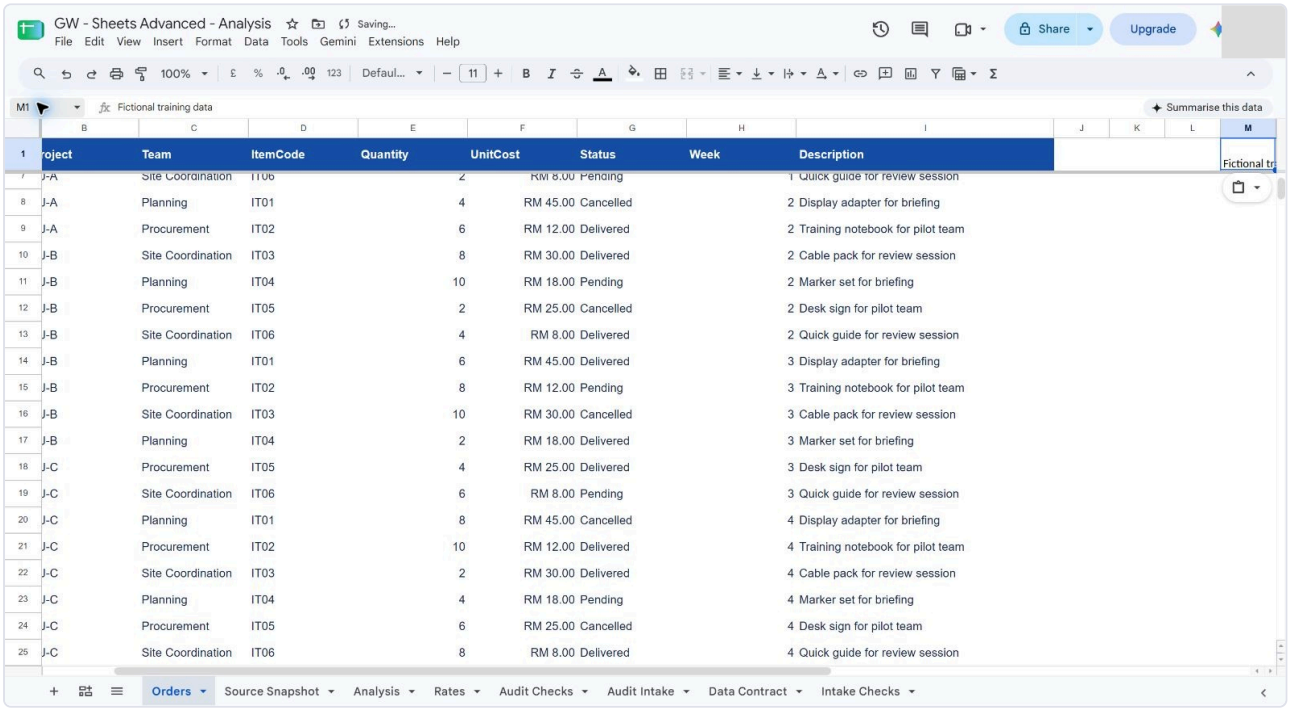
2.1 Calculate Line Amounts

1. Reopen **GW - Sheets Advanced - Analysis** from Chapter1. Inspect Orders. The supplied **K1** contains a fictional-data label, so preserve it before using K for a calculation. Keep A:I, Rates and Source Snapshot unchanged.



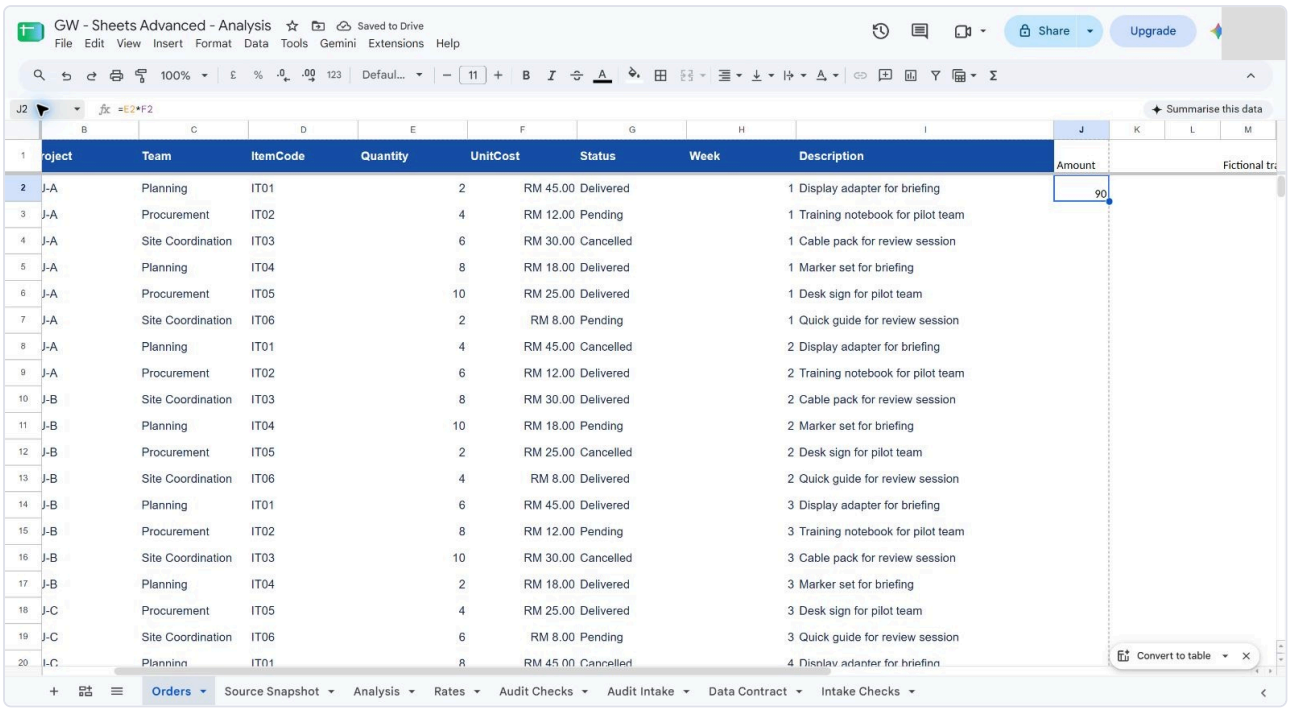
	A	B	C	D	E	F	G	H	I	J	K
1	rderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description		Fictional tr
7	O006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session		
8	O007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing		
9	O008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team		
10	O009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session		
11	O010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing		
12	O011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team		
13	O012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session		
14	O013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing		
15	O014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team		
16	O015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session		
17	O016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing		
18	O017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team		
19	O018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session		
20	O019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing		
21	O020	PJ-C	Procurement	IT02	10	RM 12.00	Delivered	4	Training notebook for pilot team		
22	O021	PJ-C	Site Coordination	IT03	2	RM 30.00	Delivered	4	Cable pack for review session		
23	O022	PJ-C	Planning	IT04	4	RM 18.00	Pending	4	Marker set for briefing		
24	O023	PJ-C	Procurement	IT05	6	RM 25.00	Cancelled	4	Desk sign for pilot team		
25	O024	PJ-C	Site Coordination	IT06	8	RM 8.00	Delivered	4	Quick guide for review session		

2. Cut **K1** and paste it into **M1**, outside the calculation columns. This preserves the label rather than silently overwriting it with ActiveAmount. Source Snapshot retains the original label at K1.

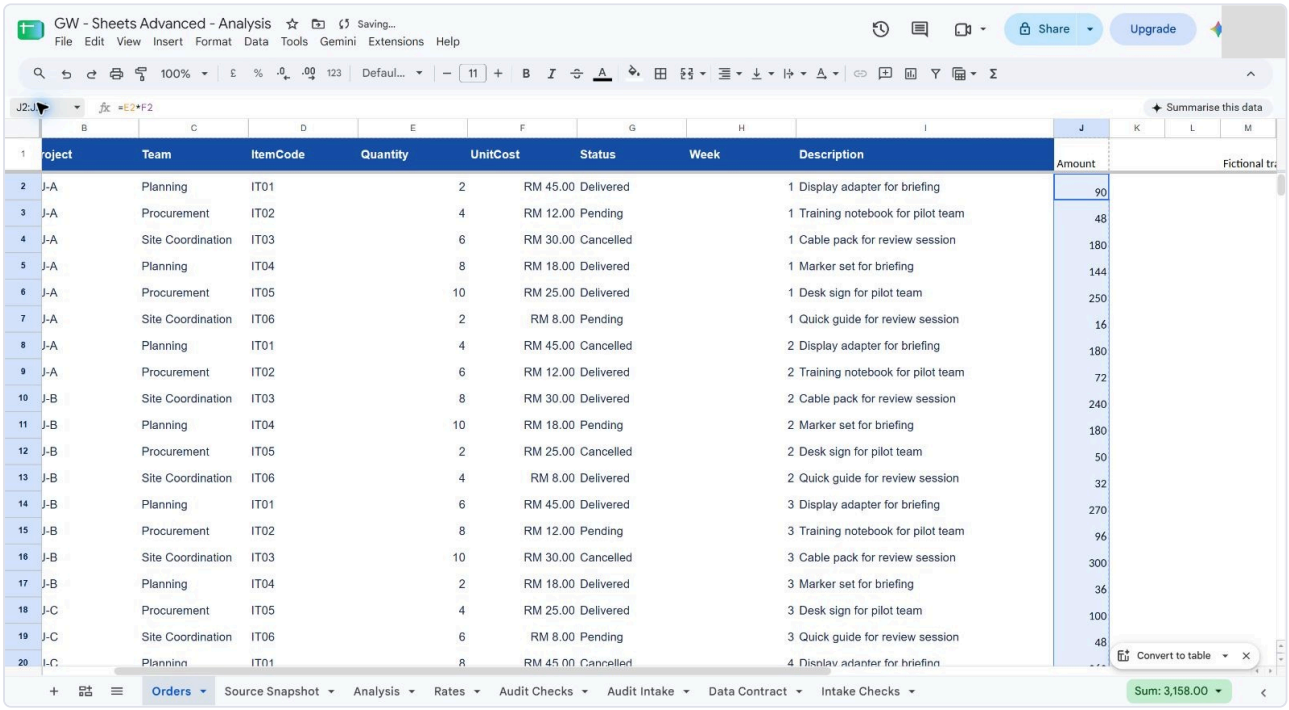


	B	C	D	E	F	G	H	I	J	K	L	M
1	oject	Team	ItemCode	Quantity	UnitCost	Status	Week	Description				Fictional tr
7	J-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session				
8	J-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing				
9	J-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team				
10	J-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session				
11	J-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing				
12	J-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team				
13	J-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session				
14	J-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing				
15	J-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team				
16	J-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session				
17	J-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing				
18	J-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team				
19	J-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session				
20	J-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing				
21	J-C	Procurement	IT02	10	RM 12.00	Delivered	4	Training notebook for pilot team				
22	J-C	Site Coordination	IT03	2	RM 30.00	Delivered	4	Cable pack for review session				
23	J-C	Planning	IT04	4	RM 18.00	Pending	4	Marker set for briefing				
24	J-C	Procurement	IT05	6	RM 25.00	Cancelled	4	Desk sign for pilot team				
25	J-C	Site Coordination	IT06	8	RM 8.00	Delivered	4	Quick guide for review session				

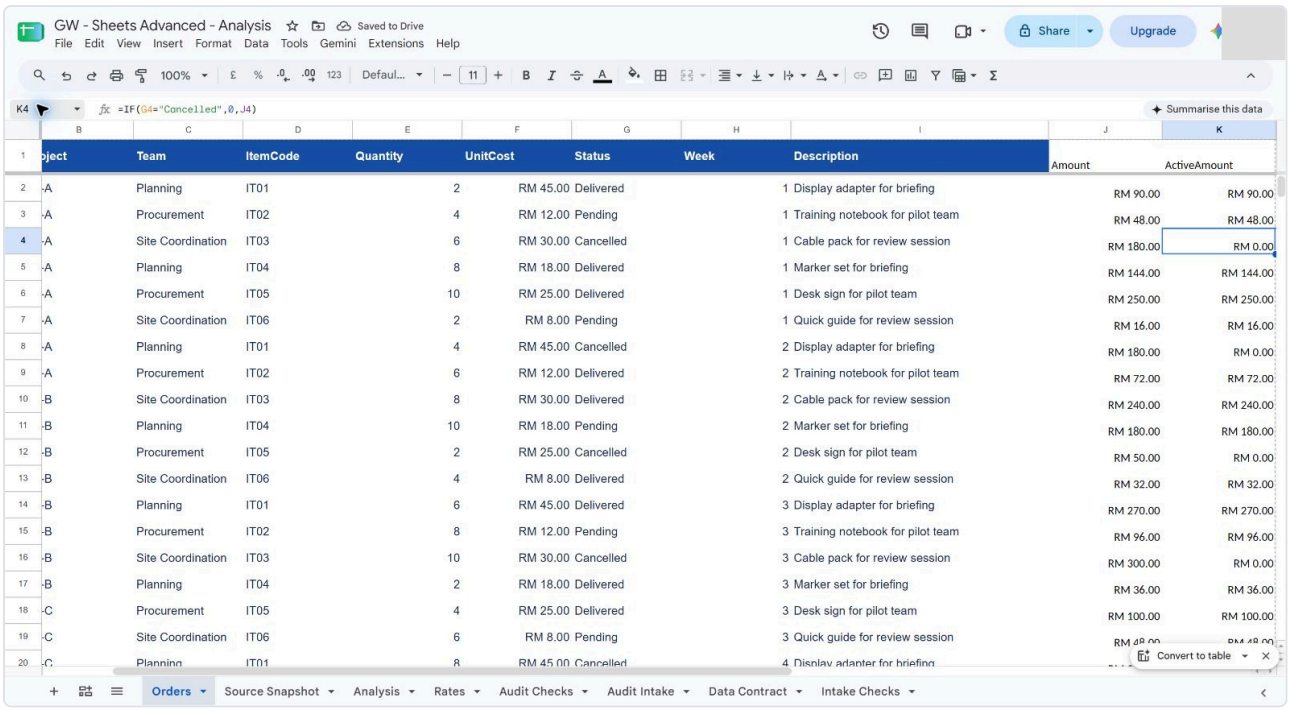
3. Enter **Amount** in J1 and **=E2*F2** in J2. For PO001, Quantity2 times UnitCost45 gives **90**.



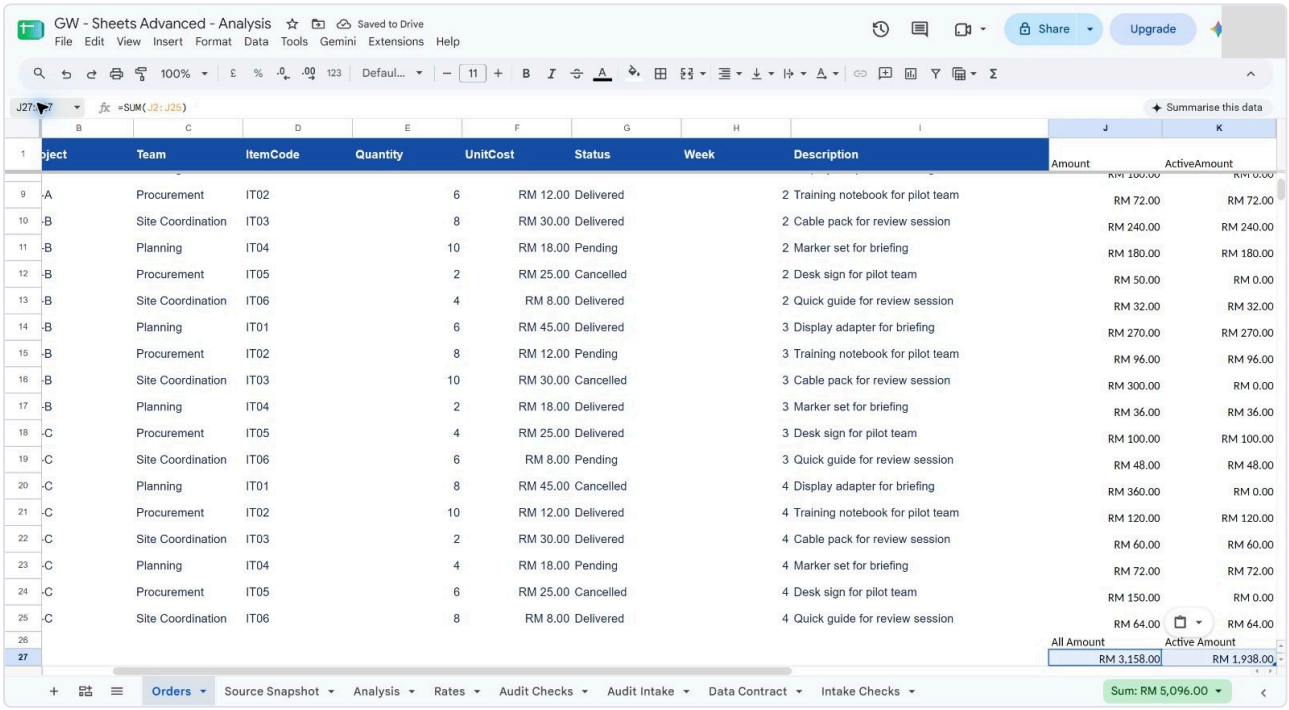
4. Select **J2:J25** and use **Fill down** (Ctrl+D). Check that every formula multiplies its own row's Quantity and UnitCost.



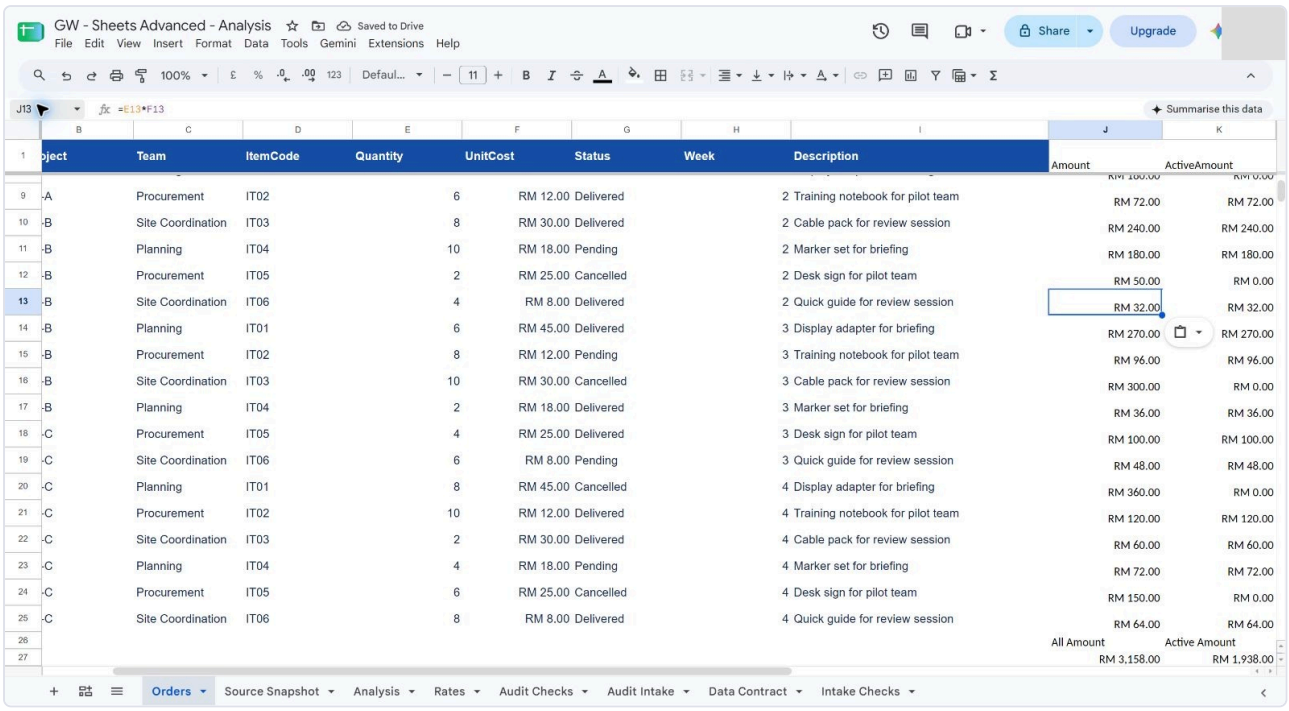
5. Enter **ActiveAmount** in K1 and **=IF(G2="Cancelled",0,J2)** in K2, then fill K2:K25 down. Inspect **K4**: PO003 remains in Orders with Amount180, while ActiveAmount is **0**. This applies a population rule; it does not delete or declare the source row corrupt. Widen new J/K columns as needed. For explicit RM display, apply custom number format **"RM" #,##0.00** to J2:K25; do not change the supplied numeric values.



6. Outside the 24-row table, put **All Amount / Active Amount** in J26:K26 and **=SUM(J2:J25) / =SUM(K2:K25)** in J27:K27. Expect **RM3158 / RM1938**. Keep these helper totals outside later bounded source calculations.

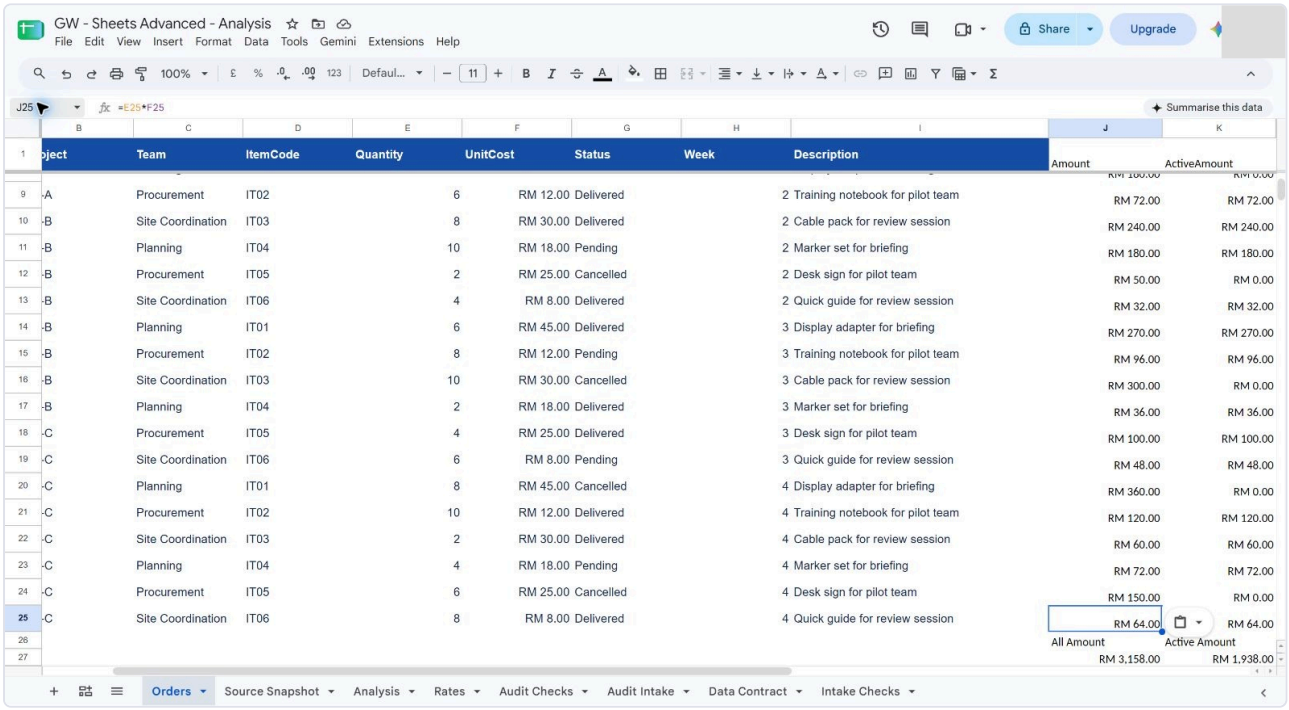


7. Inspect **J13 =E13*F13** for PO012: **4 × 8 =32**. Compare the referenced source fields, not only the displayed amount.



	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount
-A	Procurement	IT02	6	RM 12.00	Delivered		2 Training notebook for pilot team	RM 72.00	RM 72.00
-B	Site Coordination	IT03	8	RM 30.00	Delivered		2 Cable pack for review session	RM 240.00	RM 240.00
-B	Planning	IT04	10	RM 18.00	Pending		2 Marker set for briefing	RM 180.00	RM 180.00
-B	Procurement	IT05	2	RM 25.00	Cancelled		2 Desk sign for pilot team	RM 50.00	RM 0.00
-B	Site Coordination	IT06	4	RM 8.00	Delivered		2 Quick guide for review session	RM 32.00	RM 32.00
-B	Planning	IT01	6	RM 45.00	Delivered		3 Display adapter for briefing	RM 270.00	RM 270.00
-B	Procurement	IT02	8	RM 12.00	Pending		3 Training notebook for pilot team	RM 96.00	RM 96.00
-B	Site Coordination	IT03	10	RM 30.00	Cancelled		3 Cable pack for review session	RM 300.00	RM 0.00
-B	Planning	IT04	2	RM 18.00	Delivered		3 Marker set for briefing	RM 36.00	RM 36.00
-C	Procurement	IT05	4	RM 25.00	Delivered		3 Desk sign for pilot team	RM 100.00	RM 100.00
-C	Site Coordination	IT06	6	RM 8.00	Pending		3 Quick guide for review session	RM 48.00	RM 48.00
-C	Planning	IT01	8	RM 45.00	Cancelled		4 Display adapter for briefing	RM 360.00	RM 0.00
-C	Procurement	IT02	10	RM 12.00	Delivered		4 Training notebook for pilot team	RM 120.00	RM 120.00
-C	Site Coordination	IT03	2	RM 30.00	Delivered		4 Cable pack for review session	RM 60.00	RM 60.00
-C	Planning	IT04	4	RM 18.00	Pending		4 Marker set for briefing	RM 72.00	RM 72.00
-C	Procurement	IT05	6	RM 25.00	Cancelled		4 Desk sign for pilot team	RM 150.00	RM 0.00
-C	Site Coordination	IT06	8	RM 8.00	Delivered		4 Quick guide for review session	RM 64.00	RM 64.00
								All Amount	Active Amount
								RM 3,158.00	RM 1,938.00

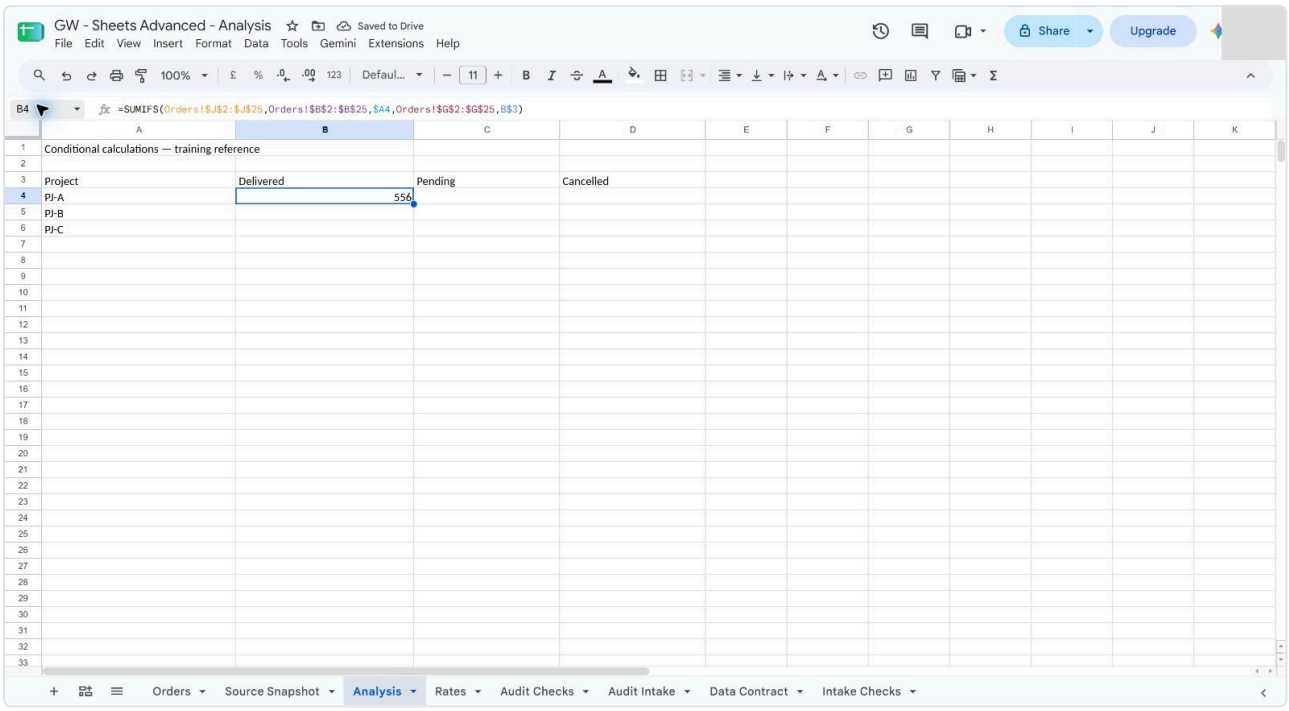
8. Inspect **J25 =E25*F25** for PO024: **8 × 8 =64**. The last row must be included in both filled columns and totals.



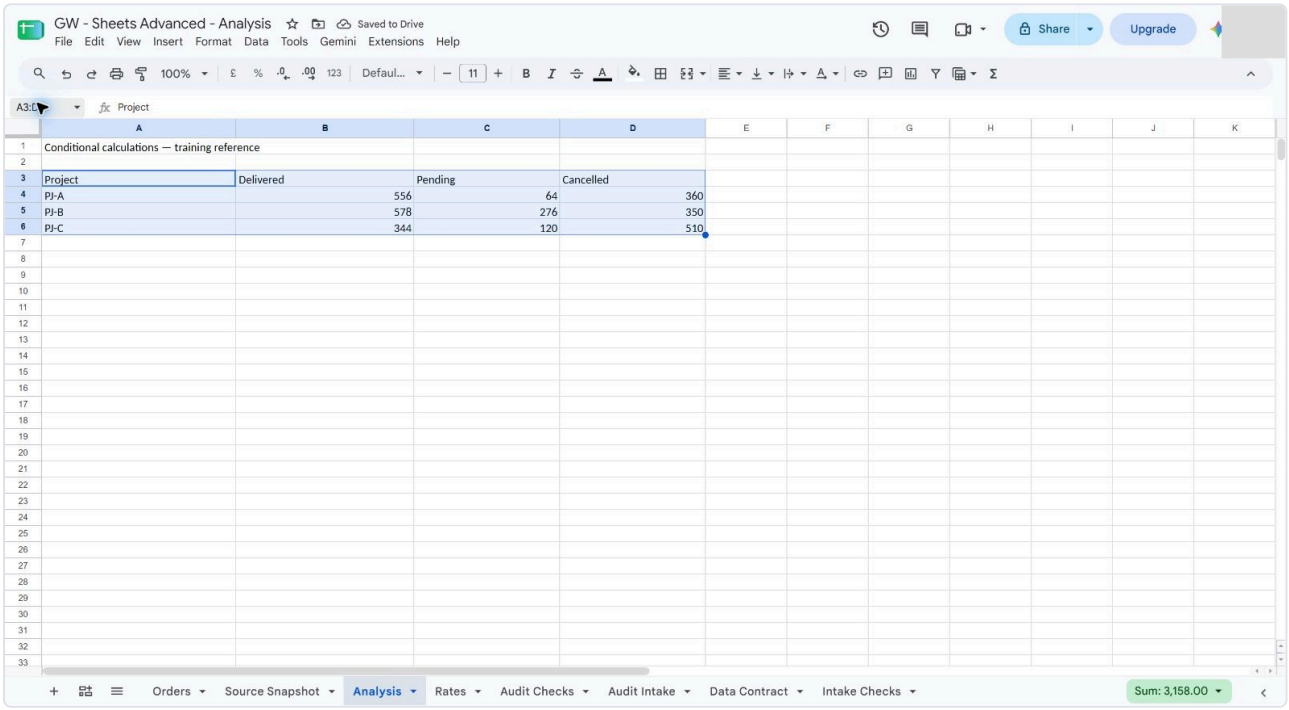
	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount
-A	Procurement	IT02	6	RM 12.00	Delivered		2 Training notebook for pilot team	RM 72.00	RM 72.00
-B	Site Coordination	IT03	8	RM 30.00	Delivered		2 Cable pack for review session	RM 240.00	RM 240.00
-B	Planning	IT04	10	RM 18.00	Pending		2 Marker set for briefing	RM 180.00	RM 180.00
-B	Procurement	IT05	2	RM 25.00	Cancelled		2 Desk sign for pilot team	RM 50.00	RM 0.00
-B	Site Coordination	IT06	4	RM 8.00	Delivered		2 Quick guide for review session	RM 32.00	RM 32.00
-B	Planning	IT01	6	RM 45.00	Delivered		3 Display adapter for briefing	RM 270.00	RM 270.00
-B	Procurement	IT02	8	RM 12.00	Pending		3 Training notebook for pilot team	RM 96.00	RM 96.00
-B	Site Coordination	IT03	10	RM 30.00	Cancelled		3 Cable pack for review session	RM 300.00	RM 0.00
-B	Planning	IT04	2	RM 18.00	Delivered		3 Marker set for briefing	RM 36.00	RM 36.00
-C	Procurement	IT05	4	RM 25.00	Delivered		3 Desk sign for pilot team	RM 100.00	RM 100.00
-C	Site Coordination	IT06	6	RM 8.00	Pending		3 Quick guide for review session	RM 48.00	RM 48.00
-C	Planning	IT01	8	RM 45.00	Cancelled		4 Display adapter for briefing	RM 360.00	RM 0.00
-C	Procurement	IT02	10	RM 12.00	Delivered		4 Training notebook for pilot team	RM 120.00	RM 120.00
-C	Site Coordination	IT03	2	RM 30.00	Delivered		4 Cable pack for review session	RM 60.00	RM 60.00
-C	Planning	IT04	4	RM 18.00	Pending		4 Marker set for briefing	RM 72.00	RM 72.00
-C	Procurement	IT05	6	RM 25.00	Cancelled		4 Desk sign for pilot team	RM 150.00	RM 0.00
-C	Site Coordination	IT06	8	RM 8.00	Delivered		4 Quick guide for review session	RM 64.00	RM 64.00
								All Amount	Active Amount
								RM 3,158.00	RM 1,938.00

2.2 Create a Summary Grid

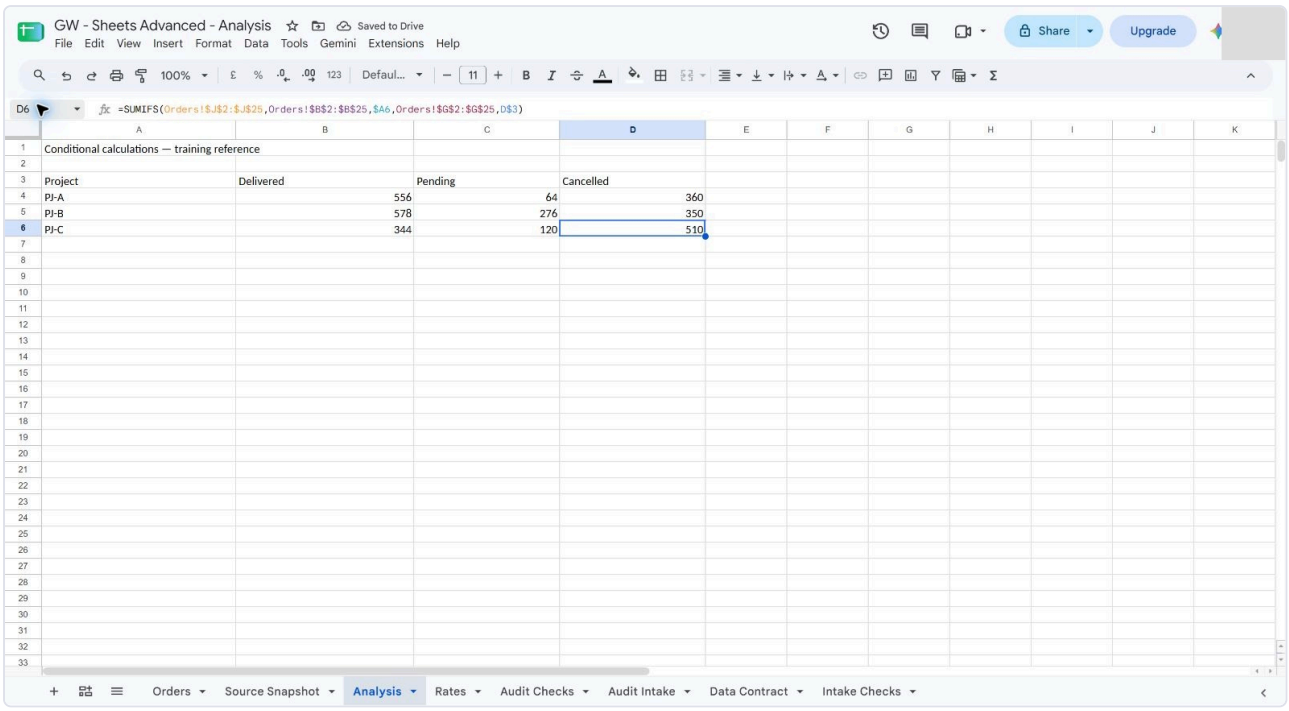
1. On **Analysis**, enter Project in A3, Delivered/Pending/Cancelled in B3:D3, and PJ-A/PJ-B/PJ-C in A4:A6. In B4 enter **=SUMIFS(Orders!$J$2:$J$25,Orders!$B$2:$B$25,$A4,Orders!$G$2:$G$25,B$3)**. Its first result is **556**. The source ranges are fully anchored; the project column is fixed but its row can move; the status row is fixed but its column can move.



2. Select **B4:D4** and **Fill right** (Ctrl+R), then select **B4:D6** and **Fill down** (Ctrl+D). Expected rows are **556/64/360**, **578/276/350** and **344/120/510**. Label the amount grid as RM. These nine intersections sum to **3158** across every status.

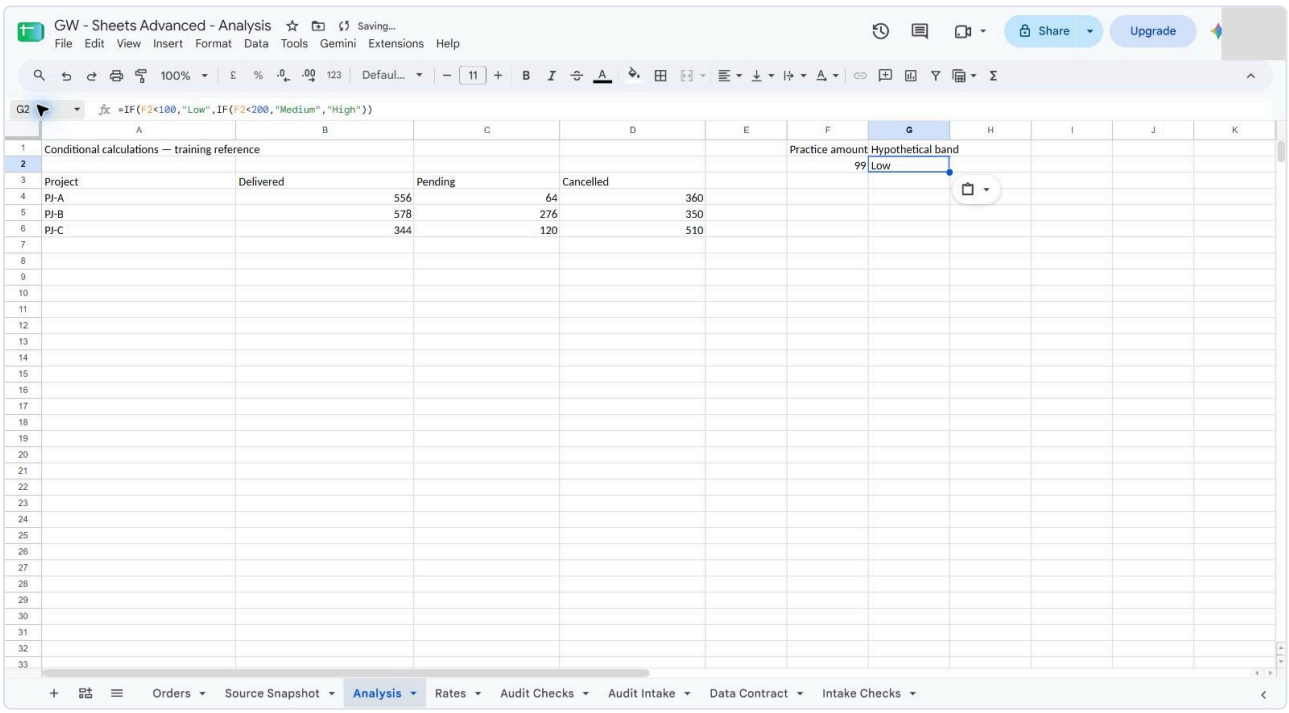


3. Inspect **D6**. It must use **$A6** for PJ-C and **D$3** for Cancelled, while both source criteria ranges and the sum range remain fixed. Its value is **510**. After populating Analysis, update the Chapter1 Audit Checks notes that previously described Analysis as empty; preserve the Chapter1 audit evidence as a historical checkpoint.

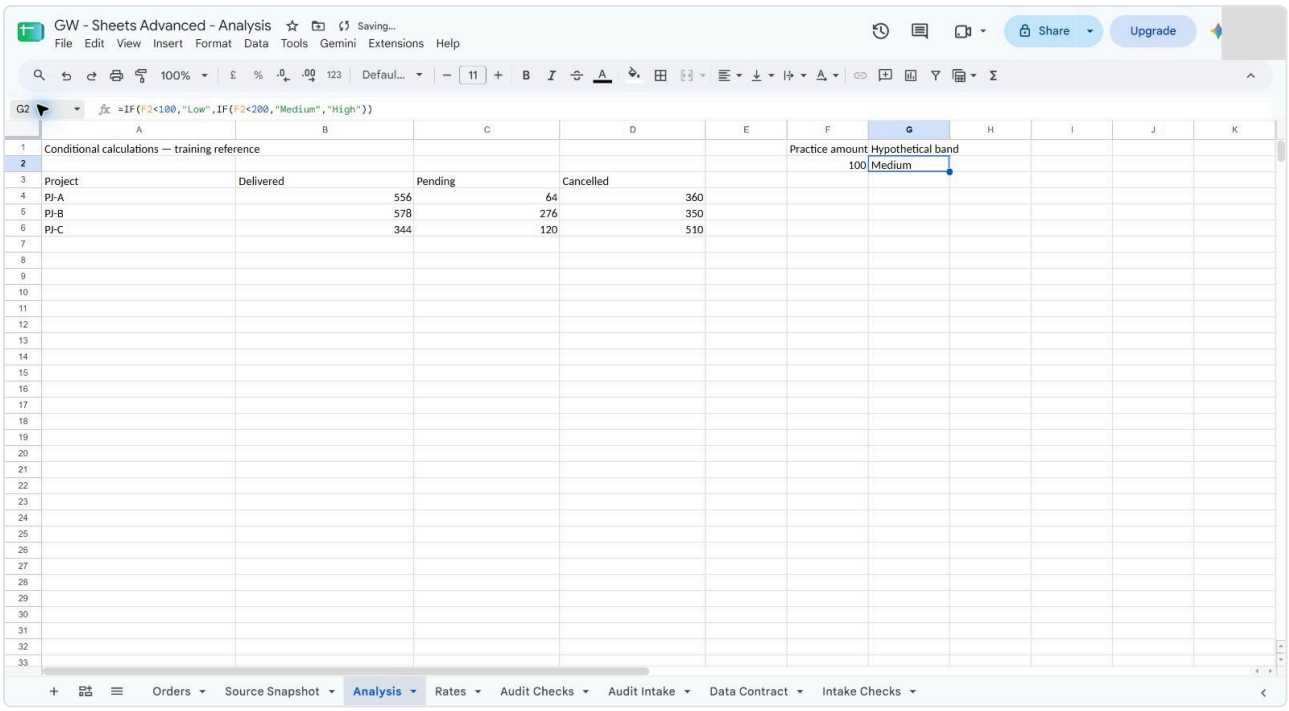


2.3 Test Logic Boundaries

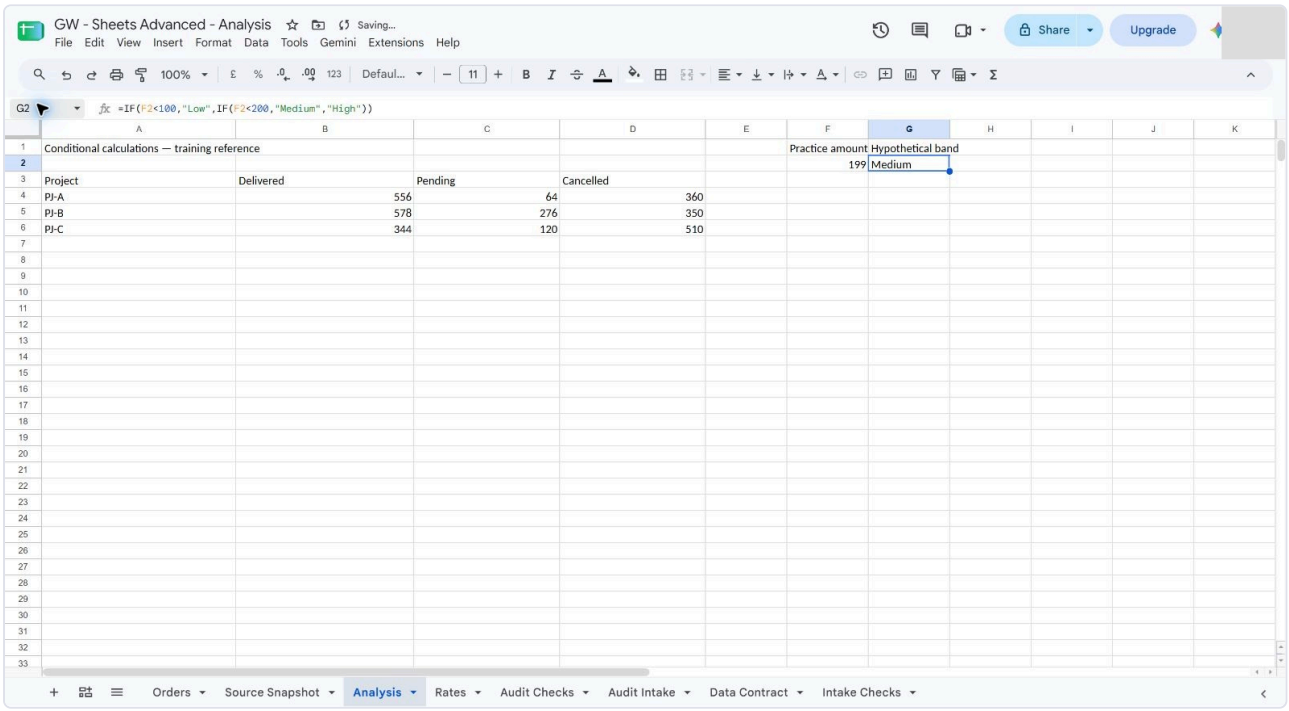
1. In the separate practice area on Analysis, label **F1/G1** Practice amount / Hypothetical band. Put **99** in F2 and **=IF(F2<100,"Low",IF(F2<200,"Medium","High"))** in G2. Expect **Low**. These are fictional training bands, not procurement policy.



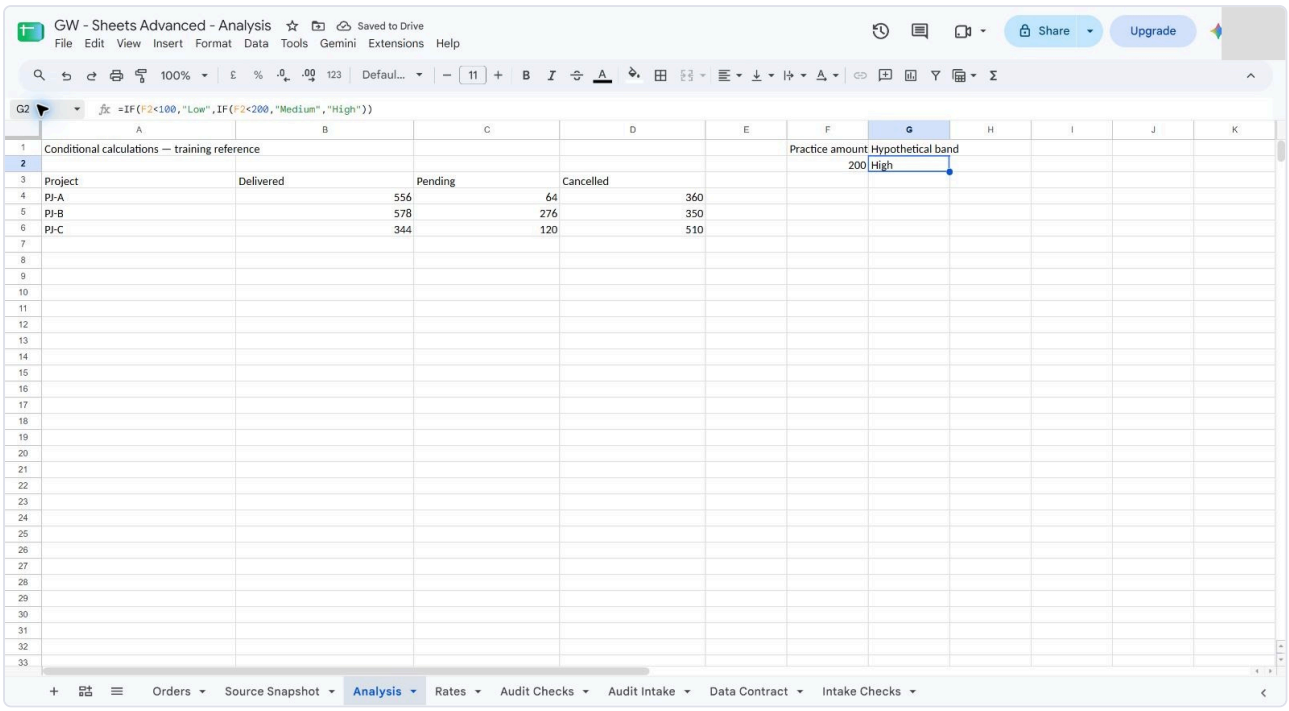
2. Change only F2 to **100**. The lower Medium boundary is inclusive: expect **Medium**.



3. Change F2 to **199**. It remains below200: expect **Medium**.

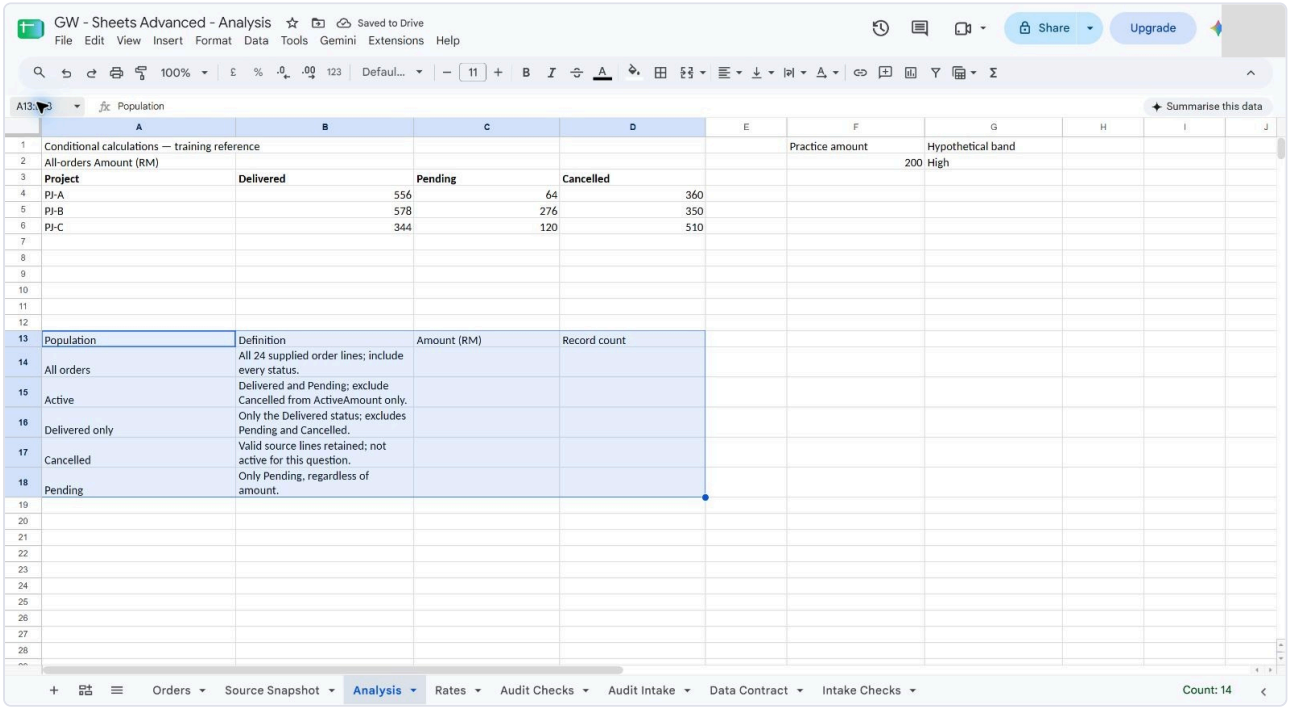


4. Change F2 to **200**. The High boundary is inclusive: expect **High**. Retain the tested formula; the cross-stage captures record the four separate results.

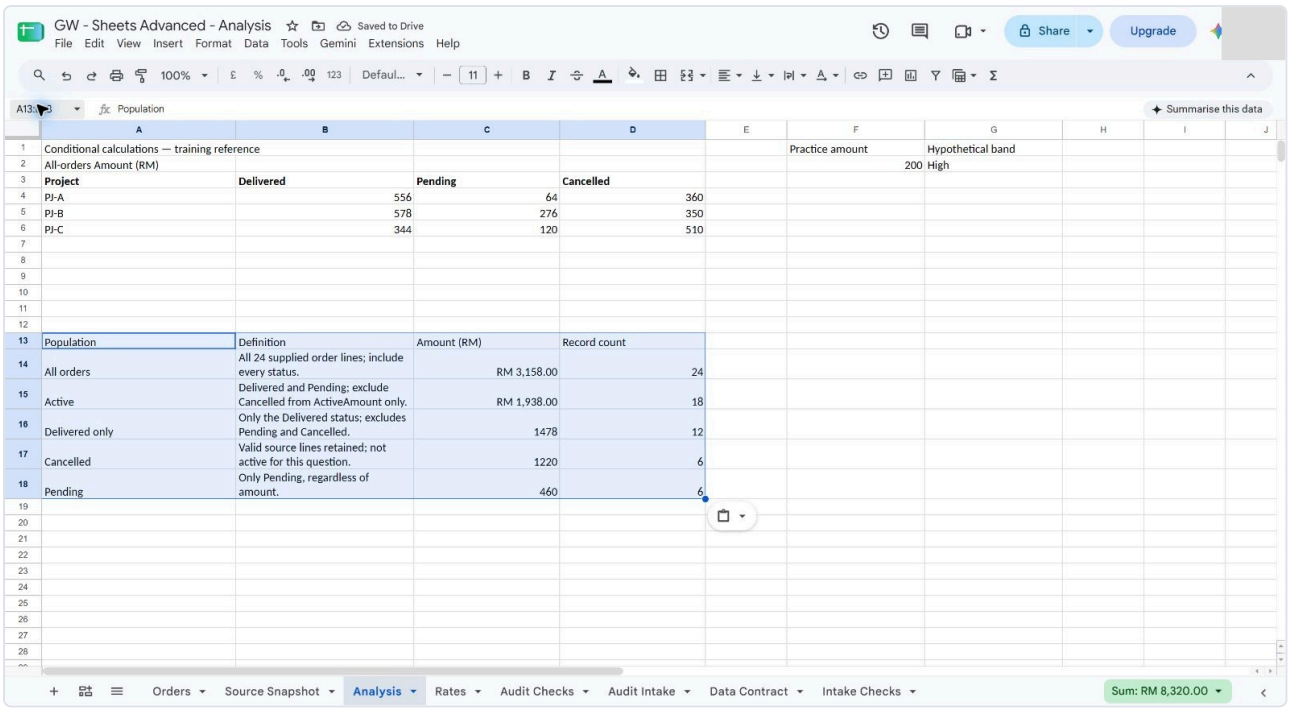


2.4 Reconcile Three Populations and Test Conditional Boundaries

1. Write the population definitions on **Analysis A13:D18 before calculating**: all24 order lines; Active excludes Cancelled; Delivered-only excludes both Pending and Cancelled. Keep Cancelled and Pending controls explicit too. Do not treat totals for different populations as interchangeable.

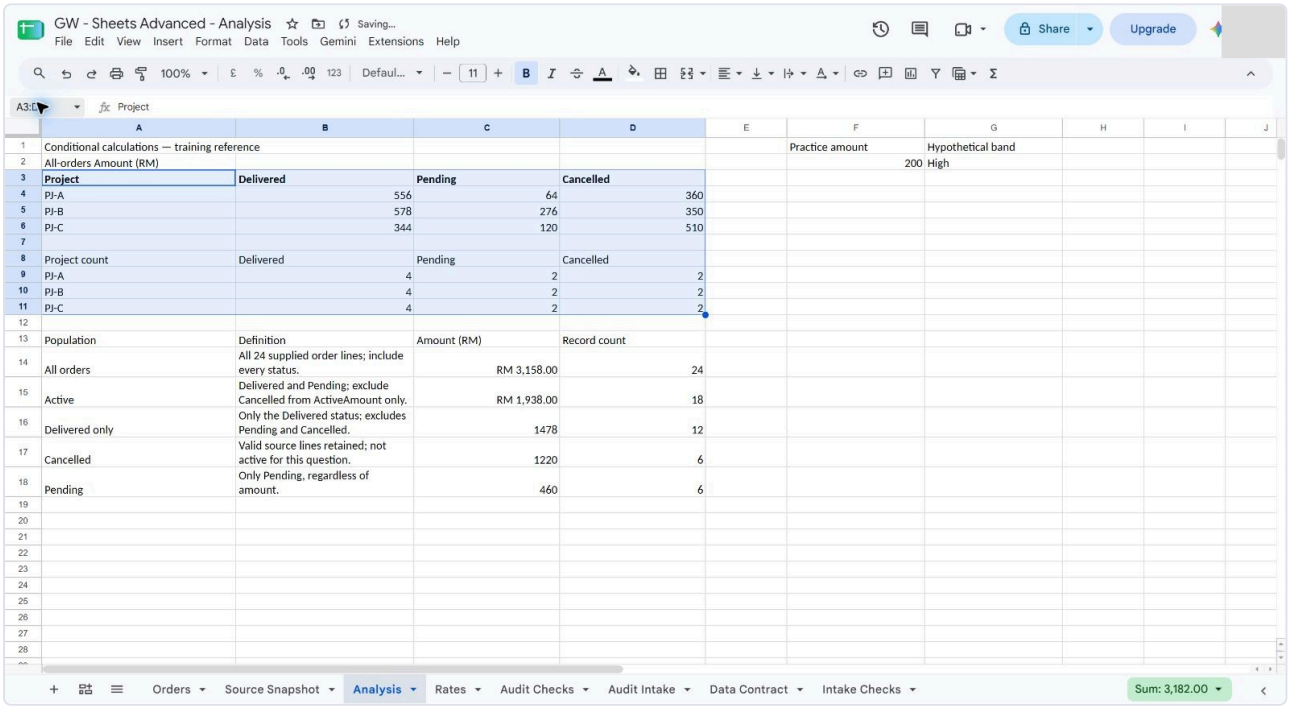


2. Fill the amount and count controls using the card on the copy-paste page. Expected amount/count pairs are **All3158/24**, **Active1938/18**, **Delivered1478/12**, **Cancelled1220/6**, **Pending460/6**. The SUMIF/COUNTIF checks use the exact status vocabulary; ActiveAmount is summed separately from all-orders Amount.



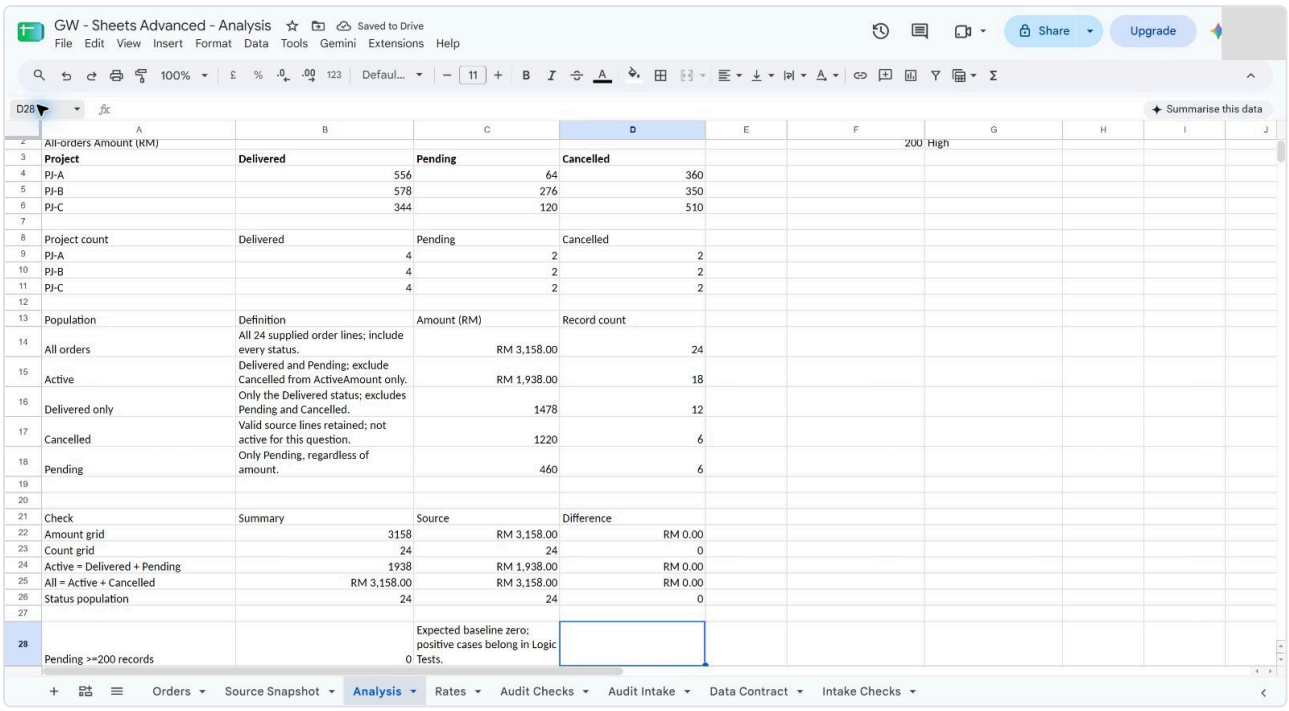
	A	B	C	D	E	F	G	H	I	J
1	Conditional calculations — training reference					Practice amount	Hypothetical band			
2	All-orders Amount (RM)					200	High			
3	**Project**	**Delivered**	**Pending**	**Cancelled**						
4	PJ-A	556	64	360						
5	PJ-B	578	276	350						
6	PJ-C	344	120	510						
7										
8										
9										
10										
11										
12										
13	Population	Definition	Amount (RM)	Record count						
14	All orders	All 24 supplied order lines; include every status.	RM 3,158.00	24						
15	Active	Delivered and Pending; exclude Cancelled from ActiveAmount only.	RM 1,938.00	18						
16	Delivered only	Only the Delivered status; excludes Pending and Cancelled.	1478	12						
17	Cancelled	Valid source lines retained; not active for this question.	1220	6						
18	Pending	Only Pending, regardless of amount.	460	6						
19										
20										
21										
22										
23										
24										
25										
26										
27										
28										

3. Build the matching COUNTIFS grid in **A8:D11**. Use Project count / Delivered / Pending / Cancelled in row8 and the three projects in A9:A11. Enter **=COUNTIFS(Orders!$B$2:$B$25,$A9,Orders!$G$2:$G$25,B$8)** in B9, fill right through D9 and down through row11. Each project has **4/2/2** records. Keep the amount and count grids separate, with clear units.

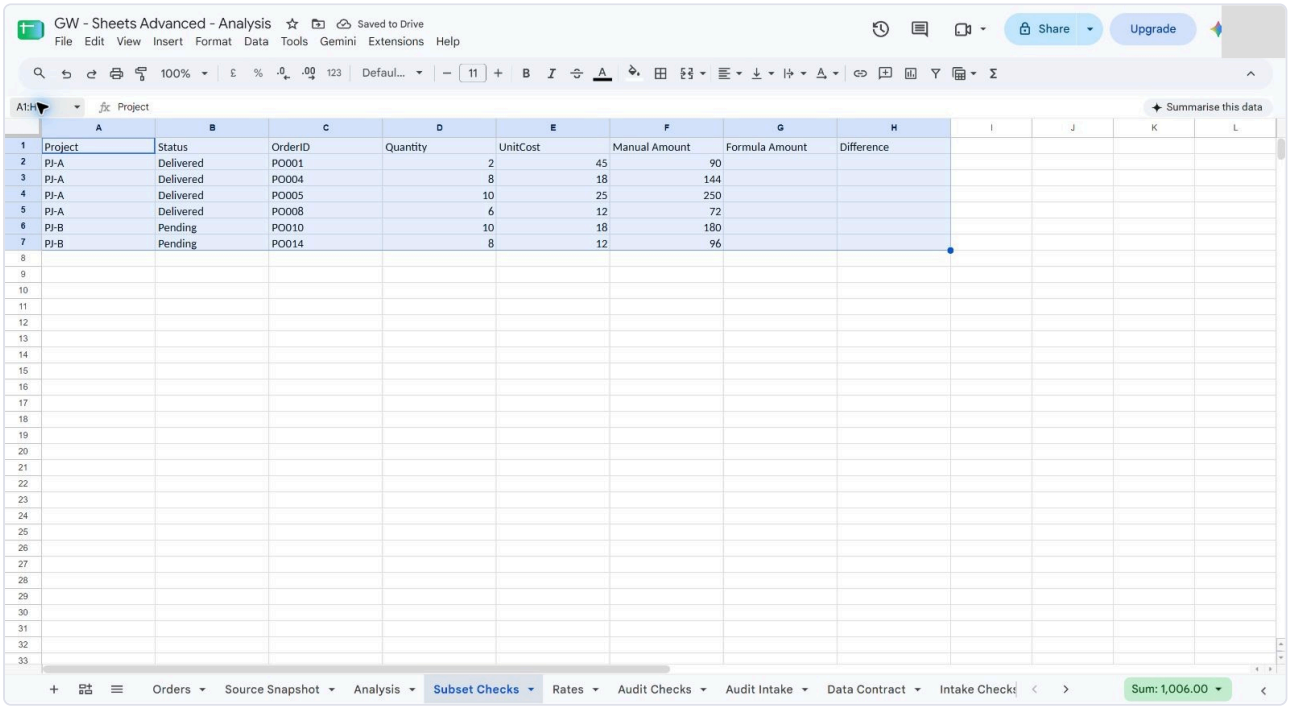


	A	B	C	D	E	F	G	H	I	J
1	Conditional calculations — training reference					Practice amount	Hypothetical band			
2	All-orders Amount (RM)					200	High			
3	**Project**	**Delivered**	**Pending**	**Cancelled**						
4	PJ-A	556	64	360						
5	PJ-B	578	276	350						
6	PJ-C	344	120	510						
7										
8	Project count	Delivered	Pending	Cancelled						
9	PJ-A	4	2	2						
10	PJ-B	4	2	2						
11	PJ-C	4	2	2						
12										
13	Population	Definition	Amount (RM)	Record count						
14	All orders	All 24 supplied order lines; include every status.	RM 3,158.00	24						
15	Active	Delivered and Pending; exclude Cancelled from ActiveAmount only.	RM 1,938.00	18						
16	Delivered only	Only the Delivered status; excludes Pending and Cancelled.	1478	12						
17	Cancelled	Valid source lines retained; not active for this question.	1220	6						
18	Pending	Only Pending, regardless of amount.	460	6						
19										
20										
21										
22										
23										
24										
25										
26										
27										
28										

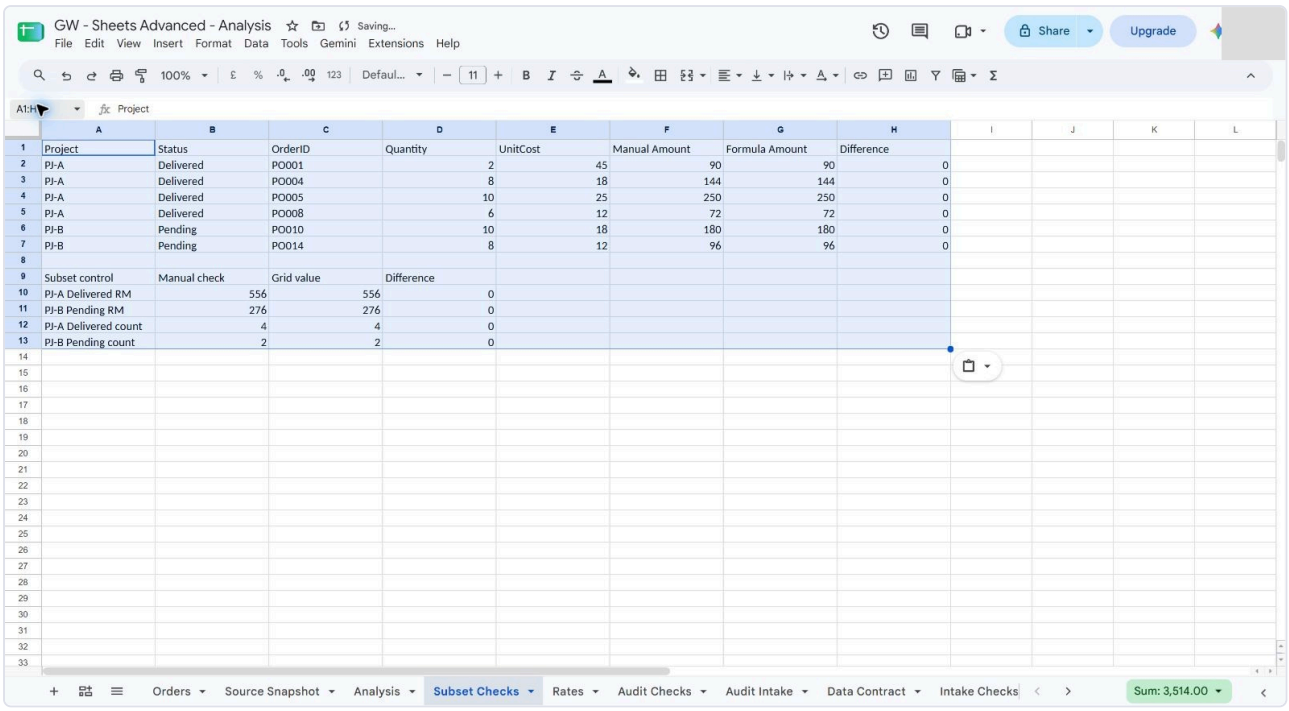
4. In **A21:D26**, reconcile the amount grid to3158 and count grid to24. Verify **Active = Delivered + Pending** and **All = Active + Cancelled**, with zero differences. In B28, **=COUNTIFS(Orders!G2:G25,"Pending",Orders!J2:J25,">=200")** returns **0**. This is a valid baseline result; positive-branch tests belong in the separate synthetic matrix.



5. Create **Subset Checks** and list the qualifying OrderIDs for two intersections. PJ-A Delivered is **PO001/PO004/PO005/PO008**, with manual products **90/144/250/72**. PJ-B Pending is **PO010/PO014**, with **180/96**. Inspect their actual source Quantity and UnitCost values. Record the manually checked products before entering the formula comparisons.



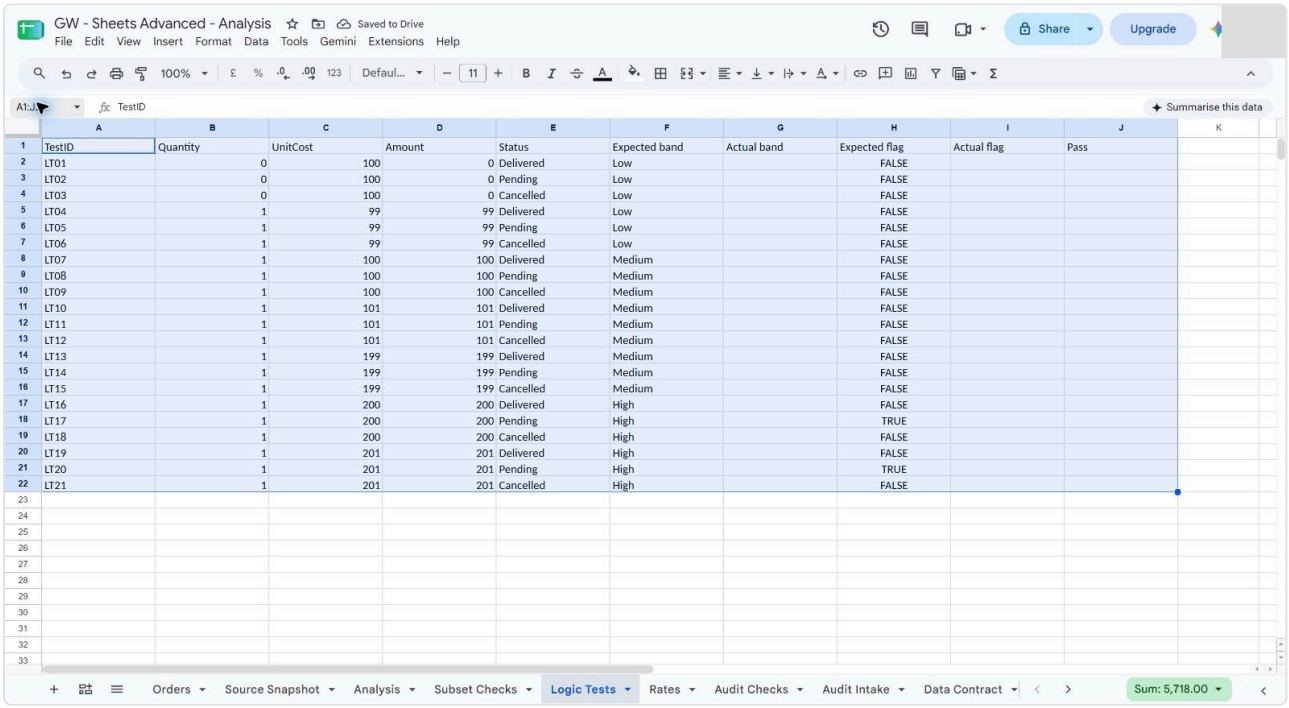
6. In Subset Checks G2 enter **=D2*E2**, H2 **=G2-F2**, and fill G2:H7 down. Compare the two manual totals to the maintained grid: **556** to Analysis B4, **276** to Analysis C5; compare qualifying-ID counts **4** and **2** to Analysis B9/C10. Every line and subset difference is zero. This is author-supported manual casework, not an independent learner validation.



Project	Status	OrderID	Quantity	UnitCost	Manual Amount	Formula Amount	Difference
PJ-A	Delivered	PO001	2	45	90	90	0
PJ-A	Delivered	PO004	8	18	144	144	0
PJ-A	Delivered	PO005	10	25	250	250	0
PJ-A	Delivered	PO008	6	12	72	72	0
PJ-B	Pending	PO010	10	18	180	180	0
PJ-B	Pending	PO014	8	12	96	96	0

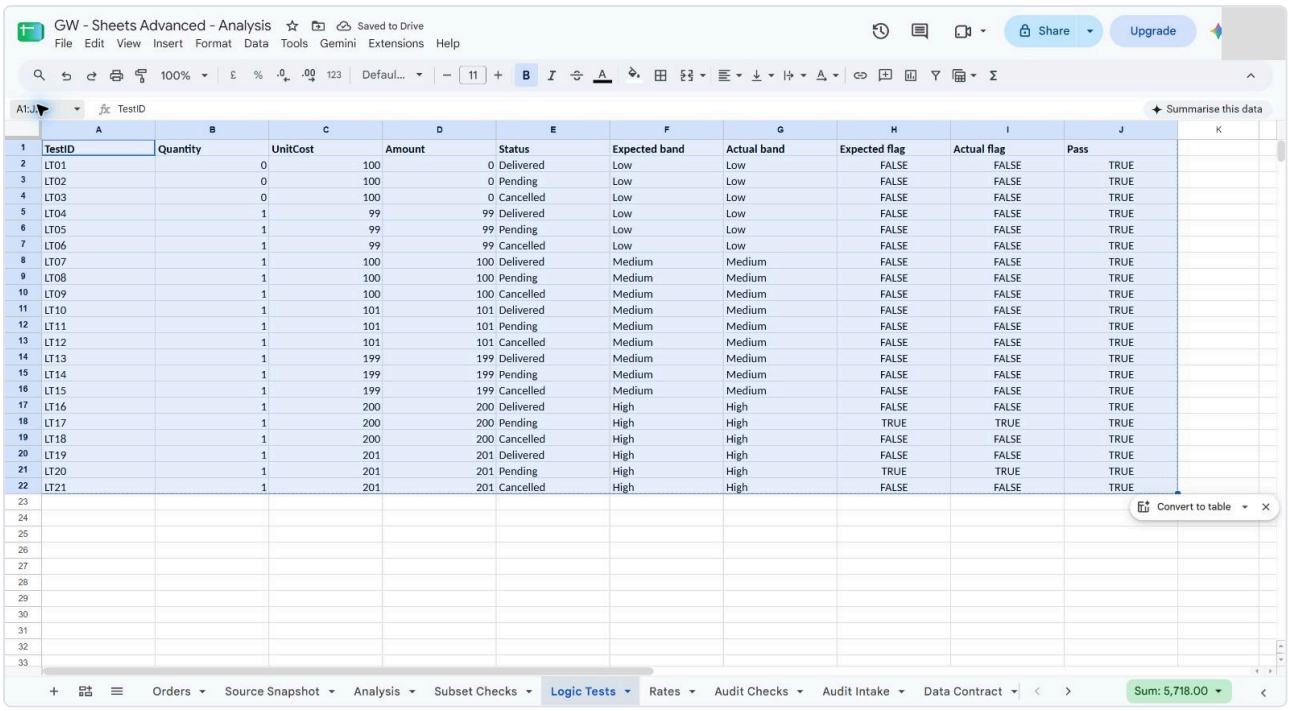
Subset control	Manual check	Grid value	Difference
PJ-A Delivered RM	556	556	0
PJ-B Pending RM	276	276	0
PJ-A Delivered count	4	4	0
PJ-B Pending count	2	2	0

7. Create **Logic Tests**. Paste the21-case expected-input card, leaving actual band/flag/pass columns empty initially. It crosses amounts **0,99,100,101,199,200,201** with all three statuses. The zero cases use **Quantity0 × known UnitCost100**, rather than an unknown rate disguised as0. Type the expected classifications and flags before running the actual formulas.



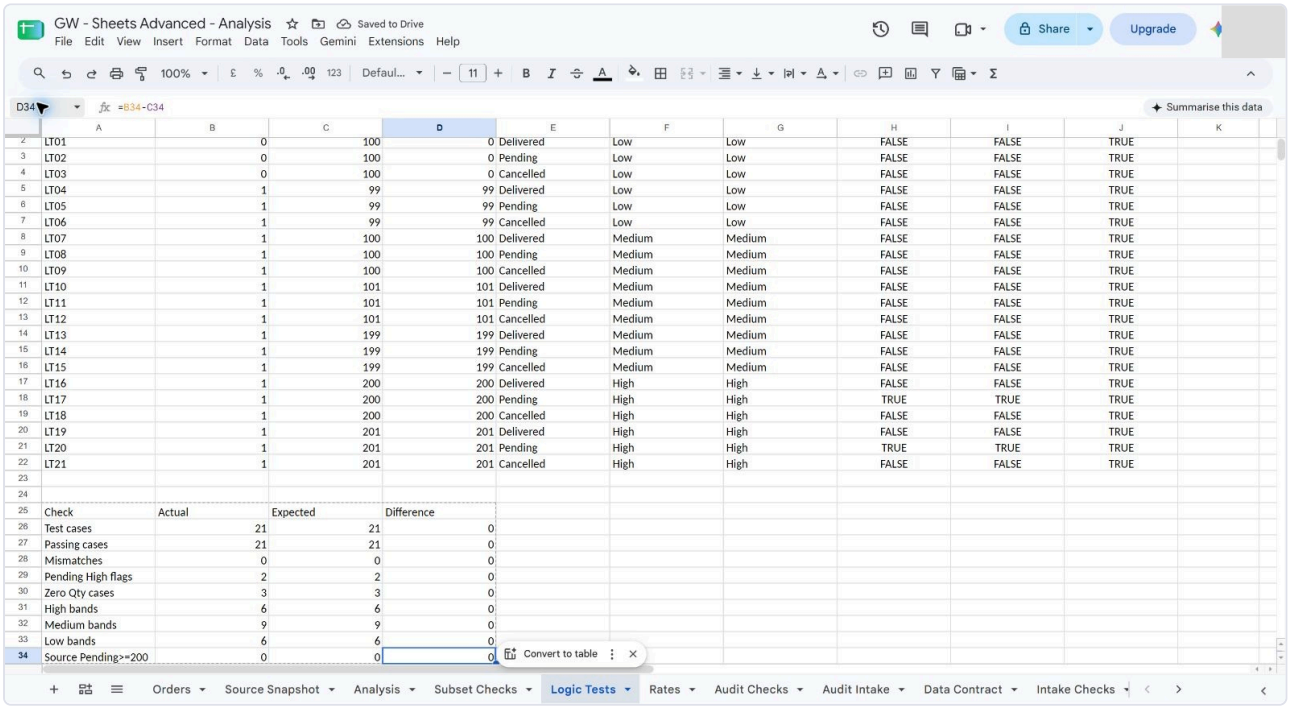
TestID	Quantity	UnitCost	Amount	Status	Expected band	Actual band	Expected flag	Actual flag	Pass
LT01	0	100	0	Delivered	Low		FALSE		
LT02	0	100	0	Pending	Low		FALSE		
LT03	0	100	0	Cancelled	Low		FALSE		
LT04	1	99	99	Delivered	Low		FALSE		
LT05	1	99	99	Pending	Low		FALSE		
LT06	1	99	99	Cancelled	Low		FALSE		
LT07	1	100	100	Delivered	Medium		FALSE		
LT08	1	100	100	Pending	Medium		FALSE		
LT09	1	100	100	Cancelled	Medium		FALSE		
LT10	1	101	101	Delivered	Medium		FALSE		
LT11	1	101	101	Pending	Medium		FALSE		
LT12	1	101	101	Cancelled	Medium		FALSE		
LT13	1	199	199	Delivered	Medium		FALSE		
LT14	1	199	199	Pending	Medium		FALSE		
LT15	1	199	199	Cancelled	Medium		FALSE		
LT16	1	200	200	Delivered	High		FALSE		
LT17	1	200	200	Pending	High		TRUE		
LT18	1	200	200	Cancelled	High		FALSE		
LT19	1	201	201	Delivered	High		FALSE		
LT20	1	201	201	Pending	High		TRUE		
LT21	1	201	201	Cancelled	High		FALSE		

8. Enter **G2 =IF(D2<100,"Low",IF(D2<200,"Medium","High"))**, **I2 =AND(E2="Pending",D2>=200)** and **J2 =AND(F2=G2,H2=I2)**. Fill each actual-formula column through row22. Compare expected and actual results; inspect the boundary cases rather than trusting only a total pass count.



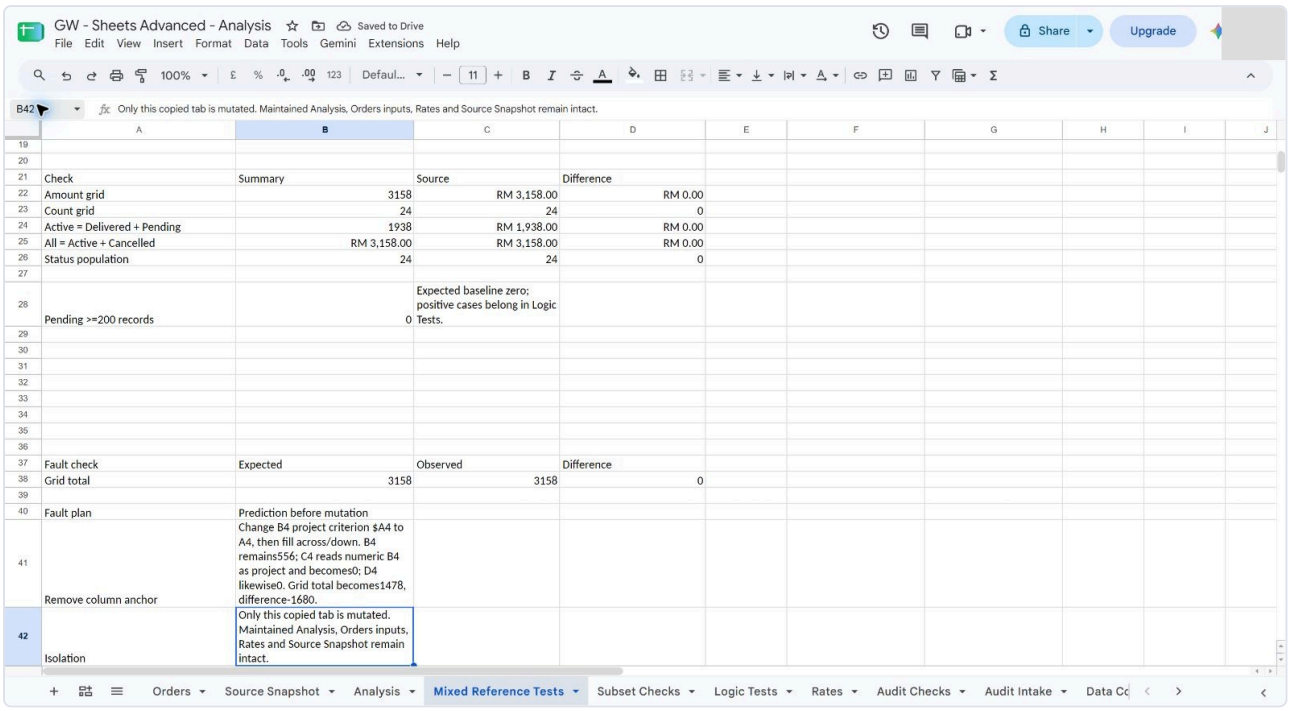
TestID	Quantity	UnitCost	Amount	Status	Expected band	Actual band	Expected flag	Actual flag	Pass
LT01	0	100	0	Delivered	Low	Low	FALSE	FALSE	TRUE
LT02	0	100	0	Pending	Low	Low	FALSE	FALSE	TRUE
LT03	0	100	0	Cancelled	Low	Low	FALSE	FALSE	TRUE
LT04	1	99	99	Delivered	Low	Low	FALSE	FALSE	TRUE
LT05	1	99	99	Pending	Low	Low	FALSE	FALSE	TRUE
LT06	1	99	99	Cancelled	Low	Low	FALSE	FALSE	TRUE
LT07	1	100	100	Delivered	Medium	Medium	FALSE	FALSE	TRUE
LT08	1	100	100	Pending	Medium	Medium	FALSE	FALSE	TRUE
LT09	1	100	100	Cancelled	Medium	Medium	FALSE	FALSE	TRUE
LT10	1	101	101	Delivered	Medium	Medium	FALSE	FALSE	TRUE
LT11	1	101	101	Pending	Medium	Medium	FALSE	FALSE	TRUE
LT12	1	101	101	Cancelled	Medium	Medium	FALSE	FALSE	TRUE
LT13	1	199	199	Delivered	Medium	Medium	FALSE	FALSE	TRUE
LT14	1	199	199	Pending	Medium	Medium	FALSE	FALSE	TRUE
LT15	1	199	199	Cancelled	Medium	Medium	FALSE	FALSE	TRUE
LT16	1	200	200	Delivered	High	High	FALSE	FALSE	TRUE
LT17	1	200	200	Pending	High	High	TRUE	TRUE	TRUE
LT18	1	200	200	Cancelled	High	High	FALSE	FALSE	TRUE
LT19	1	201	201	Delivered	High	High	FALSE	FALSE	TRUE
LT20	1	201	201	Pending	High	High	TRUE	TRUE	TRUE
LT21	1	201	201	Cancelled	High	High	FALSE	FALSE	TRUE

9. Reconcile the matrix controls in **A25:D34**: **21 cases /21 passes /0 mismatches**, two Pending High flags, three zero-quantity cases, **6 High /9 Medium /6 Low**. The two positive flag cases are synthetic Pending amounts200 and201; the maintained source still has0 qualifying records. No baseline row is changed to force a positive result.



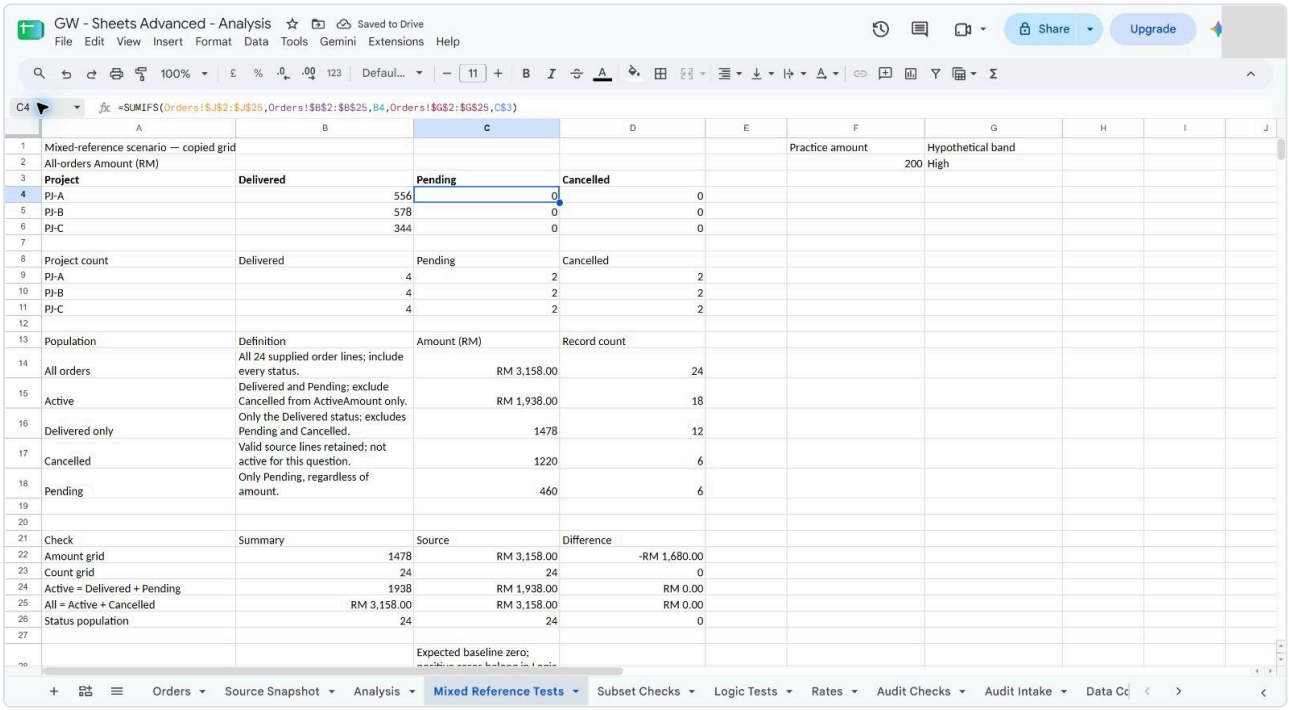
Check	Actual	Expected	Difference
Test cases	21	21	0
Passing cases	21	21	0
Mismatches	0	0	0
Pending High flags	2	2	0
Zero Qty cases	3	3	0
High bands	6	6	0
Medium bands	9	9	0
Low bands	6	6	0
Source Pending>=200	0	0	0

10. Duplicate the **Analysis** tab as **Mixed Reference Tests**, a named scenario copy. Record the fault prediction before editing: removing the column anchor from **$A4** will leave B4 at556, make C4 use numeric B4 as its project criterion and return0, and reduce the copied grid total to **1478**, a **-1680** discrepancy from3158. Add the copied-grid control **A37:D38** and prediction notes outside the working grid.



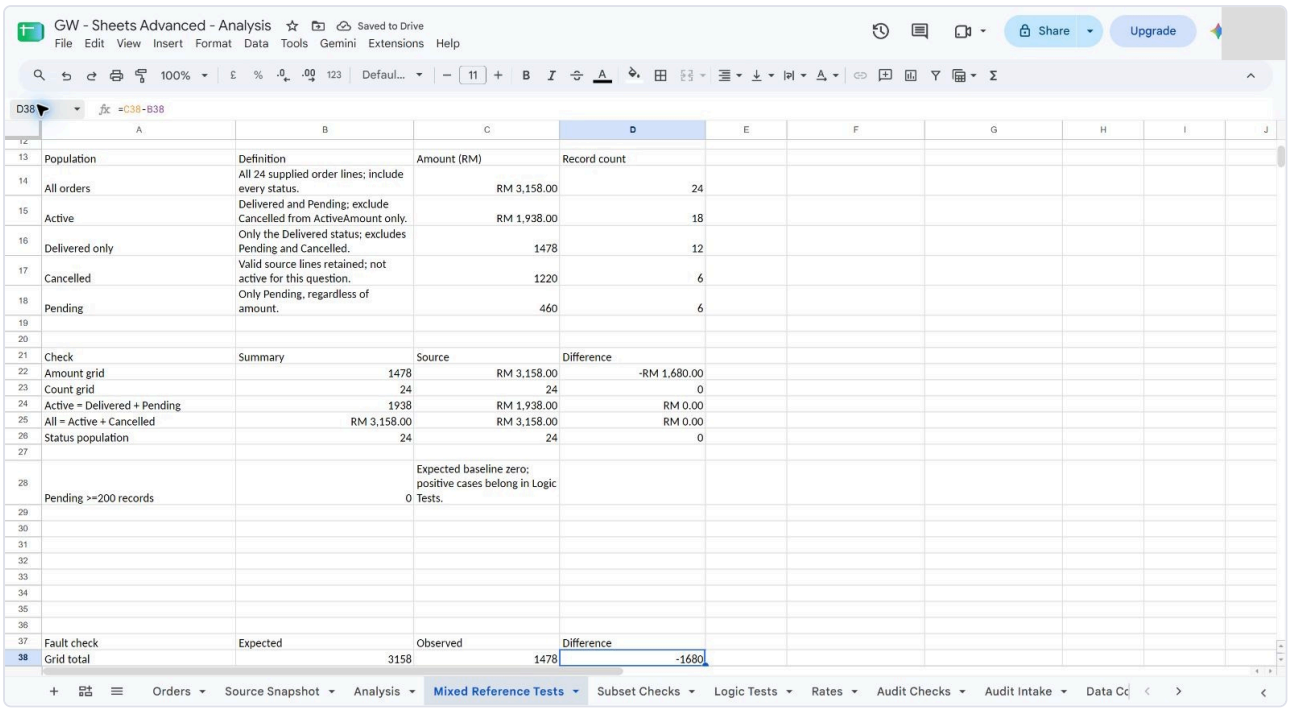
	A	B	C	D
21	Check	Summary	Source	Difference
22	Amount grid	3158	RM 3,158.00	RM 0.00
23	Count grid	24	24	0
24	Active = Delivered + Pending	1938	RM 1,938.00	RM 0.00
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00
26	Status population	24	24	0
27				
28	Pending >=200 records	0	Expected baseline zero; positive cases belong in Logic Tests.	
29				
30				
31				
32				
33				
34				
35				
36				
37	Fault check	Expected	Observed	Difference
38	Grid total	3158	3158	0
39				
40	Fault plan	Prediction before mutation		
41	Remove column anchor	Change B4 project criterion $A4 to A4, then fill across/down. B4 remains556; C4 reads numeric B4 as project and becomes0; D4 likewise0. Grid total becomes1478, difference-1680.		
42	Isolation	Only this copied tab is mutated. Maintained Analysis, Orders inputs, Rates and Source Snapshot remain intact.		

11. On the copied tab only, change B4's project criterion from **$A4** to **A4**. Fill right and down again. Inspect **C4**: the project criterion has shifted to **B4**, a numeric amount rather than PJ-A. The first incorrect intersection is **0**. Do not damage the maintained Analysis grid.



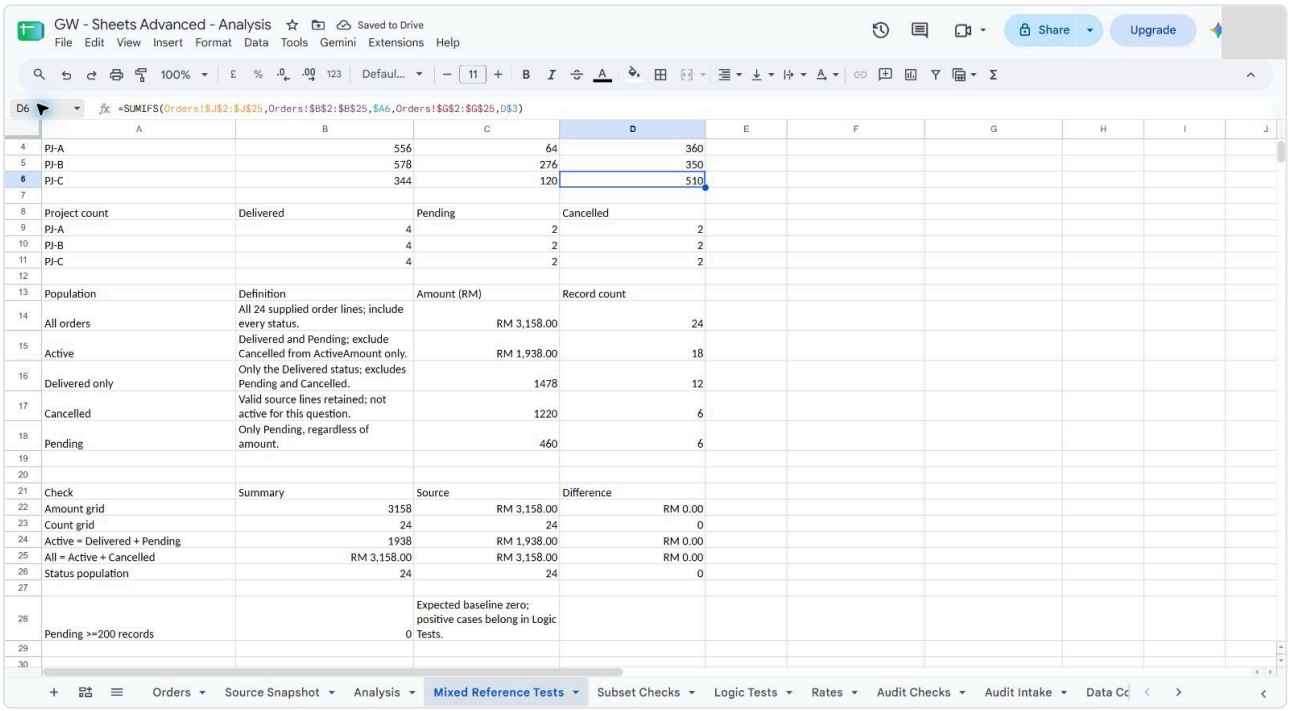
	A	B	C	D	E	F	G
1	Mixed-reference scenario — copied grid					Practice amount	Hypothetical band
2	All-orders Amount (RM)					200	High
3	**Project**	**Delivered**	**Pending**	**Cancelled**			
4	PJ-A	556	0	0			
5	PJ-B	578	0	0			
6	PJ-C	344	0	0			
7							
8	Project count	Delivered	Pending	Cancelled			
9	PJ-A	4	2	2			
10	PJ-B	4	2	2			
11	PJ-C	4	2	2			
12							
13	Population	Definition	Amount (RM)	Record count			
14	All orders	All 24 supplied order lines; include every status.	RM 3,158.00	24			
15	Active	Delivered and Pending; exclude Cancelled from ActiveAmount only.	RM 1,938.00	18			
16	Delivered only	Only the Delivered status; excludes Pending and Cancelled.	1478	12			
17	Cancelled	Valid source lines retained; not active for this question.	1220	6			
18	Pending	Only Pending, regardless of amount.	460	6			
19							
20							
21	Check	Summary	Source	Difference			
22	Amount grid	1478	RM 3,158.00	-RM 1,680.00			
23	Count grid	24	24	0			
24	Active = Delivered + Pending	1938	RM 1,938.00	RM 0.00			
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00			
26	Status population	24	24	0			
27							
28			Expected baseline zero;				

12. Inspect the copied-grid total control: expected3158, observed **1478**, difference **-1680**. The actual fault state was captured and read back before repair; a correct first cell alone did not prove that copying preserved the logic.



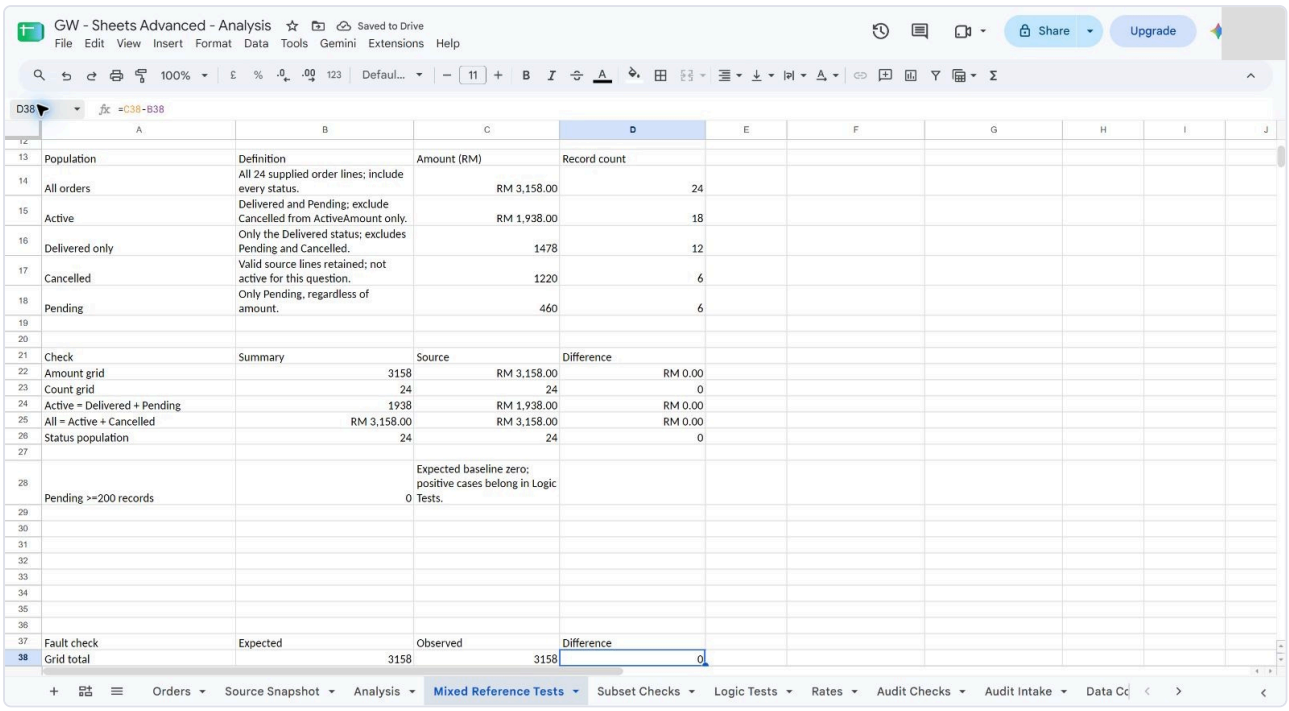
	A	B	C	D
13	Population	Definition	Amount (RM)	Record count
14	All orders	All 24 supplied order lines; include every status.	RM 3,158.00	24
15	Active	Delivered and Pending; exclude Cancelled from ActiveAmount only.	RM 1,938.00	18
16	Delivered only	Only the Delivered status; excludes Pending and Cancelled.	1478	12
17	Cancelled	Valid source lines retained; not active for this question.	1220	6
18	Pending	Only Pending, regardless of amount.	460	6
19				
20				
21	Check	Summary	Source	Difference
22	Amount grid	1478	RM 3,158.00	-RM 1,680.00
23	Count grid	24	24	0
24	Active = Delivered + Pending	1938	RM 1,938.00	RM 0.00
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00
26	Status population	24	24	0
27				
28	Pending >=200 records	0	Expected baseline zero; positive cases belong in Logic Tests.	
29				
30				
31				
32				
33				
34				
35				
36				
37	Fault check	Expected	Observed	Difference
38	Grid total	3158	1478	-1680

13. Restore **$A4** in copied B4 and refill right/down. Inspect the repaired D6 using **$A6 / D$3**, value510.



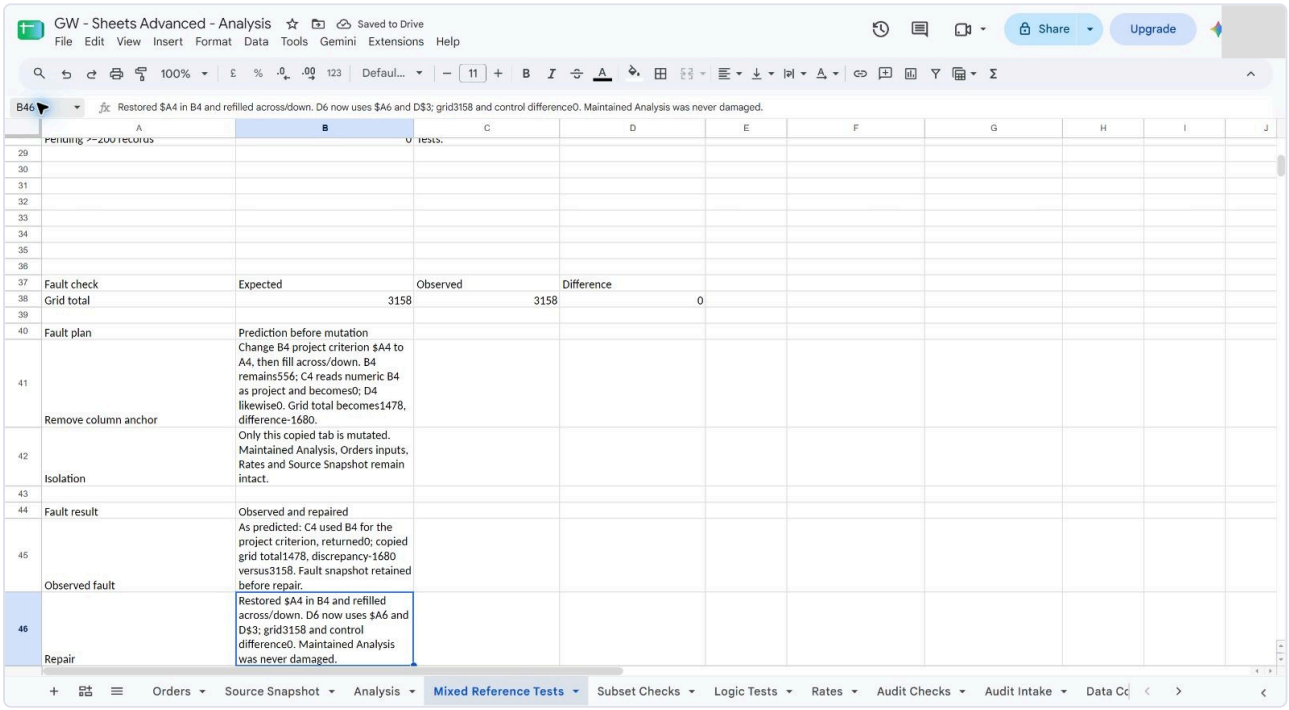
	A	B	C	D
4	PJ-A	556	64	360
5	PJ-B	578	276	350
6	PJ-C	344	120	510
7				
8	Project count	Delivered	Pending	Cancelled
9	PJ-A	4	2	2
10	PJ-B	4	2	2
11	PJ-C	4	2	2
12				
13	Population	Definition	Amount (RM)	Record count
14	All orders	All 24 supplied order lines; include every status.	RM 3,158.00	24
15	Active	Delivered and Pending; exclude Cancelled from ActiveAmount only.	RM 1,938.00	18
16	Delivered only	Only the Delivered status; excludes Pending and Cancelled.	1478	12
17	Cancelled	Valid source lines retained; not active for this question.	1220	6
18	Pending	Only Pending, regardless of amount.	460	6
19				
20				
21	Check	Summary	Source	Difference
22	Amount grid	3158	RM 3,158.00	RM 0.00
23	Count grid	24	24	0
24	Active = Delivered + Pending	1938	RM 1,938.00	RM 0.00
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00
26	Status population	24	24	0
27				
28	Pending >=200 records	0	Expected baseline zero; positive cases belong in Logic Tests.	
29				
30				

14. Recheck the copied total: **3158**, difference **0**. Keep the scenario clearly named as a repaired copy.



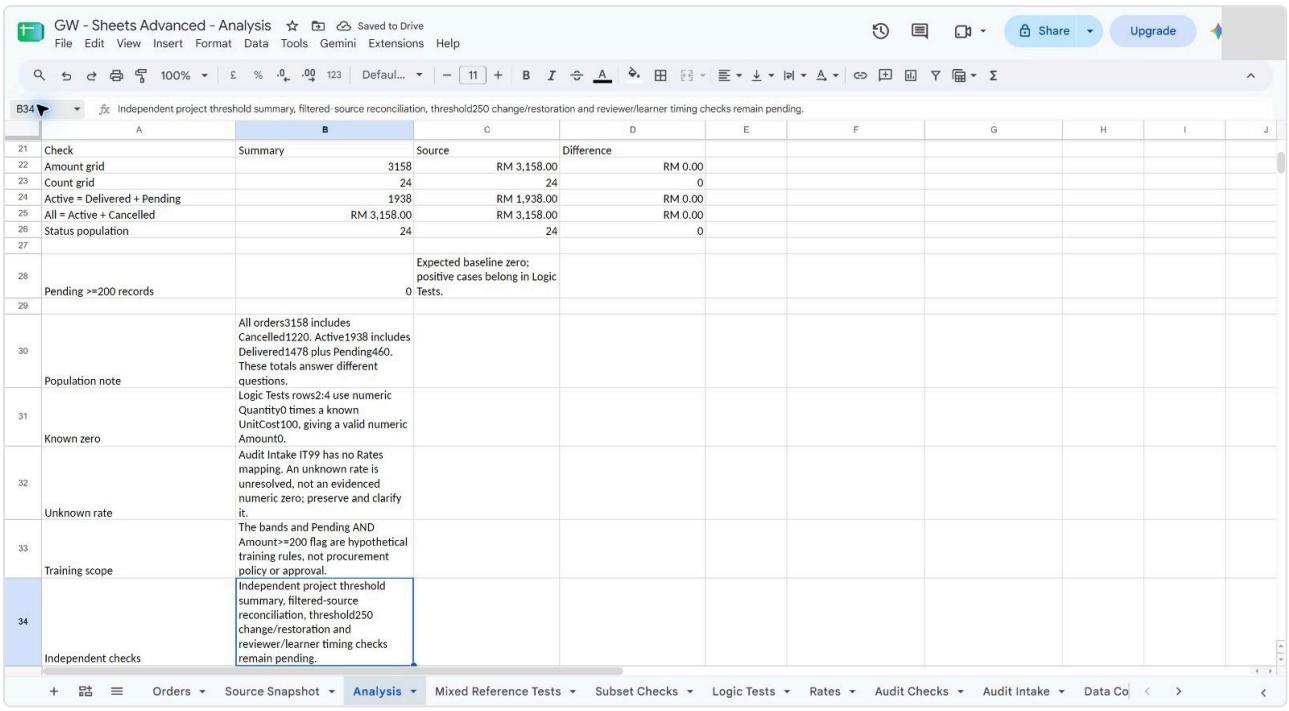
	A	B	C	D
13	Population	Definition	Amount (RM)	Record count
14	All orders	All 24 supplied order lines; include every status.	RM 3,158.00	24
15	Active	Delivered and Pending; exclude Cancelled from ActiveAmount only.	RM 1,938.00	18
16	Delivered only	Only the Delivered status; excludes Pending and Cancelled.	1478	12
17	Cancelled	Valid source lines retained; not active for this question.	1220	6
18	Pending	Only Pending, regardless of amount.	460	6
19				
20				
21	Check	Summary	Source	Difference
22	Amount grid	3158	RM 3,158.00	RM 0.00
23	Count grid	24	24	0
24	Active = Delivered + Pending	1938	RM 1,938.00	RM 0.00
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00
26	Status population	24	24	0
27				
28	Pending >=200 records	0	Expected baseline zero; positive cases belong in Logic Tests.	
29				
30				
31				
32				
33				
34				
35				
36				
37	Fault check	Expected	Observed	Difference
38	Grid total	3158	3158	0

15. Record the predicted error, observed criterion/value and repair in **A40:B46**. Label the tab's title cell **Mixed-reference scenario — repaired copy**. The maintained Analysis never required repair.



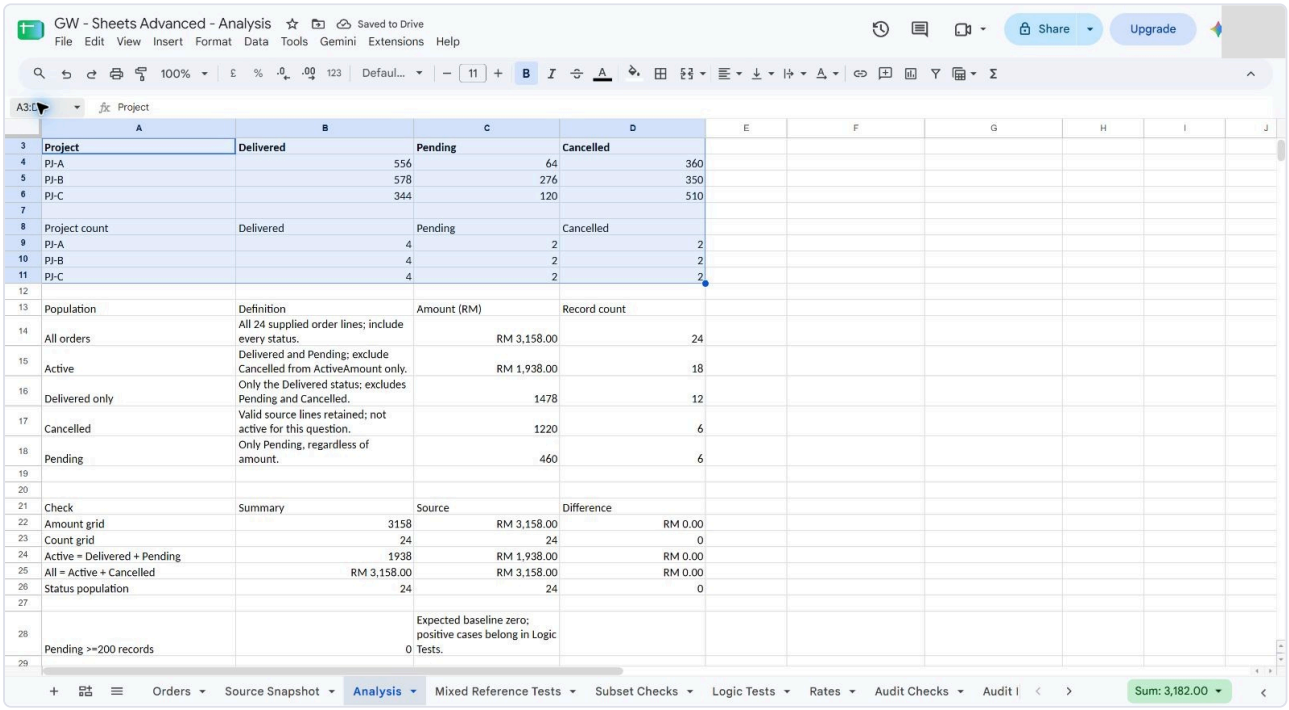
	A	B	C	D
37	Fault check	Expected	Observed	Difference
38	Grid total	3158	3158	0
39				
40	Fault plan	Prediction before mutation		
41	Remove column anchor	Change B4 project criterion $A4 to A4, then fill across/down. B4 remains556; C4 reads numeric B4 as project and becomes0; D4 likewise0. Grid total becomes1478, difference-1680.		
42	Isolation	Only this copied tab is mutated. Maintained Analysis, Orders inputs, Rates and Source Snapshot remain intact.		
43				
44	Fault result	Observed and repaired		
45	Observed fault	As predicted: C4 used B4 for the project criterion, returned0; copied grid total1478, discrepancy-1680 versus3158. Fault snapshot retained before repair.		
46	Repair	Restored $A4 in B4 and refilled across/down. D6 now uses $A6 and D$3; grid3158 and control difference0. Maintained Analysis was never damaged.		

16. On maintained Analysis, record why a **known numeric zero** differs from an unresolved **IT99** rate mapping, and why the all-orders, active and delivered-only totals answer different questions. Keep the training bands separate from source policy. Label the independent project threshold summary, filtered-source reconciliation and250/restoration challenge as pending.



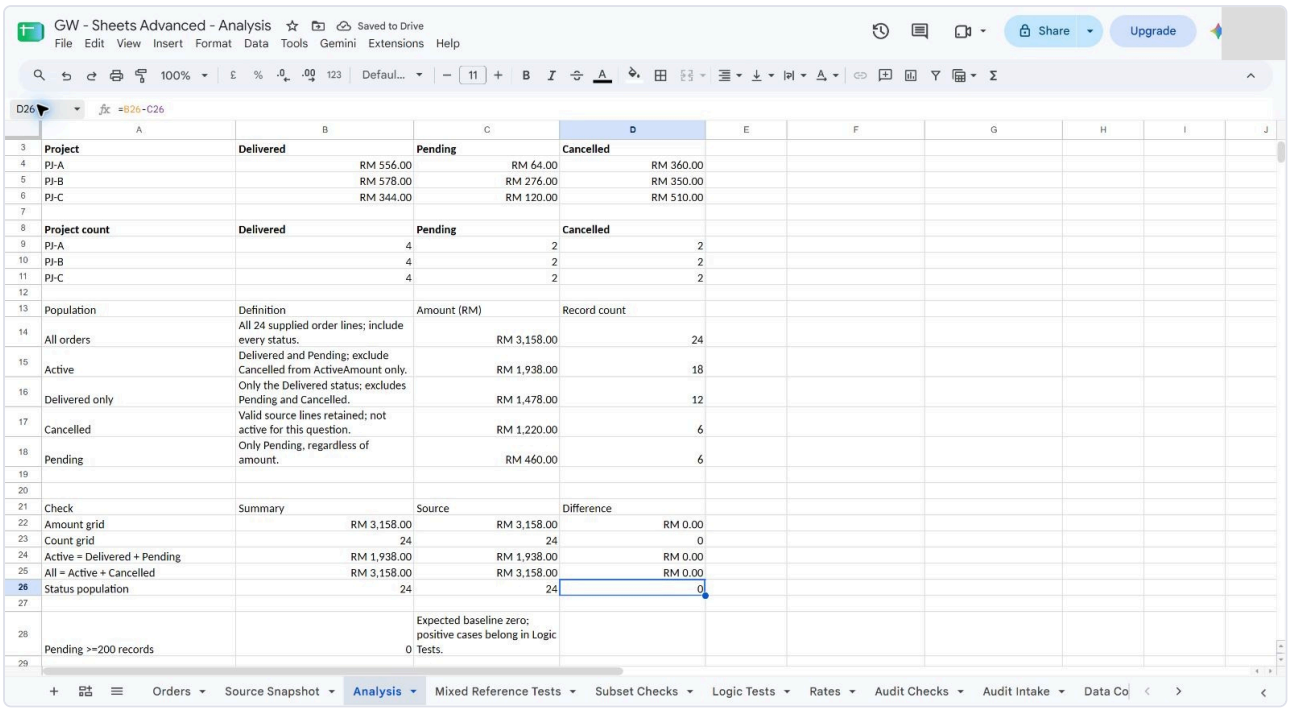
	A	B	C	D
21	Check	Summary	Source	Difference
22	Amount grid	3158	RM 3,158.00	RM 0.00
23	Count grid	24	24	0
24	Active = Delivered + Pending	1938	RM 1,938.00	RM 0.00
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00
26	Status population	24	24	0
27				
28	Pending >=200 records	0	Expected baseline zero; positive cases belong in Logic Tests.	
29				
30	Population note	All orders3158 includes Cancelled1220. Active1938 includes Delivered1478 plus Pending460. These totals answer different questions.		
31	Known zero	Logic Tests rows2:4 use numeric Quantity0 times a known UnitCost100, giving a valid numeric Amount0.		
32	Unknown rate	Audit Intake IT99 has no Rates mapping. An unknown rate is unresolved, not an evidenced numeric zero; preserve and clarify it.		
33	Training scope	The bands and Pending AND Amount>=200 flag are hypothetical training rules, not procurement policy or approval.		
34	Independent checks	Independent project threshold summary, filtered-source reconciliation, threshold250 change/restoration and reviewer/learner timing checks remain pending.		

17. Reinspect the maintained amount and count grids. The nine intersections still match the source, with amount total3158 and count24.

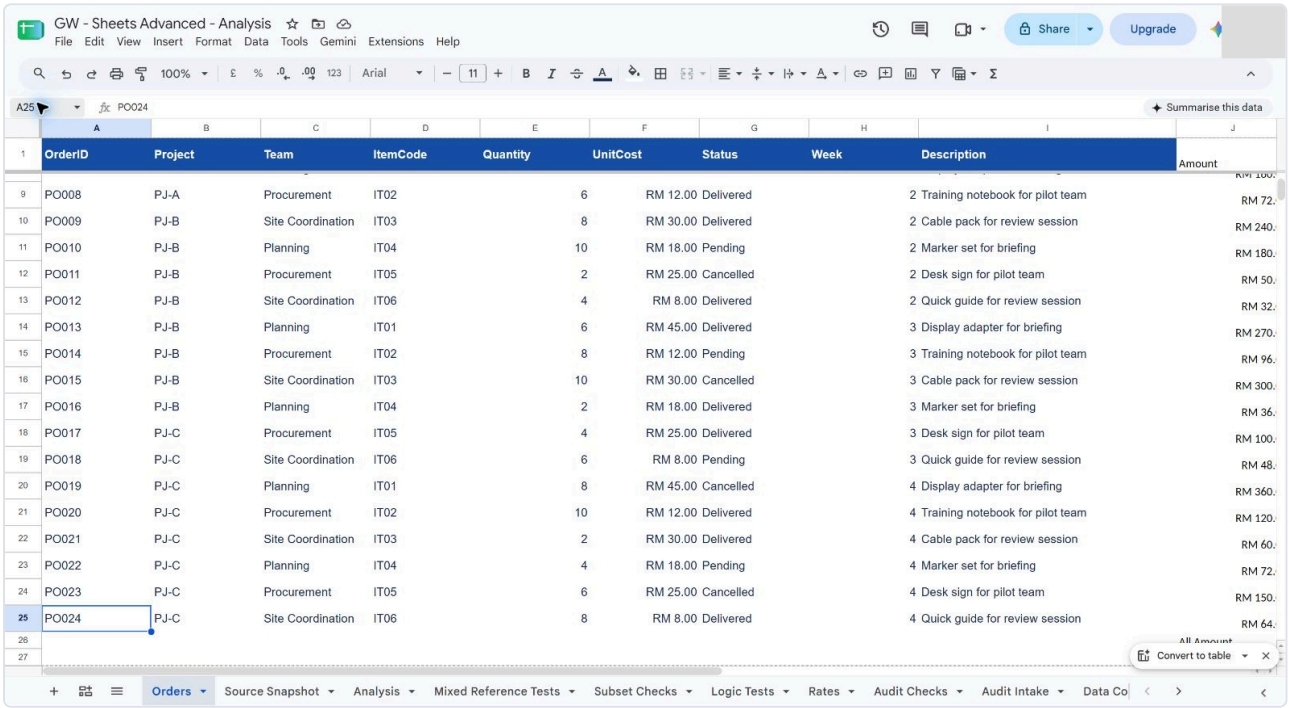


	A	B	C	D
3	**Project**	**Delivered**	**Pending**	**Cancelled**
4	PJ-A	556	64	360
5	PJ-B	578	276	350
6	PJ-C	344	120	510
7				
8	Project count	Delivered	Pending	Cancelled
9	PJ-A	4	2	2
10	PJ-B	4	2	2
11	PJ-C	4	2	2
12				
13	Population	Definition	Amount (RM)	Record count
14	All orders	All 24 supplied order lines; include every status.	RM 3,158.00	24
15	Active	Delivered and Pending; exclude Cancelled from ActiveAmount only.	RM 1,938.00	18
16	Delivered only	Only the Delivered status; excludes Pending and Cancelled.	1478	12
17	Cancelled	Valid source lines retained; not active for this question.	1220	6
18	Pending	Only Pending, regardless of amount.	460	6
19				
20				
21	Check	Summary	Source	Difference
22	Amount grid	3158	RM 3,158.00	RM 0.00
23	Count grid	24	24	0
24	Active = Delivered + Pending	1938	RM 1,938.00	RM 0.00
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00
26	Status population	24	24	0
27				
28	Pending >=200 records	0	Expected baseline zero; positive cases belong in Logic Tests.	
29				

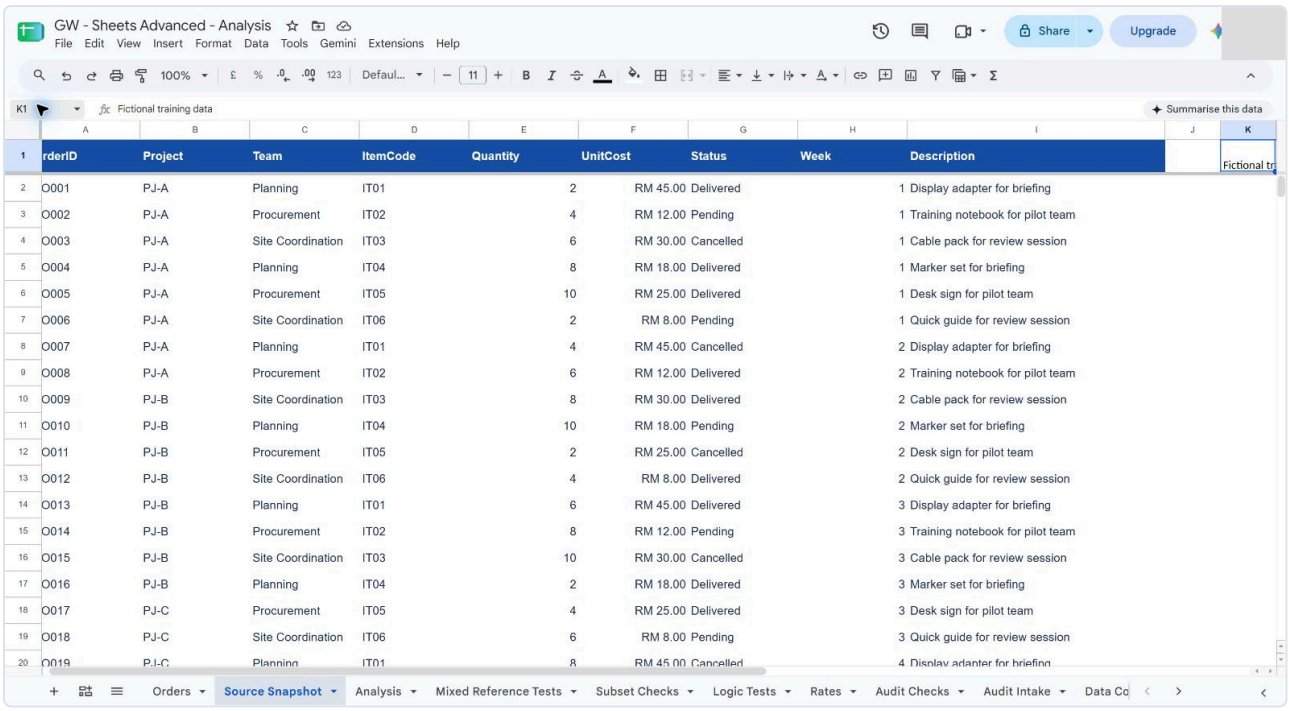
18. Finish readable amount formats and inspect all five zero-difference control rows. In the captured Analysis layout A/B/C/D are approximately240/220/180/180pixels, with wrapped definition and note cells; money formats are applied only to amount ranges, not the count grid.



19. Recheck Orders A:I through PO024. Original input values and formats, Rates, intake evidence and the training contract remain unchanged; only the new calculation columns, outside-table controls and preserved label location were added to Orders.

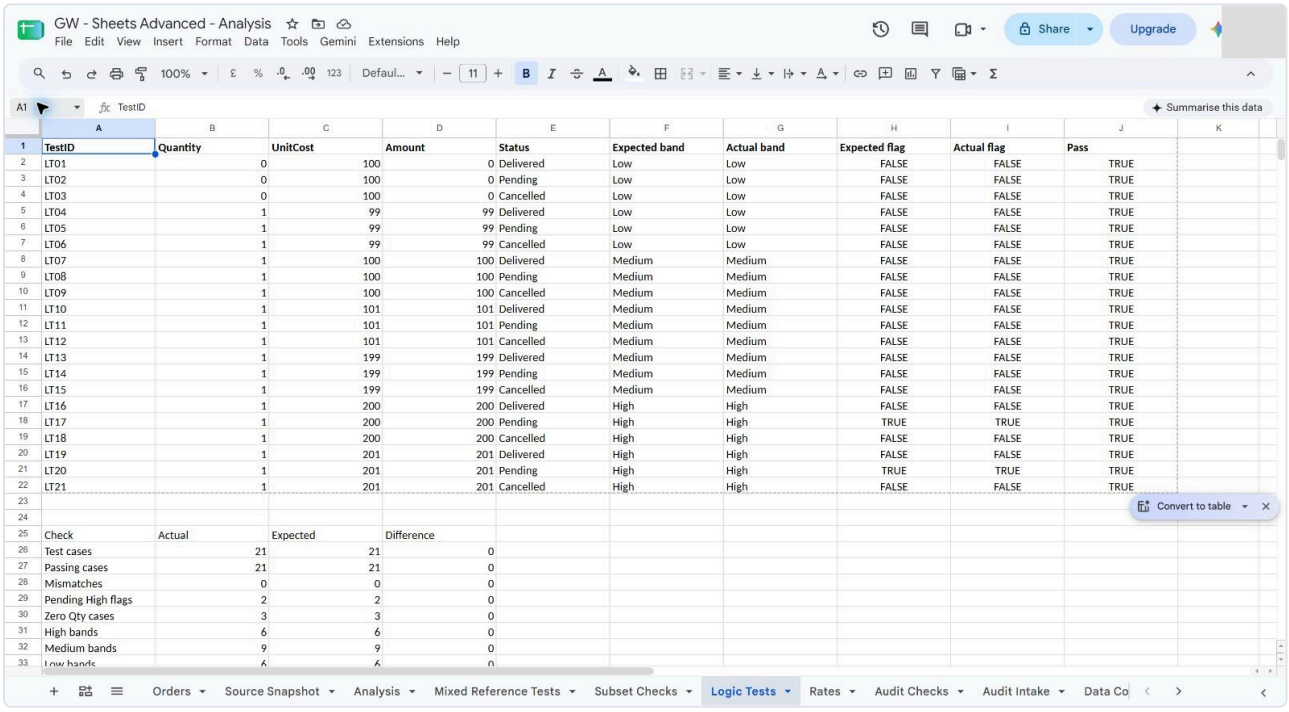


20. Inspect **Source Snapshot**. It retains the original source values and original K1 label, without the new calculation columns. It remains a comparison copy, not the maintained calculation source.



	rderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description
2	O001	PJ-A	Planning	IT01	2	RM 45.00	Delivered	1	Display adapter for briefing
3	O002	PJ-A	Procurement	IT02	4	RM 12.00	Pending	1	Training notebook for pilot team
4	O003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancelled	1	Cable pack for review session
5	O004	PJ-A	Planning	IT04	8	RM 18.00	Delivered	1	Marker set for briefing
6	O005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered	1	Desk sign for pilot team
7	O006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session
8	O007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing
9	O008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team
10	O009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session
11	O010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing
12	O011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team
13	O012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session
14	O013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing
15	O014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team
16	O015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session
17	O016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing
18	O017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team
19	O018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session
20	O019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing

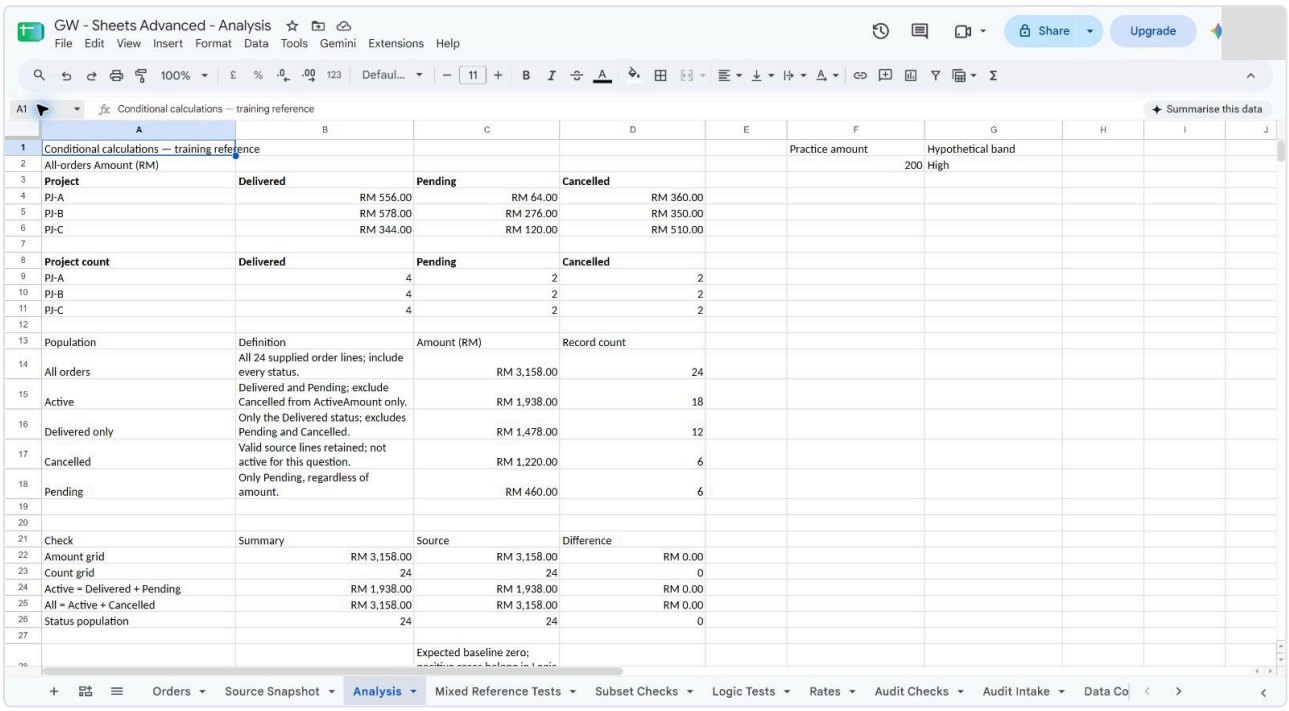
21. Leave all21 Logic Tests passing. Preserve the synthetic cases separately from Orders and the repaired reference scenario separately from Analysis.



	TestID	Quantity	UnitCost	Amount	Status	Expected band	Actual band	Expected flag	Actual flag	Pass
2	LT01	0	100	0	Delivered	Low	Low	FALSE	FALSE	TRUE
3	LT02	0	100	0	Pending	Low	Low	FALSE	FALSE	TRUE
4	LT03	0	100	0	Cancelled	Low	Low	FALSE	FALSE	TRUE
5	LT04	1	99	99	Delivered	Low	Low	FALSE	FALSE	TRUE
6	LT05	1	99	99	Pending	Low	Low	FALSE	FALSE	TRUE
7	LT06	1	99	99	Cancelled	Low	Low	FALSE	FALSE	TRUE
8	LT07	1	100	100	Delivered	Medium	Medium	FALSE	FALSE	TRUE
9	LT08	1	100	100	Pending	Medium	Medium	FALSE	FALSE	TRUE
10	LT09	1	100	100	Cancelled	Medium	Medium	FALSE	FALSE	TRUE
11	LT10	1	101	101	Delivered	Medium	Medium	FALSE	FALSE	TRUE
12	LT11	1	101	101	Pending	Medium	Medium	FALSE	FALSE	TRUE
13	LT12	1	101	101	Cancelled	Medium	Medium	FALSE	FALSE	TRUE
14	LT13	1	199	199	Delivered	Medium	Medium	FALSE	FALSE	TRUE
15	LT14	1	199	199	Pending	Medium	Medium	FALSE	FALSE	TRUE
16	LT15	1	199	199	Cancelled	Medium	Medium	FALSE	FALSE	TRUE
17	LT16	1	200	200	Delivered	High	High	FALSE	FALSE	TRUE
18	LT17	1	200	200	Pending	High	High	TRUE	TRUE	TRUE
19	LT18	1	200	200	Cancelled	High	High	FALSE	FALSE	TRUE
20	LT19	1	201	201	Delivered	High	High	FALSE	FALSE	TRUE
21	LT20	1	201	201	Pending	High	High	TRUE	TRUE	TRUE
22	LT21	1	201	201	Cancelled	High	High	FALSE	FALSE	TRUE

	Check	Actual	Expected	Difference
26	Test cases	21	21	0
27	Passing cases	21	21	0
28	Mismatches	0	0	0
29	Pending High flags	2	2	0
30	Zero Qty cases	3	3	0
31	High bands	6	6	0
32	Medium bands	9	9	0
33	Low bands	6	6	0

22. Leave the saved maintained **Analysis** ready for the next lab. Source inputs are preserved; hypothetical tests and historical audit notes are clearly scoped. Independent threshold work, reviewer checks and learner timing remain unfinished.



	A	B	C	D	E	F	G	H	I	J
1	Conditional calculations — training reference					Practice amount	Hypothetical band			
2	All-orders Amount (RM)					200	High			
3	**Project**	**Delivered**	**Pending**	**Cancelled**						
4	PJ-A	RM 556.00	RM 64.00	RM 360.00						
5	PJ-B	RM 578.00	RM 276.00	RM 350.00						
6	PJ-C	RM 344.00	RM 120.00	RM 510.00						
7										
8	**Project count**	**Delivered**	**Pending**	**Cancelled**						
9	PJ-A	4	2	2						
10	PJ-B	4	2	2						
11	PJ-C	4	2	2						
12										
13	Population	Definition	Amount (RM)	Record count						
14	All orders	All 24 supplied order lines; include every status.	RM 3,158.00	24						
15	Active	Delivered and Pending; exclude Cancelled from ActiveAmount only.	RM 1,938.00	18						
16	Delivered only	Only the Delivered status; excludes Pending and Cancelled.	RM 1,478.00	12						
17	Cancelled	Valid source lines retained; not active for this question.	RM 1,220.00	6						
18	Pending	Only Pending, regardless of amount.	RM 460.00	6						
19										
20										
21	Check	Summary	Source	Difference						
22	Amount grid	RM 3,158.00	RM 3,158.00	RM 0.00						
23	Count grid	24	24	0						
24	Active = Delivered + Pending	RM 1,938.00	RM 1,938.00	RM 0.00						
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00						
26	Status population	24	24	0						
27										
28			Expected baseline zero;							

> **Validation boundary:** guided2.1–2.3 and author casework2.4 are verified through actual learner UI and bounded read-only native-cell evidence. The independent project/threshold summary, filtered-source reconciliation,250/restoration test and human checks were not performed. Planned75-minute classroom timing remains unmeasured.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Build a second summary answering count and amount for Pending orders of at least 200 by project. Keep threshold logic explicit.
2. Use a filtered source list as an independent reconciliation. A zero-match baseline is a valid result; create a separate synthetic qualifying case and a non-qualifying boundary case to test the rule.
3. Change the threshold to 250 in a labelled assumption cell, repair references so the grid follows it, then restore 200 and verify restoration.

Evidence to Hand Over

- Reconciled count/amount grids for clearly defined populations.
- Expected-versus-actual boundary matrix and repaired mixed-reference defect.
- Independent threshold summary with supporting OrderIDs and a restored assumption.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check

Correct totals conceal the wrong population	Reconcile counts and contributing IDs as well as the monetary total.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Visible row calculations and explicit criteria make advanced logic checkable.

Check yourself: Which references in the summary grid must stay fixed when copied across and down?

Zero-Match Check

The supplied baseline has no Pending record with Amount at least 200. Record that result explicitly. Use a separately labelled synthetic case to prove the positive branch; do not edit baseline records just to obtain a highlight.

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
Amount
```

```
=E2*F2
```

```
ActiveAmount
```

```
=IF(G2="Cancelled",0,J2)
```

```
=SUMIFS(Orders!$J$2:$J$25,Orders!$B$2:$B$25,$A4,Orders!$G$2:$G$25,B$3)
```

```
=IF(F2<100,"Low",IF(F2<200,"Medium","High"))
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Logic matrix must cover below/on/above thresholds, zero quantity, Delivered, Pending and
Cancelled. Practice bands: Low <100; Medium 100 to <200; High ⩾200. Exception: Pending AND
Amount ⩾200. These are training rules, not procurement policy.
```

Captured Conditional-Calculation Reference

Preserve the supplied K1 fictional-data label in M1 before using K for ActiveAmount. Original A:I inputs, Rates, Source Snapshot and intake audit remain intact. J2=E2*F2 and K2=IF(G2="Cancelled",0,J2), filled to25; source totals3158 and1938. Spot checks J2/J13/J25 are90/32/64.

Amount grid B4:D6:556/64/360;578/276/350;344/120/510. D6 uses $A6 and D$3. Count grid B9:D11:4/2/2 for each project. B9 seed:

```
=COUNTIFS(Orders!$B$2:$B$25,$A9,Orders!$G$2:$G$25,B$8)
```

Population control card — Analysis A13:D18

Write each population definition before calculating the final two columns.

```
Population      Definition      Amount (RM)     Record count
All orders      All 24 supplied order lines; include every status.      =SUM(Orders!J2:J25)
=COUNTA(Orders!A2:A25)
Active  Delivered and Pending; exclude Cancelled from ActiveAmount only.
=SUM(Orders!K2:K25)     =COUNTIF(Orders!G2:G25,"<>Cancelled")
Delivered only  Only the Delivered status; excludes Pending and Cancelled.
=SUMIF(Orders!G2:G25,"Delivered",Orders!J2:J25) =COUNTIF(Orders!G2:G25,"Delivered")
Cancelled       Valid source lines retained; not active for this question.
=SUMIF(Orders!G2:G25,"Cancelled",Orders!J2:J25) =COUNTIF(Orders!G2:G25,"Cancelled")
Pending Only Pending, regardless of amount.     =SUMIF(Orders!G2:G25,"Pending",Orders!J2:J25)
=COUNTIF(Orders!G2:G25,"Pending")
```

Reconciliation card — Analysis A21:D26

```
Check   Summary Source  Difference
Amount grid     =SUM(B4:D6)     =C14    =B22-C22
Count grid      =SUM(B9:D11)    =D14    =B23-C23
Active = Delivered + Pending    =C16+C18        =C15    =B24-C24
All = Active + Cancelled        =C15+C17        =C14    =B25-C25
Status population       =SUM(D16:D18)   =D14    =B26-C26
```

B28 baseline check **=COUNTIFS(Orders!G2:G25,"Pending",Orders!J2:J25,">=200")** gives0. All3158/24, Active1938/18, Delivered1478/12, Cancelled1220/6, Pending460/6. Every reconciliation difference0.

Manual subset inputs — Subset Checks A1:H7

Record the six manually checked products before G/H formula comparisons. Inputs match the named source OrderIDs; all amounts use training RM.

```
Project Status  OrderID Quantity        UnitCost        Manual Amount   Formula Amount
Difference
PJ-A    Delivered       PO001   2       45      90
PJ-A    Delivered       PO004   8       18      144
PJ-A    Delivered       PO005   10      25      250
PJ-A    Delivered       PO008   6       12      72
PJ-B    Pending PO010   10      18      180
PJ-B    Pending PO014   8       12      96
```

G2 **=D2*E2**, H2 **=G2-F2**, fill to7. The two subsets total556 and276; their counts4 and2 also match the maintained grids.

Logic matrix input/expectations — Logic Tests A1:J22

Actual columns G/I/J are deliberately empty on this input card. D calculates Quantity times a known UnitCost. Zero Quantity is numeric0; it is not an unresolved rate lookup. Expected flags are Boolean TRUE/FALSE.

```
TestID  Quantity        UnitCost        Amount  Status  Expected band   Actual band
Expected flag   Actual flag     Pass
LT01    0       100     =B2*C2  Delivered       Low             FALSE
LT02    0       100     =B3*C3  Pending Low             FALSE
LT03    0       100     =B4*C4  Cancelled       Low             FALSE
LT04    1       99      =B5*C5  Delivered       Low             FALSE
LT05    1       99      =B6*C6  Pending Low             FALSE
```

```
LT06    1       99      =B7*C7  Cancelled       Low             FALSE
LT07    1       100     =B8*C8  Delivered       Medium          FALSE
LT08    1       100     =B9*C9  Pending Medium          FALSE
LT09    1       100     =B10*C10        Cancelled       Medium          FALSE
LT10    1       101     =B11*C11        Delivered       Medium          FALSE
LT11    1       101     =B12*C12        Pending Medium          FALSE
LT12    1       101     =B13*C13        Cancelled       Medium          FALSE
LT13    1       199     =B14*C14        Delivered       Medium          FALSE
LT14    1       199     =B15*C15        Pending Medium          FALSE
LT15    1       199     =B16*C16        Cancelled       Medium          FALSE
LT16    1       200     =B17*C17        Delivered       High            FALSE
LT17    1       200     =B18*C18        Pending High            TRUE
LT18    1       200     =B19*C19        Cancelled       High            FALSE
LT19    1       201     =B20*C20        Delivered       High            FALSE
LT20    1       201     =B21*C21        Pending High            TRUE
LT21    1       201     =B22*C22        Cancelled       High            FALSE
```

G2 **=IF(D2<100,"Low",IF(D2<200,"Medium","High"))**; I2 **=AND(E2="Pending",D2>=200)**; J2 **=AND(F2=G2,H2=I2)**. Fill each to22. Expected21passes/0mismatches,2positiveflags,3zero-quantitycases,6High/9Medium/6Low. Synthetic Pending200/201 exercise the positive flag branch; baseline remains0.

Repaired mixed-reference scenario

Duplicate Analysis as Mixed Reference Tests. On that copy only, remove the column anchor from B4's $A4 criterion, then refill across/down. C4 incorrectly uses B4 as project criterion and gives0; total1478, discrepancy-1680 from3158. Restore $A4 and refill: grid3158, difference0; D6 uses $A6 and D$3. The actual fault snapshot was retained before repair. The maintained Analysis grid was never damaged.

Independent project threshold summary, filtered-source reconciliation, threshold250 change/restoration and reviewer/learner-timing checks remain pending. Practice bands and flags are fictional training rules, not procurement policy.

03

CHAPTER 3

Validate Entries and Flag Exceptions

In this chapter you will use dropdowns, custom validation and formula-based conditional formatting.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 65 minutes: 15 explanation/demonstration, 22 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 50 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Validation rules and a tested pending-order exception highlight.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Validate categorical and numeric inputs.
- Highlight exceptions using a rule.
- Test an invalid and boundary value.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use a practice copy of Orders. Restore all temporary tests before completing the chapter.

3.1 Validate New Entries

1. Select G2:G25 and add a dropdown for Delivered, Pending and Cancelled.
2. Add a validation rule to E2:E25 requiring a whole number at least 0. A zero quantity is permitted for this training dataset.
3. Inspect whether invalid values are rejected or warned about.

What should you see?

The rules reflect the stated training choices and do not silently invent a positive-only restriction.

3.2 Flag Pending High Amounts

1. Select A2:K25 and open Format, Conditional formatting.
2. Use a custom formula `=AND($G2="Pending",$J2≥200)` and a readable light fill.
3. Check the rule starts at source row 2 and applies across the intended rows.

What should you see?

Only pending rows with Amount at least 200 are highlighted; delivered or cancelled rows do not qualify.

3.3 Test the Rules

1. On one practice row, try an invalid status and a fractional quantity, then undo.
2. Test a pending row at amount 199 and 200 by temporarily changing its numeric inputs.
3. Restore the source values and inspect that the highlighting recalculates.

What should you see?

You have tested both rejection behaviour and the conditional-format threshold.

3.4 Create a Rule Test Harness and Reconcile Exceptions

1. Create **Validation Tests** with expected result and observed result before trying each input. Include whole zero, positive integer, negative, fractional and text quantities.
2. For whole nonnegative Quantity, inspect a custom rule such as `=AND(ISNUMBER(E2),E2≥0,MOD(E2,1)=0)` on a practice range. Decide and document how a blank should be handled separately.
3. Test rejection behaviour for each quantity and status case. Do not equate a coloured cell with an enforced validation rule.
4. Build an explicit exception flag column on a practice Orders copy using the same Pending/Amount logic as the conditional format.
5. Filter the flag and compare its qualifying OrderIDs with the highlighted rows. Check both the first and final source row for rule coverage.
6. On test records compare Pending at 199, 200 and 201, plus Delivered at 200 and Cancelled at 200. Explain which logical condition excludes each non-qualifying row.
7. Restore all temporary values and record the exact data-validation and conditional-format ranges used.

What should you see?

Eight-case input validation matrix and five-case exception matrix. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Append a synthetic pending record only to the test copy and diagnose whether input validation, flag formulas and conditional formatting extend to it.
2. Repair coverage and produce a test report containing at least eight input cases and five exception cases.
3. State whether blanks and pasted values were actually tested and list any behaviour still requiring confirmation in the customer account.

Evidence to Hand Over

- Eight-case input validation matrix and five-case exception matrix.
- Agreement between explicit flags, highlighted rows and filtered IDs.
- New-row coverage repair and restored source values.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The colour rule flags the wrong row	Check the apply-to start row and the relative row in the custom formula.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Data rules constrain entry; conditional formatting draws attention to defined exceptions.

Check yourself: How does a validation rule differ from a coloured warning?

Zero-Match Check

The supplied baseline has no Pending record with Amount at least 200. Record that result explicitly. Use a separately labelled synthetic case to prove the positive branch; do not edit baseline records just to obtain a highlight.

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
=AND($G2="Pending",$J2⩾200)
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Synthetic coverage record, test copy only: PO-TEST | PJ-A | Planning | IT01 | Quantity 5 |
UnitCost 45 | Pending | week 4 | Coverage test. Quantity candidates: 0, 2, -1, 1.5, text
“two”, blank and numeric text imported from a CSV.
```

04

CHAPTER 4

Combine and Clean Text

In this chapter you will use LEFT, RIGHT, MID and concatenation; check AI categorisation against explicit rules.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 60 minutes: 15 explanation/demonstration, 17 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 45 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Text extraction and labels with checked category mappings.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Extract parts of identifiers.
- Combine readable labels.
- Check AI categories against a visible mapping.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Orders and Rates. The supplied IDs follow PO plus three digits; this is a training convention.

4.1 Extract Identifier Parts

1. On a separate Text Practice tab copy OrderID and ItemCode from a few rows.
2. Use `=LEFT(A2,2)` for the prefix and `=RIGHT(A2,3)` for the numeric text.
3. Use `=MID(A2,3,3)` to extract the same three characters and compare the results. Keep leading zeros as text.

> **What should you see?**
>
> PO001 yields prefix PO and sequence text 001 rather than a newly assigned key.

4.2 Combine and Clean Text

1. Copy a space-padded team name from DirtyText.csv into a spare cell and use TRIM to clean it.
2. Use CONCATENATE or & to build a label from an order ID and project code, for example `=A2&" - "&B2` on Orders.
3. Check the underlying separate fields remain available for filtering and analysis.

What should you see?

The readable label does not replace the source columns or destroy identifier zeros.

4.3 Check Categorisation

1. Read the six-item Rates table and its Category column.
2. If Gemini is available, ask it to propose categories for the item names using only Equipment, Stationery and Print and the supplied mapping.
3. Compare every proposed category with Rates. Use the mapping as the reference; record ambiguous or unmapped inputs.

What should you see?

AI suggestions have been checked against an explicit six-item rule rather than accepted by appearance.

4.4 Build a Reversible Text Normalisation Pipeline

1. Import DirtyText.csv into **Text Audit** and preserve its raw columns. Add Normalised team, Normalised status and Decision columns.
2. Apply TRIM for spaces and use explicit case handling only where it preserves the known vocabulary. Record any non-breaking or unusual spaces that the observed function does not remove.
3. Build an identifier test table from the card. Apply LEFT, RIGHT and MID, then compare extracted values with the expected identifier pattern.
4. Flag malformed or unknown keys rather than allowing a correctly extracted substring to prove a valid mapping. Retain leading zeros as text.
5. Build a display label from OrderID and Project using CONCATENATE or & while retaining the original separate key fields.
6. Use the Rates mapping to assign the six known item categories. If AI proposes categories, compare every proposal with that mapping and retain unmapped keys as explicit exceptions.
7. Write a pipeline note showing raw → normalised → validated → category/display output. Name which transformations are reversible and which require the raw source for audit.

What should you see?

Raw/clean text audit and identifier extraction tests. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Independently process the mixed identifier intake into accepted keys and an exception list. Do not rely on string length alone for validity.
2. Demonstrate that IT99 is syntactically plausible but absent from the supplied mapping, and that a blank key is not a category.
3. Have a reviewer reconstruct one clean value's origin from your raw and transformation columns. Repair any missing trace.

Evidence to Hand Over

- Raw/clean text audit and identifier extraction tests.
- Separate syntax and mapping validation with explicit exceptions.
- Traceable category/display outputs and an independent mixed-intake result.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
Text extraction succeeds on a bad identifier	Validate the complete pattern and reference mapping separately from substring extraction.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Text functions reshape fields while source keys and mappings preserve meaning.

Check yourself: Why should an ID suffix such as 001 remain text?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
=LEFT(A2,2)
```

```
=RIGHT(A2,3)
```

```
=MID(A2,3,3)
```

```
=A2&" - "&B2
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Identifier intake: PO001; PO024; PO1; blank; IT01; IT06; IT99; “ IT03 ”. Order IDs require PO plus three digits; mapped item keys are the six supplied Rates codes. Do not treat every extracted suffix as a valid key.
```

05

CHAPTER 5

Match Records and Handle Errors

In this chapter you will compare exact-match VLOOKUP, HLOOKUP and INDEX/MATCH; distinguish a missing key from bad input.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 80 minutes: 15 explanation/demonstration, 37 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 65 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Exact-match lookup examples and an explicit missing-item result.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Use exact-match lookups.
- Compare vertical and horizontal lookup layouts.
- Distinguish missing keys from broken formulas.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Orders and Rates. Rates A2:D7 has unique item codes and numeric unit costs.

5.1 Look Up a Rate

1. In a spare Orders column labelled `LookupRate`, enter `=VLOOKUP(D2,Rates!$A$2:$D$7,3,FALSE)` and fill down.
2. Compare each looked-up rate with the original F column.
3. In another labelled practice column use `=INDEX(Rates!$C$2:$C$7,MATCH(D2,Rates!$A$2:$A$7,0))` and reconcile the same result.

What should you see?

IT01 returns 45 and IT06 returns 8. Both formulas use exact matches.

5.2 Practise a Horizontal Lookup

1. On Analysis G1:L1 enter IT01 through IT06. In G2:L2 enter their unit costs 45, 12, 30, 18, 25 and 8.
2. Put IT03 in F4 and enter `=HLOOKUP(F4,$G$1:$L$2,2,FALSE)` in G4.
3. Change F4 to IT06, check 8, then restore IT03.

What should you see?

The horizontal lookup returns 30 for IT03 and uses the same explicit matching principle.

5.3 Handle an Expected Missing Key

1. In a spare test cell use item code IT99. Observe the lookup's missing-match error.
2. Wrap only the intended missing-match case, for example `=IFNA(VLOOKUP("IT99",Rates!$A$2:$D$7,3,FALSE),"Unmapped item")`.
3. Deliberately misspell a function in another spare cell, observe the different error and repair it. Discuss how a blanket IFERROR can hide defects as well as expected missing matches.

What should you see?

An unknown item is labelled Unmapped item. A broken formula is repaired rather than treated as a zero price.

5.4 Challenge Exact Lookups With Missing and Duplicate Keys

1. Create a **Lookup Tests** table containing all six known ItemCodes, IT99, a blank and a space-padded key. Preserve raw input next to any normalised key.
2. For each known key compare exact VLOOKUP with INDEX/MATCH and the supplied Rates value. Count and investigate disagreements instead of hiding them.
3. Create **Rates Duplicate Test** as a disposable Rates copy. Append a second IT03 with a different UnitCost. Observe which result a first-match lookup returns.
4. Add a key-count diagnostic so multiple matches are flagged before accepting a looked-up amount. Record that lookup success does not prove uniqueness.
5. Reorder a copied Rates table and compare exact versus omitted-match-mode VLOOKUP. Keep approximate-match experiments clearly separated from the authoritative mapping.
6. Compare a missing-key error, a bad function name and a bad range reference. Apply IFNA to the intended missing-match case and explain how broad IFERROR could conceal other defects.
7. Reconcile a known item's lookups across vertical, horizontal and INDEX/MATCH methods and write the range/column assumptions of each.

What should you see?

Known/missing/blank/padded-key test matrix with exact-method reconciliation. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Design an independent lookup output with Rate, MatchCount and Decision for the mixed intake. A zero rate, unknown key and duplicate key must remain distinguishable.
2. Move a column only on a disposable mapping copy and inspect which lookup formula needs repair. Explain why the original column-index assumption failed.
3. Hand over a rejected-key list and source-mapping check, leaving baseline Rates unchanged.

Evidence to Hand Over

- Known/missing/blank/padded-key test matrix with exact-method reconciliation.
- Duplicate-key and column-move diagnosis on labelled copies.
- Independent lookup result that never silently turns unknown rates into zero.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A lookup returns a plausible but wrong rate	Inspect exact-match mode, key uniqueness, whitespace and the lookup column position.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Reliable lookups depend on clean keys, explicit match behaviour and deliberate error handling.

Check yourself: Does an exact-match lookup prove that a matched source value is populated?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
LookupRate
```

```
=VLOOKUP(D2,Rates!$A$2:$D$7,3,FALSE)
```

```
=INDEX(Rates!$C$2:$C$7,MATCH(D2,Rates!$A$2:$A$7,0))
```

```
=HLOOKUP(F4,$G$1:$L$2,2,FALSE)
```

```
=IFNA(VLOOKUP("IT99",Rates!$A$2:$D$7,3,FALSE),"Unmapped item")
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Lookup challenge: Known IT01 and IT06; unknown IT99; blank; space-padded IT03. Duplicate-test mapping adds IT03 with UnitCost 99 only on a disposable copy. Baseline IT03 remains 30. Keep the supplied six-key Rates source intact.
```

06

CHAPTER 6

Complete the Formula Clinic

In this chapter you will debug supplied defects, explain the correction and reconcile a checked sample.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 65 minutes: 15 explanation/demonstration, 22 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 50 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Three repaired defects with independent baseline reconciliations.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Diagnose formula defects.
- Test zero and missing values.
- Reconcile source totals independently.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use a challenge copy of your advanced workbook. Keep the baseline original untouched.

6.1 Create Three Test Defects

1. On the challenge copy, remove FALSE from a copied VLOOKUP, shift one multiplication reference to the next row, and shorten one SUM range by a row.
2. Record the intended formula behaviour before fixing anything.
3. Inspect the formula bar, not just the visible result, for each defect.

What should you see?

You have three clearly identified test defects; a plausible number is not assumed correct.

6.2 Repair and Test

1. Restore exact matching and the correct same-row multiplication. Extend the total to every source row.
2. Test a temporary zero quantity and check that its amount becomes a valid zero. Restore the quantity.
3. Test an unknown item code in a separate practice cell and confirm it remains an explicit missing match.

What should you see?

The repair handles ordinary rows, valid zero and a missing key distinctly.

6.3 Reconcile and Explain

1. Sum line amounts from all 24 rows and reconcile to 3,158. Filter Delivered rows and reconcile their amount to 1,478.
2. Compare the conditional summary with a manually inspected subset.
3. Write one sentence per defect explaining the cause and why your test detects it.

What should you see?

The baseline is restored and both independent control totals agree.

6.4 Run an Independent Formula Fault Clinic

1. Create a defect register for wrong match mode, shifted row reference and omitted total row. Include expected behaviour, observed symptom, hypothesis and proof.
2. Preserve faulty and repaired formulas side by side on the clinic copy so a reviewer can see the specific change.
3. Use a small manually calculated subset to distinguish a logic error from a display-format difference. Do not repair by typing the desired result.
4. Add the duplicate-key and numeric-text cases from the card as separate tests. Decide whether each problem belongs in source cleaning, lookup validation or formula logic.
5. For every repair, test one normal input and one boundary/exception input. Record actual results and any unresolved issue.
6. Reconcile baseline all-orders and Delivered totals after restoration, plus a record-count control. Keep cancelled rows in the source.
7. Write a brief troubleshooting decision path: inspect source → define population → inspect references → test boundary → reconcile result.

What should you see?

Defect register with preserved faulty/repaired formulas and test evidence. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Ask a partner to introduce a defect on a separate clinic copy without telling you its cell; if alone, use the card and hide your defect note before diagnosis.
2. Locate it through controls, isolate the cause, repair it and run a regression check on a previously correct population.
3. Present the evidence trail in two minutes, including why your chosen control would catch the defect on a future refresh.

Evidence to Hand Over

- Defect register with preserved faulty/repaired formulas and test evidence.
- Normal/boundary checks for every repair and restored baseline controls.
- Independent diagnosis, regression check and an explainable troubleshooting path.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The total is corrected but the defect remains	Require a formula/source repair and a new input test rather than a hard-coded total.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

A useful formula check tests meaning and boundaries rather than merely the absence of an error message.

Check yourself: Which of your checks would catch a missing final row?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

This chapter uses the supplied files and visible controls. No additional prompt or typed code is required.

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Extra fault choices, clinic copy only: change a numeric quantity to text “four”; insert a duplicate lookup key; replace a blank/missing-rate outcome with 0; shorten one delivered summary range by two rows. Diagnose the cause before restoring from Source Snapshot.
```

07

CHAPTER 7

Summarise with Pivot Tables

In this chapter you will group project costs and status; inspect filters, calculated fields and source coverage.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 75 minutes: 15 explanation/demonstration, 32 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 60 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A native pivot with checked filters and a calculated-field demonstration or recorded limitation.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Create a native pivot summary.
- Choose filters and aggregation deliberately.
- Check a calculated field against row calculations.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Orders A1:K25 after calculating Amount in J and ActiveAmount in K. A recovery route is to repeat Advanced Chapter 2 on the supplied workbook.

7.1 Build the Pivot

1. Select Orders A1:K25 and use Insert, Pivot table. Place it on a new tab named `Pivot Review`.
2. Add Project as rows and Status as columns. Add Amount as values, summarised by SUM.
3. Inspect totals and confirm the grand total is 3,158.

What should you see?

The pivot shows all projects and statuses. It has not discarded cancelled records without a stated filter.

7.2 Filter and Compare

1. Filter the pivot to Delivered only and check the total becomes 1,478.
2. Compare the result with the earlier SUMIFS or filtered source calculation.
3. Remove the pivot filter and verify the all-orders total returns.

What should you see?

Filtered and unfiltered results answer different documented questions.

7.3 Inspect a Calculated Field

1. Create a separate practice pivot using the same source. Group by Project.
2. Add a calculated field for average line amount, using `=SUM(Amount)/COUNTA(OrderID)` and **Summarise by**, **Custom** if supported by the target UI. Compare with a source calculation. Retain SUM of the row-level Amount as the authoritative total.
3. If that calculated-field expression is rejected, retain the working Amount source column and record the limitation for the trainer; do not replace it with an unrelated metric.

What should you see?

The calculated-field demonstration is reconciled to the existing row calculation, or explicitly recorded as unavailable.

7.4 Reconcile Pivot Semantics and New-Row Coverage

1. Create a second pivot counting OrderID by Project and Status. Compare its population count with the 24-row source and its category counts with COUNTIFS.
2. Double-click a populated pivot value where supported to inspect its contributing source rows. Reconcile the amount from those rows independently.
3. Use a safe calculated field such as `=SUM(Amount)/COUNTA(OrderID)` with **Summarise by**, **Custom**, where the target UI supports it. Label this as average line amount and compare with a source calculation.
4. Do not replace SUM of row Amount with SUM(Quantity)*SUM(UnitCost). Calculate both on a small two-row practice table and explain the different meanings.
5. On a separate pivot challenge copy append the supplied PO025 record. Inspect whether the fixed pivot source range includes it.
6. Extend the source range and inspect both the count and amount pivots. Compare them with a repaired formula summary.
7. Record pivot source range, filters, aggregation and calculated-field definition in an analysis note. If the calculated field is unavailable, retain source Amount and document that specific UI check as pending.

What should you see?

Amount and count pivots reconciled to formulas and contributing source rows. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Build a fresh pivot independently answering Delivered amount by Project, with a count of contributing order lines.
2. On the challenge copy include PO025 and show the source row, count, total and changed project contribution. Leave the baseline pivot unchanged.
3. Explain one aggregation trap and use contributing rows to prove the correct result.

Evidence to Hand Over

- Amount and count pivots reconciled to formulas and contributing source rows.
- Safe calculated-field comparison or documented unavailable route.
- Separate new-record pivot with repaired coverage and demonstrated aggregation semantics.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A pivot total is wrong even though the row appears	Inspect aggregation, filters, source range and calculated-field meaning separately.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Native pivots summarise a defined source. Their filters and formulas need the same checks as ordinary calculations.

Check yourself: Why must the pivot's source range include every record and calculated column?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
Pivot Review
```

```
=SUM(Quantity*UnitCost)
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
PO025, challenge copy only: PJ-A | Planning | IT01 | Quantity 2 | UnitCost 45 | Delivered |
week 4 | Additional practice line. Amount is quantity times cost. Extend both formulas and
pivot source range; do not alter the baseline workbook.
```

08

CHAPTER 8

Query Structured Criteria

In this chapter you will use DSUM, DCOUNT and DAVERAGE with a visible criteria range and reconcile against a manual subset.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 60 minutes: 15 explanation/demonstration, 17 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 45 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A visible criteria area and reconciled database function results.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Build a visible database criteria range.
- Use DSUM, DCOUNT and DAVERAGE.
- Reconcile criteria-based results.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Orders A1:J25, including Amount. Keep criteria outside the source table.

8.1 Build the Criteria

1. On Analysis type Status in A10 and Delivered in A11. Match the Orders header exactly.
2. In C10 type `Delivered amount`. In D10 enter `=DSUM(Orders!A1:J25,"Amount",A10:A11)`.
3. Compare the result with the delivered control total.

What should you see?

DSUM returns 1,478 using the visible Delivered criterion.

8.2 Count and Average

1. In C11 type `Delivered numeric quantity records`; in D11 enter `=DCOUNT(Orders!A1:J25,"Quantity",A10:A11)`.
2. In C12 type `Average delivered unit cost`; in D12 enter `=DAVERAGE(Orders!A1:J25,"UnitCost",A10:A11)`.
3. Discuss why DCOUNT of the text OrderID field would not count order IDs like a text-aware count.

What should you see?

DCOUNT counts numeric Quantity entries in qualifying records. The average is of unit costs, not a quantity-weighted price.

8.3 Check Criteria Semantics

1. Filter the source to Delivered and manually inspect a small sample contributing to each result.
2. Add Project as a second criteria header beside Status and use PJ-A on the same criteria row to mean both conditions.
3. Compare with SUMIFS for the same project/status pair. Restore or clearly label the expanded criteria.

What should you see?

The database function and ordinary conditional formula agree when their criteria describe the same subset.

8.4 Prove AND, OR and Overlapping Criteria

1. Create three separate criteria blocks with exact source headers: Delivered AND PJ-A; Delivered OR Pending; and Delivered OR PJ-A. Label each natural-language population.
2. Use same-row criteria for the AND block and separate criteria rows for OR blocks. Keep blocks spatially separate from result cells.
3. Calculate DSUM, DCOUNT of numeric Quantity and DAVERAGE of UnitCost for each block. Record how blanks or text could affect the latter two metrics.
4. For the AND block reconcile amount with SUMIFS and list matching OrderIDs using a filter view.
5. For Delivered OR Pending reconcile with the sum of two disjoint status totals. For Delivered OR PJ-A inspect the overlap so a record satisfying both conditions is not counted twice.
6. Compare an unweighted mean UnitCost with a quantity-weighted cost measure on a small explicit test. State which question each answers.
7. Deliberately misspell one criteria header on a disposable block. Record the observed error or result and repair it using the exact source heading.

What should you see?

Three labelled criteria blocks with source-ID reconciliation. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Write a new analysis question requiring two conditions and design its database criteria without copying a finished block.
2. Create a second version that uses OR instead of AND, predict the population difference and verify it through contributing IDs.
3. Explain the numeric field counted by DCOUNT, the average's denominator and one overlap risk.

Evidence to Hand Over

- Three labelled criteria blocks with source-ID reconciliation.
- AND/OR and overlap proof, plus weighted/unweighted meaning comparison.
- Independent question and corrected criteria-header test.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
An OR total double-counts overlapping rows	Compare the contributing ID sets and count each qualifying source record once.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Database functions are useful when their header matching, criteria logic and field meaning are explicit.

Check yourself: Why is an average unit cost different from a quantity-weighted average?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
Delivered amount
```

```
=DSUM(Orders!A1:J25,"Amount",A10:A11)
```

```
Delivered numeric quantity records
```

```
=DCOUNT(Orders!A1:J25,"Quantity",A10:A11)
```

```
Average delivered unit cost
```

```
=DAVERAGE(Orders!A1:J25,"UnitCost",A10:A11)
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Criteria cases: Status=Delivered AND Project=PJ-A; Status=Delivered OR Status=Pending;
Status=Delivered OR Project=PJ-A. The last case overlaps. DCOUNT uses numeric Quantity;
OrderID is text and must not be treated as a numeric record count.
```

09

CHAPTER 9

Build a Dashboard

In this chapter you will choose charts, sparklines and filter views; distinguish source-linked charts from generated analysis snapshots.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 70 minutes: 15 explanation/demonstration, 27 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 55 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A compact source-linked dashboard with a tested update path.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Choose source-linked visuals.
- Use a sparkline for a defined time sequence.
- Explain how a dashboard refreshes.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use the checked advanced workbook with all-orders and delivered controls reconciled.

9.1 Build a Compact View

1. Create a Dashboard tab. Link a small project summary to the checked analysis or create a fresh pivot from the same source.
2. Add a chart comparing Amount by Project and label whether cancelled orders are included.
3. Add a separate status count view with clear labels. Keep detailed source records in Orders.

What should you see?

The dashboard identifies the question and population behind each number.

9.2 Show the Weekly Sequence

1. On Analysis create week labels 1, 2, 3 and 4 and calculate Amount totals for each with SUMIFS.
2. On Dashboard create a labelled sparkline using the four weekly result cells, for example `=SPARKLINE(Analysis!B16:E16)` after placing the totals there.
3. Check that the sequence is chronological and its totals reconcile to all-orders Amount.

What should you see?

The sparkline describes the four supplied practice weeks, not an invented real calendar period.

9.3 Test a Refresh

1. Temporarily change one source quantity. Check the line amount, summary, chart and sparkline.
2. Inspect whether a pivot has updated and whether its source still includes all rows.
3. Restore the source quantity. Document any manual refresh step and keep filters visible or clearly named.

What should you see?

The dashboard returns to the reconciled baseline after restoration.

9.4 Design a Dashboard With a Visible Refresh Contract

1. Write a dashboard brief specifying audience, population, units, three useful questions and the source range. Do not mix all-orders and active totals without labels.
2. Add KPI cells for record count, all-orders Amount, ActiveAmount and Delivered Amount. Reconcile each to the corresponding checked analysis.
3. Show project amount comparison, status count and weekly sequence with explicit filter scope. Keep week order numeric and label it as practice week.
4. Create a visible **Checks** block comparing KPI totals with source controls and indicating any mismatch. It must recalculate from cells rather than contain typed “OK” text.
5. On a scenario copy change PO001 Quantity from its baseline to 4. Predict the amount delta and check the source line, summary, chart and weekly sequence.
6. Restore the quantity and prove all affected values return. Record which components follow source-cell changes and which depend on range extension for new rows.
7. Prepare a refresh contract listing source bounds, pivot bounds, formula bounds, chart ranges, filter scope and snapshot/export steps.

What should you see?

Defined dashboard with distinct populations, units and recalculating controls. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Build an independent dashboard view answering the three audience questions on the card, with at least one visible reconciliation check.
2. Append PO025 only on a challenge copy and repair source-range coverage across KPIs, pivots, charts and the weekly sequence.
3. Ask a reviewer to identify the population and a stale component using the labels/checks, then revise any misleading element.

Evidence to Hand Over

- Defined dashboard with distinct populations, units and recalculating controls.
- Source-change/restoration trace across line, summary and visuals.
- Independent new-row coverage test and a usable refresh contract.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A dashboard looks current but omits the new line	Trace range coverage through every KPI and visual rather than checking only one total.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

A dashboard is useful when its totals, populations and update path are understandable.

Check yourself: Which dashboard objects depend on which source calculations?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
=SPARKLINE(Analysis!B16:E16)
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Audience questions: Which project has the largest Delivered amount? How much Amount remains
Pending? Which practice week has the largest all-orders Amount? PO025 extension: PJ-A,
Planning, IT01, Quantity 2, UnitCost 45, Delivered, week 4.
```

10

CHAPTER 10

Use Gemini as an Analysis Partner

In this chapter you will generate and inspect formula proposals and chart summaries; test exceptions before accepting.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 55 minutes: 10 explanation/demonstration, 25 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 45 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A checked AI formula proposal and evidence-based findings.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Ask AI for a bounded formula proposal.
- Verify a generated finding.
- Distinguish a source-linked result from a generated snapshot.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use the native checked workbook and usable Gemini in Sheets access. If unavailable, critique the supplied examples manually and record the AI objective as incomplete.

10.1 Request a Formula Proposal

1. Open Ask Gemini. Paste the Part 2 prompt describing the actual source columns and range.
2. Read the proposed formula before applying it. Check its row bounds, exact status spelling and use of Amount.
3. Test the formula in a spare labelled cell and compare with the known Delivered total of 1,478.

What should you see?

The proposal is accepted only after it matches the actual schema and control total.

10.2 Check an Interpretation

1. Ask for two findings using the checked project/status summary, without web research.
2. For each finding, identify the cells that support it. Reject unsupported causal claims, forecasts or benefit percentages.
3. Write a final finding in your own words with its population and units.

What should you see?

Each retained finding is traceable to values rather than AI confidence.

10.3 Inspect Generated Visuals

1. If asking Gemini to generate a chart, inspect where its data was written and what range the chart references.
2. Temporarily change a source quantity and observe whether that generated analysis follows the original source.
3. Restore the quantity. Label a static/generated analysis snapshot appropriately or use your already checked source-linked chart.

What should you see?

The final dashboard does not imply automatic refresh where it has not been demonstrated.

10.4 Evaluate AI Proposals Against Known Test Cases

1. Prepare a test matrix before prompting: Delivered total, Pending >=200, unknown ItemCode and a new record outside the original bounds.
2. Ask Gemini for a formula and explanation using the actual headers and range. If unavailable, use the deliberately flawed proposals on the card for the same audit.
3. Inspect every proposed formula's population, field, match mode, bounds and copied-reference behaviour before using it.
4. Run the proposal in labelled spare cells against known controls and a manually verified subset. Record pass/fail and the cause of each failure.
5. Request a correction for one identified defect, or repair it manually. Keep the original proposal alongside the checked replacement so improvement is demonstrable.
6. For any generated chart inspect its data source and test a source-cell change. Record whether it follows the maintained data, separate generated cells or a static snapshot.
7. Write a final two-finding commentary grounded in checked values, rejecting causal claims, forecasts and unsupported ROI.

What should you see?

Prewritten test matrix and observed proposal results. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Design an independent prompt that includes the population, fields, intended output and tests, then run or manually review its result.
2. Challenge it with IT99 and PO025 in separate test contexts. Keep missing mapping distinct from zero and inspect new-row coverage.
3. Produce a short audit comparing original proposal, correction, evidence and remaining limitation.

Evidence to Hand Over

- Prewritten test matrix and observed proposal results.
- Original/corrected formula trace with independent verification.
- Generated-chart source check or stated unavailable route, and supported final findings.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The AI explanation sounds right while the formula differs	Audit the actual formula and contributing rows, not its prose description.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

AI can propose analysis; source tests establish whether it is suitable.

Check yourself: What evidence would establish that a generated chart follows the maintained source?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 2: Propose a Delivered Total

Session: new chat | **Grounding:** explicit Orders schema

```
Goal: Propose a formula for delivered order amount.
Context: Fictional Google Sheets analysis.
Source: Orders rows 2 to 25; Status is G and Amount is J. Header row is 1.
Expectations: Reply with a formula and explanation only. Do not edit cells. Include only
Status equal to Delivered. Use bounded ranges. I will test the formula against the source.
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Flawed formula proposals to audit if AI is unavailable: `=SUM(Orders!J2:J24)` described as all
24 rows; `=IFERROR(VLOOKUP("IT99",Rates!A2:D7,3,FALSE),0)` described as a confirmed zero rate;
`=SUMIFS(Orders!J2:J25,Orders!G2:G25,"Delivered")` described as active amount excluding only
Cancelled.
```

11

CHAPTER 11

Complete the Analysis Capstone

In this chapter you will answer a new project question, substantiate the finding and deliver a checked dashboard.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 80 minutes: 15 explanation/demonstration, 37 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 65 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A source-traceable capstone analysis and checked presentation.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Answer a new analysis question.
- Reconcile a result independently.
- Communicate assumptions and limitations.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Create a challenge copy. Use the same 24-order source and six-item Rates table; do not import real purchasing data.

11.1 Define and Calculate

1. Answer: which project has the largest Delivered amount, and which Pending records have Amount at least 200?
2. Write the population and criteria before choosing formulas or a pivot.
3. Calculate the result and retain a visible trace to qualifying order IDs.

What should you see?

The ranking and exception list are based on explicit delivered/pending definitions.

11.2 Challenge the Result

1. Reconcile the delivered total to 1,478 and all-orders total to 3,158.
2. If no baseline pending row qualifies, record zero matches explicitly. Inspect a non-qualifying baseline row and use a separate synthetic row to test the positive boundary without altering the baseline.
3. Test a duplicate or missing lookup key on a disposable copy and explain its effect without changing the baseline workbook.

What should you see?

Your result survives an independent check and identifies meaningful data assumptions.

11.3 Present the Analysis

1. Show one clear chart or table with project names and currency units.
2. State the source population, result and one limitation. Do not equate order amount with profit or budget approval.
3. Explain which sources and calculations must update for a future record.

What should you see?

A reviewer can trace the finding to source records and understand its limits.

11.4 Complete a Decision Analysis With Independent Evidence

1. Write a one-page analysis specification for the card's three questions, naming population, metric, threshold and exclusions.
2. Build answers with formulas and a pivot or filtered-source reconciliation. Retain the contributing OrderIDs for the exception list.
3. Compare project Delivered amounts and identify the largest using checked numbers. Do not infer profitability or project performance from order amount alone.
4. For Pending Amount >=200, first report the actual baseline match count, including zero if none qualify. Inspect a non-qualifying baseline row and use a separate synthetic positive/boundary test. Keep synthetic tests outside the baseline analysis.
5. Run a separate PO025 scenario and describe its effect on ranking, delivered totals and week 4. Do not overwrite the baseline result.
6. Prepare an executive table/chart and an audit appendix showing source range, assumptions, controls and unresolved data issues.
7. Ask a reviewer to challenge one answer and prove it from the source rows and a second calculation method.

What should you see?

Three-question specification and independently reconciled results. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Independently compare all-orders and active populations and explain how excluding Cancelled changes the question and results.
2. Stress-test an unknown rate key and a duplicate lookup key in a scenario copy. Demonstrate the warning/disposition rather than forcing a numeric answer.
3. Deliver a three-minute recommendation limited to what the data supports, plus one next question requiring additional evidence.

Evidence to Hand Over

- Three-question specification and independently reconciled results.
- Separate added-row and key-quality scenarios with audit evidence.
- Executive output, reviewer challenge response and a bounded recommendation.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A recommendation claims savings or profitability	Return to dataset meaning and request missing cost/outcome evidence before making that claim.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

The capstone combines reliable data, explicit calculations and a defensible explanation.

Check yourself: Would the same result follow if Cancelled records were included?

Zero-Match Check

The supplied baseline has no Pending record with Amount at least 200. Record that result explicitly. Use a separately labelled synthetic case to prove the positive branch; do not edit baseline records just to obtain a highlight.

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

This chapter uses the supplied files and visible controls. No additional prompt or typed code is required.

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Capstone questions: Largest Delivered amount by project; Pending records with Amount ⩾200 and
their IDs; highest all-orders Amount by practice week. Scenario PO025: PJ-A, Planning, IT01,
Quantity 2, UnitCost 45, Delivered, week 4. No actual budget or profit data is supplied.
```

12

CHAPTER 12

Review and Handover

In this chapter you will explain assumptions, refresh steps and known data limitations.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 50 minutes: 10 explanation/demonstration, 20 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 40 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A restored baseline and an update-tested workbook handover.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Explain the workbook update path.
- Identify future maintenance work.
- Leave a checked final state.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use your baseline workbook and separate capstone copy.

12.1 Review the Baseline

1. Clear temporary test entries and restore all source values changed during exercises.
2. Confirm the 24-record count and control totals.
3. Inspect filter states and remove or name any view that could mislead the next reader.

What should you see?

The baseline is consistent and temporary tests no longer affect it.

12.2 Write the Handover

1. Record where the source lives, what one row represents and which columns users may edit.

2. Describe formula, validation, pivot and chart ranges that must include a new row.
3. Document unmapped keys and other unresolved data issues rather than converting them to zero.

What should you see?

A future editor can update the analysis without silently omitting records.

12.3 Demonstrate One Update

1. On the capstone copy, add one new fictional order line and walk through the documented update steps.
2. Check the resulting calculations and visuals, then clearly label that copy as an extension test.
3. Keep the original baseline for future class recovery.

What should you see?

The handover has been tested on an actual extension in a separate copy.

12.4 Make the Analysis Operable by Another Analyst

1. Create a **Runbook** with source location, row grain, accepted inputs, editable fields, formula columns, exception handling and release controls.
2. Create a range inventory for validation, conditional formatting, SUMIFS, lookups, database functions, pivots, chart sources and weekly summaries used in your workbook.
3. On a separately named extension copy add PO025 and follow only the runbook to extend the analysis. Record every instruction that needed interpretation.
4. Reconcile the 25-record scenario by count, all-orders total, Delivered total and project/week contribution, then compare with the preserved 24-record baseline.
5. Test the unknown-key route without overwriting the supplied Rates mapping. Record the issue and leave a visible unresolved disposition.
6. Give a partner the runbook and ask them to locate a source, run a check and explain a failed key. If alone, use a fresh read-through with previous working tabs closed.
7. Repair one handover gap and issue a manifest identifying the maintained baseline, scenario test and current fixed summary.

What should you see?

Complete range inventory and a runbook tested on PO025. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Use the different PO026 line on the card in another disposable copy and update the analysis without returning to the lesson steps.

2. Verify how Pending amount and exception flags change, including any new-row coverage failure. Record and repair the cause.
3. Hand over the runbook, controls, open issues and manifest. Another analyst must be able to reproduce both a successful update and an unresolved-key outcome.

Evidence to Hand Over

- Complete range inventory and a runbook tested on PO025.
- Independent PO026 update with reconciled controls and exception coverage.
- Recorded handover repair, open-issue route and clear scenario manifest.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The next analyst must guess a range extension	Record exact current bounds, the new-row action and the expected checks in the runbook.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

You can maintain and hand over a checked analysis rather than a collection of unexplained formulas.

Check yourself: Which assumption would you check first if a future total unexpectedly drops?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

This chapter uses the supplied files and visible controls. No additional prompt or typed code is required.

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Independent PO026, separate baseline-derived copy: PJ-B | Procurement | IT03 | Quantity 10 | UnitCost 30 | Pending | week 4 | Handover exception test. Do not combine it with PO025 unless you explicitly define and reconcile that additional scenario.
```

Google Sheets Advanced

A practical 2-day course with step-by-step draft labs, copy-ready prompts, fictional inputs and expected-result checks. Use the supplied practice pack. Full lab execution, screenshots and classroom timing validation remain required before participant release.

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